

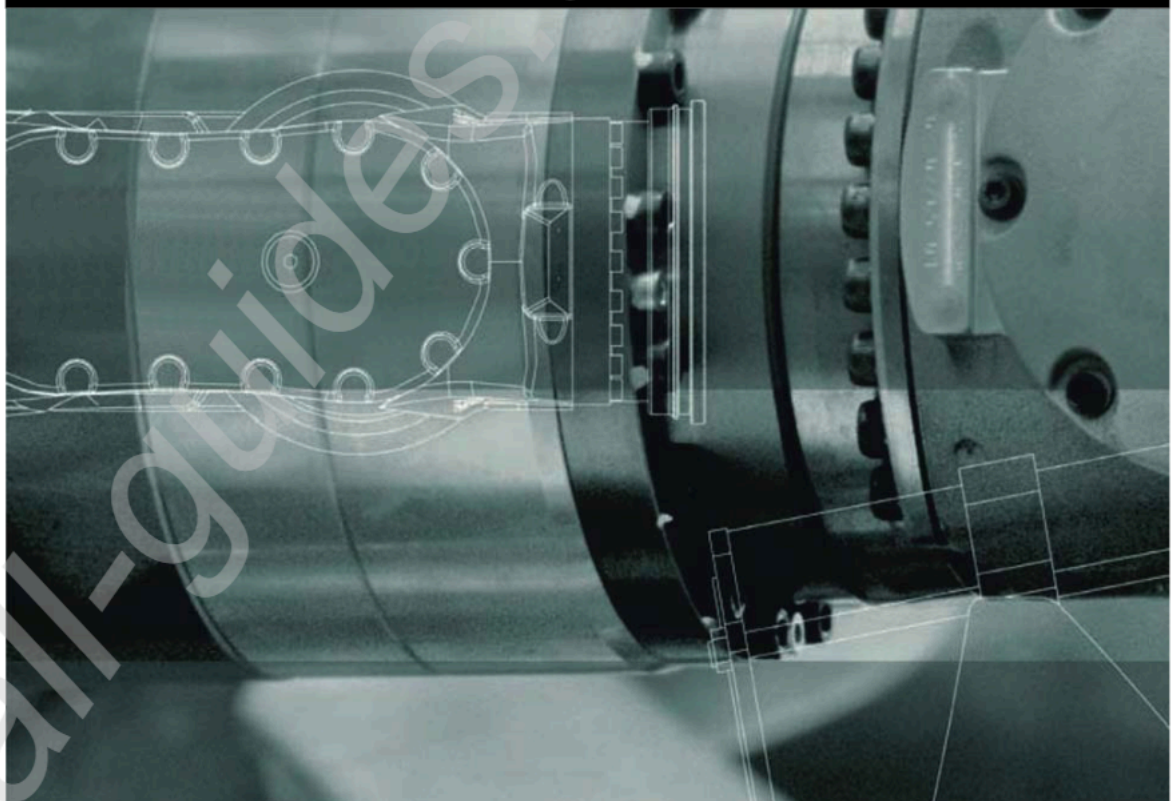
# KUKA

**Robots**

KUKA Roboter GmbH

## **KR 300-2 PA, KR 470-2 PA**

### **Assembly Instructions**



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Version: MA KR 300 470-2 PA V5



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Other functions not described in this documentation may be operable in the controller. The user has no claims to these functions, however, in the case of a replacement or service work.

We have checked the content of this documentation for conformity with the hardware and software described. Nevertheless, discrepancies cannot be precluded, for which reason we are not able to guarantee total conformity. The information in this documentation is checked on a regular basis, however, and necessary corrections will be incorporated in the subsequent edition.

Subject to technical alterations without an effect on the function.

Translation of the original documentation

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# 1 Introduction

## 1.1 Industrial robot documentation

The industrial robot documentation consists of the following parts:

- Documentation for the manipulator
- Documentation for the robot controller
- Operating and programming instructions for the System Software
- Instructions for options and accessories
- Parts catalog on storage medium

Each of these sets of instructions is a separate document.

## 1.2 Representation of warnings and notes

### Safety

These warnings are relevant to safety and **must** be observed.



**DANGER** These warnings mean that it is certain or highly probable that death or severe injuries **will** occur, if no precautions are taken.



**WARNING** These warnings mean that death or severe injuries **may** occur, if no precautions are taken.



**CAUTION** These warnings mean that minor injuries **may** occur, if no precautions are taken.



**NOTICE** These warnings mean that damage to property **may** occur, if no precautions are taken.



These warnings contain references to safety-relevant information or general safety measures.  
These warnings do not refer to individual hazards or individual precautionary measures.

This warning draws attention to procedures which serve to prevent or remedy emergencies or malfunctions:



**SAFETY INSTRUCTIONS** Procedures marked with this warning **must** be followed exactly.

### Notices

These notices serve to make your work easier or contain references to further information.



**i** Tip to make your work easier or reference to further information.

[all-guides.com](http://all-guides.com)

## 2 Purpose

### 2.1 Target group

This documentation is aimed at users with the following knowledge and skills:

- Advanced knowledge of mechanical engineering
- Advanced knowledge of electrical and electronic systems
- Knowledge of the robot controller system



For optimal use of our products, we recommend that our customers take part in a course of training at KUKA College. Information about the training program can be found at [www.kuka.com](http://www.kuka.com) or can be obtained directly from our subsidiaries.

### 2.2 Intended use

#### Use

The industrial robot is intended for handling tools and fixtures, or for processing or transferring components or products. Use is only permitted under the specified environmental conditions.

#### Misuse

Any use or application deviating from the intended use is deemed to be misuse and is not allowed. This includes e.g.:

- Transportation of persons and animals
- Use as a climbing aid
- Operation outside the permissible operating parameters
- Use in potentially explosive environments
- Use in underground mining

**NOTICE**

Changing the structure of the manipulator, e.g. by drilling holes, etc., can result in damage to the components. This is considered improper use and leads to loss of guarantee and liability entitlements.

**NOTICE**

Deviations from the operating conditions specified in the technical data or the use of special functions or applications can lead to premature wear. KUKA Roboter GmbH must be consulted.



The robot system is an integral part of a complete system and may only be operated in a CE-compliant system.



### 3 Product description

#### 3.1 Overview of the robot system

The industrial robot consists of the following components:

- Manipulator
- Robot controller
- Teach pendant
- Connecting cables
- Software
- Options, accessories

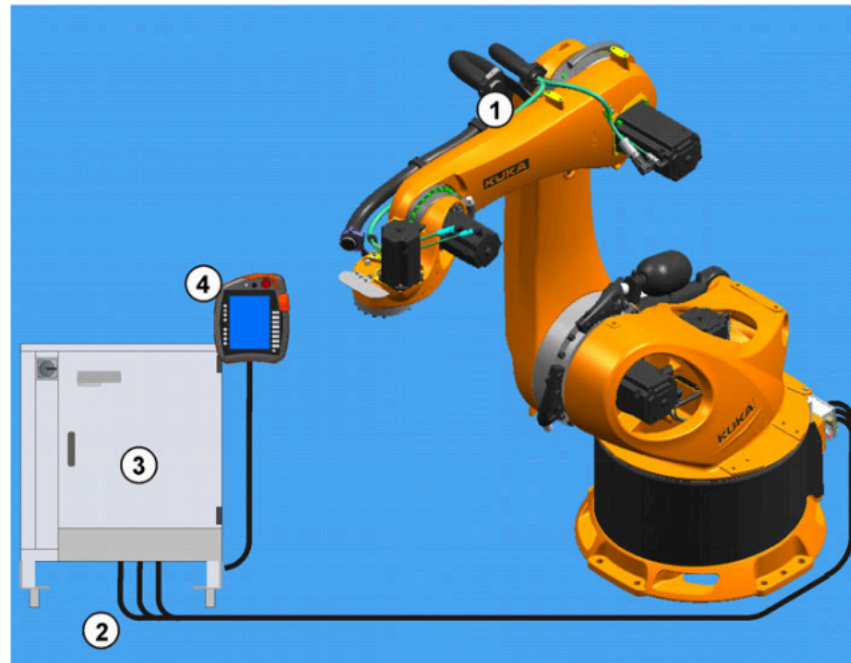


Fig. 3-1: Robot system KR 300, 470-2 PA

- |                     |                                     |
|---------------------|-------------------------------------|
| 1 Robot             | 3 KR C4 robot controller            |
| 2 Connecting cables | 4 Teach pendant KCP (KUKA smartPAD) |

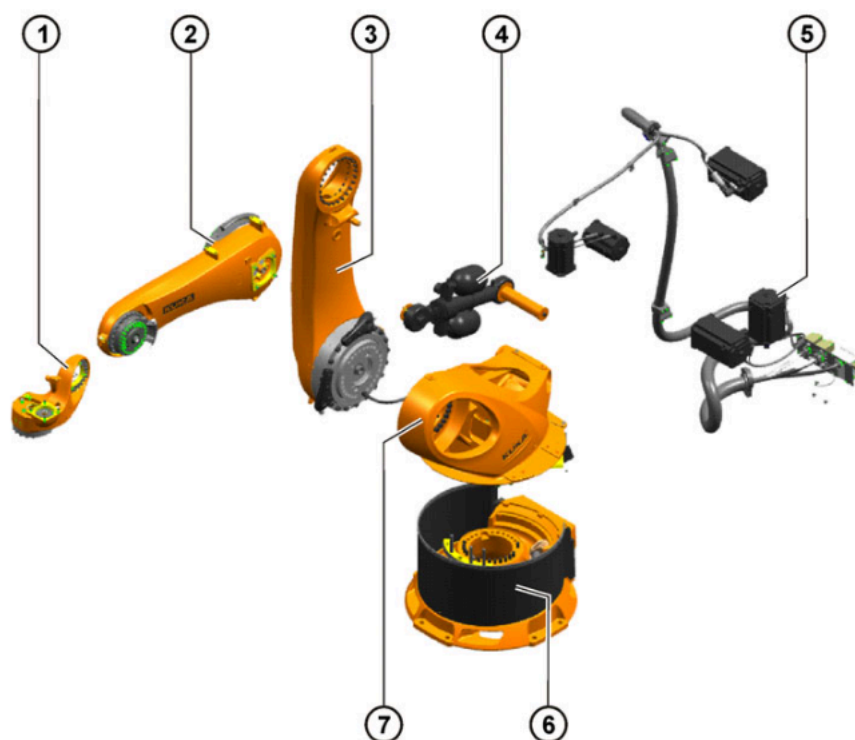
#### 3.2 Description of the robot

##### Overview

This robot is designed as a 5-axis jointed-arm kinematic system. The structural components of the robots are made of light alloy and iron castings. The axes are driven by AC servomotors. A hydropneumatic counterbalancing system is used to equalize the load moment about axis 2.

The robot consists of the following principal components (>>> Fig. 3-2 ):

- Wrist
- Arm
- Link arm
- Rotating column
- Base frame
- Counterbalancing system
- Electrical installations



**Fig. 3-2: KR 300 470-2 PA: main assemblies**

- |   |                         |   |                          |
|---|-------------------------|---|--------------------------|
| 1 | Wrist                   | 5 | Electrical installations |
| 2 | Arm                     | 6 | Base frame               |
| 3 | Link arm                | 7 | Rotating column          |
| 4 | Counterbalancing system |   |                          |

### Wrist

The robot variants KR 300 PA and KR 470 PA robots are equipped with a two-axis wrist for rated payloads of 300 kg and 470 kg, respectively. The wrist is fastened to the arm via a gear unit and motor and is driven by these. The main components of the wrist are the swing frame, axis 6 motor and the corresponding gear unit. The mounting flange embodies the output side of axis 6. The motor unit consists of a brushless AC servomotor with a permanent-magnet single-disk brake and hollow-shaft resolver, both integrated. The permanent-magnet single-disk brake performs a holding function when the servomotor is at rest and contributes to the braking of axis 6 in the event of short-circuit braking (e.g. if one or more of the enabling switches is released while in Test mode). Short-circuit braking must not be used to stop the robot under normal circumstances. End effectors can be attached to the mounting flange of axis 6. The wrist is designed as a hollow-shaft wrist and features a through-hole with a diameter of 60 mm.

The assembly also has a gauge mount with a gauge cartridge, through which the mechanical zero of the axis can be determined by means of a dial gauge or an electronic probe (accessory) and transferred to the controller.

### Arm

The arm is the transmission element between the wrist and the link arm. The swing frame of the wrist is mounted on the arm via gear unit A5. This motor/gear combination embodies axis 5, which cannot be freely controlled during operation. The arm is driven by an AC servomotor via a gear unit that is installed between the arm and the link arm. This gear unit is also the bearing for the arm. The maximum permissible swivel angle is mechanically limited by a stop for each direction, plus and minus. The buffers are attached to the arm. The corresponding stops are situated on the link arm.

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Link arm</b>                 | The link arm is the assembly located between the arm and the rotating column. It is mounted on one side of the rotating column via the gear unit of axis 2 and is driven by an AC servomotor. During motion about axis 2, the link arm moves about the stationary rotating column. The cable harness of the electrical installations is routed inside the link arm and is mounted in hinged clamps.                                                                                                                      |
| <b>Rotating column</b>          | The rotating column houses the motors of axes 1 and 2. The rotational motion of axis 1 is performed by the rotating column. It is screwed to the base frame via the gear unit of axis 1. The AC servomotor for driving axis 1 is mounted inside the rotating column. The counterbearing for the counterbalancing system is integrated into the rear of the rotating column housing. The fork slots are also screwed to the rotating column if the robot is transported using a fork lift truck.                          |
| <b>Base frame</b>               | The base frame is the base of the robot. It is screwed to the mounting base. The interfaces for the electrical installations and the energy supply systems (accessory) are housed in the base frame. The base frame and rotating column are connected via the gear unit of axis 1. The flexible tube for the electrical installations and the energy supply system is accommodated in the base frame.                                                                                                                    |
| <b>Counterbalancing system</b>  | The counterbalancing system is installed between the rotating column and the link arm and serves to minimize the moments generated about axis 2 when the robot is in motion and at rest. A closed, hydropneumatic system is used. The system consists of two accumulators, a hydraulic cylinder with associated hoses, a pressure gauge and an accumulator safety valve as a safety element to protect against overload. The accumulators correspond to category II, fluid group 2, of the Pressure Equipment Directive. |
| <b>Electrical installations</b> | The electrical installations are described in Chapter .                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Options</b>                  | The robot can be fitted and operated with various options, such as energy supply systems for axes 1 to 3, energy supply systems for axes 3 to 6, or working range limitation systems. The options are described in separate documentation.                                                                                                                                                                                                                                                                               |



## 4 Technical data

### 4.1 Basic data

#### Basic data

|                                |                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------|
| Type                           | KR 300-2 PA<br>KR 470-2 PA                                                              |
| Number of axes                 | 5                                                                                       |
| Work envelope volume           | 73.5 m <sup>3</sup>                                                                     |
| Pose repeatability (ISO 9283)  | KR 300-2 PA ±0.08 mm<br>KR 470-2 PA ±0.08 mm                                            |
| Work envelope reference point  | Intersection of axis 6 with the mounting flange face                                    |
| Reach                          | 3,150 mm                                                                                |
| Weight of robot                | 2,150 kg                                                                                |
| Weight of transport frame      | 212 kg                                                                                  |
| Principal dynamic loads        | See "Loads acting on the mounting base"                                                 |
| Protection rating of the robot | IP65<br>ready for operation, with connecting cables plugged in (according to EN 60529)  |
| Protection rating of the wrist | IP65                                                                                    |
| Sound level                    | < 72 dB (A) outside the working envelope                                                |
| Mounting position              | Ground                                                                                  |
| Surface finish, paint-work     | Base (stationary) and counterbalancing system: RAL 9005; moving parts: KUKA orange 2567 |

#### Ambient temperature, standard

|                            |                                                                                                                                                             |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operation                  | 273 K to 328 K (0 °C to +55 °C)                                                                                                                             |
| Storage and transportation | 233 K to 333 K (-40 °C to +60 °C)                                                                                                                           |
| Start-up                   | In the case of start-up in the range of 273 K to 288 K (0 °C to +15 °C), the robot may have to be warmed up. Other temperature limits available on request. |
| Ambient conditions         | DIN EN 60721-3-3,<br>Class 3K3                                                                                                                              |

Further information about temperatures, duty cycle, set-up, etc. can be obtained from KUKA Technical Support. No restrictions in operating mode T1.

#### Connecting cables

| Cable designation | Connector designation | Interface with robot            |
|-------------------|-----------------------|---------------------------------|
| Motor cable       | X20.1 - X30.1         | Harting connectors at both ends |
| Motor cable       | X20.4 - X30.4         | Harting connectors at both ends |

| Cable designation                                                                               | Connector designation | Interface with robot           |
|-------------------------------------------------------------------------------------------------|-----------------------|--------------------------------|
| Control cable                                                                                   | X21 - X31             | HAN 3A EMC at both ends        |
| Ground conductor / equipotential bonding<br>16 mm <sup>2</sup><br>(can be ordered as an option) |                       | M8 ring cable lug at both ends |

| Cable lengths |                             |
|---------------|-----------------------------|
| Standard      | 7 m, 15 m, 25 m, 35 m, 50 m |

| Minimum bending radius | 10x D |
|------------------------|-------|
|------------------------|-------|

For detailed specifications of the connecting cables, see (>>> 6.4 "Connecting cables and interfaces" Page 58).

## 4.2 Axis data

### Axis data

The following data are valid for the robot KR 300-2 PA.

| Axis | Range of motion, software-limited | Speed with rated payload |
|------|-----------------------------------|--------------------------|
| 1    | +/-185°                           | 97.5 °/s                 |
| 2    | +20° to -130°                     | 91.0 °/s                 |
| 3    | +155° to -0°*                     | 89.0 °/s                 |
| 5    | Axis not actively selectable      |                          |
| 6    | +/-350°                           | 177.0 °/s                |

The following data are valid for the robot KR 470-2 PA.

| Axis | Range of motion, software-limited | Speed with rated payload |
|------|-----------------------------------|--------------------------|
| 1    | +/-185°                           | 84.0 °/s                 |
| 2    | +20° to -130°                     | 78.0 °/s                 |
| 3    | +155° to -0°*                     | 73.0 °/s                 |
| 5    | Axis not actively selectable      |                          |
| 6    | +/-350°                           | 177.0 °/s                |

\* Maximum value, referred to the link arm, depending on the position of axis 2

The direction of motion and the arrangement of the individual axes may be noted from the diagram (>>> Fig. 4-1 ).

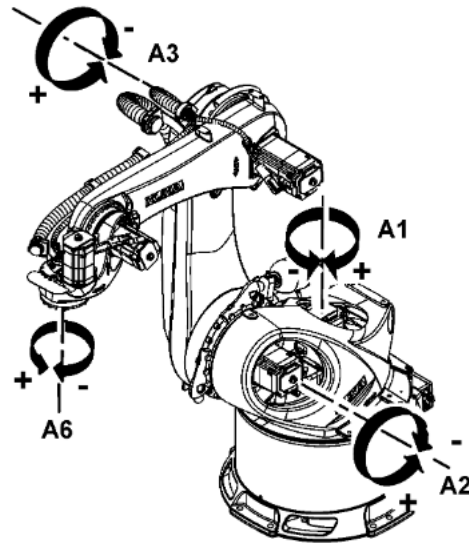


Fig. 4-1: Direction of rotation of robot axes

#### Working envelope

The diagram (>>> Fig. 4-2 ) shows the shape and size of the working envelope.

The reference point for the working envelope is the intersection of axis 6 with the mounting flange face.

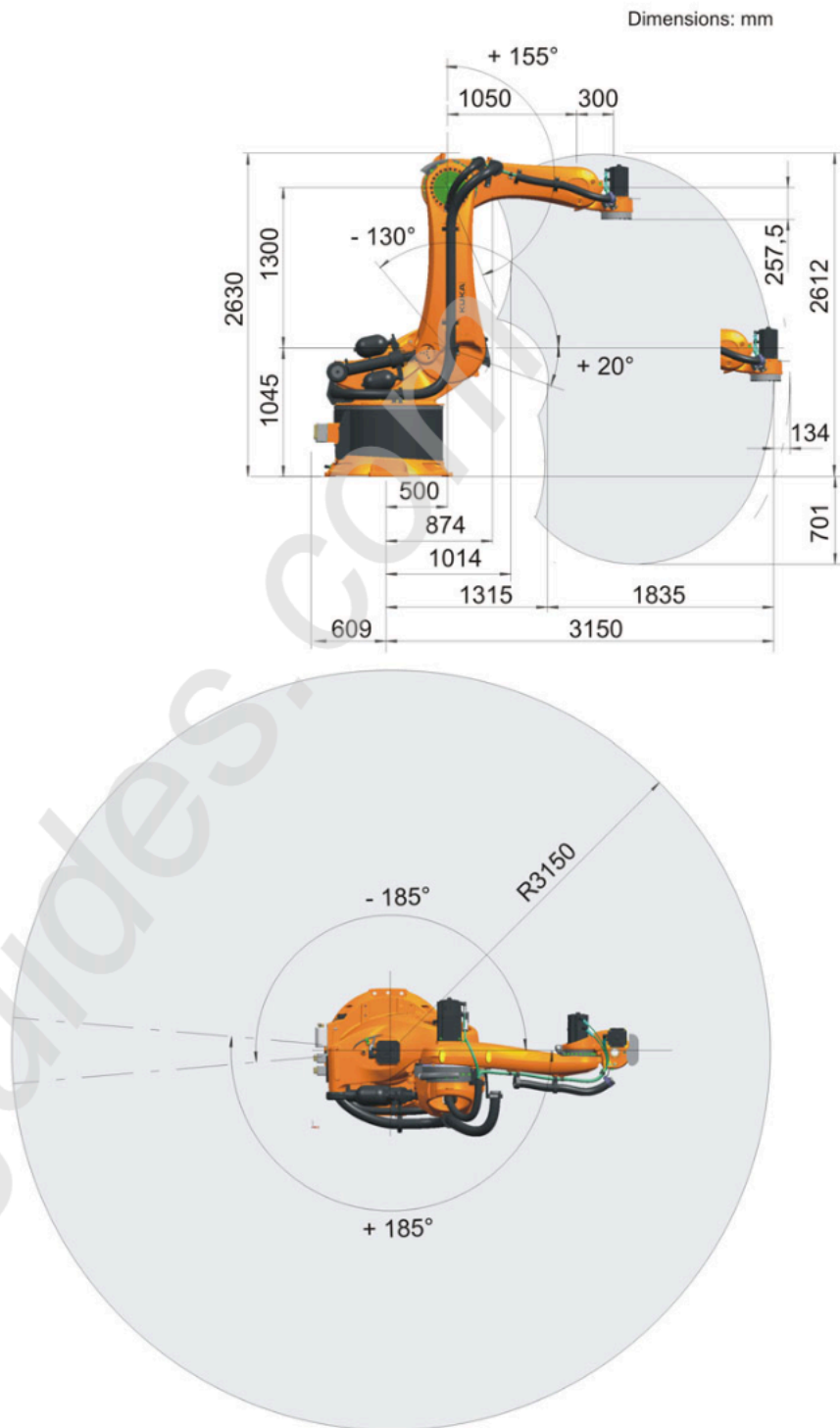


Fig. 4-2: Working envelope: KR 300 470 PA

### 4.3 Payloads

#### Payloads

|       |                    |                    |
|-------|--------------------|--------------------|
| Robot | KR 300-2 PA        | KR 470-2 PA        |
| Wrist | Hollow-shaft wrist | Hollow-shaft wrist |

|                                                 |                      |                      |
|-------------------------------------------------|----------------------|----------------------|
| Rated payload                                   | 300 kg               | 470 kg               |
| Distance of the load center of gravity $L_z$    | 300 mm               | 300 mm               |
| Distance of the load center of gravity $L_{xy}$ | 100 mm               | 100 mm               |
| Permissible moment of inertia                   | 150 kgm <sup>2</sup> | 235 kgm <sup>2</sup> |
| Max. total load                                 | 350 kg               | 520 kg               |
| Supplementary load, arm                         | 50 kg                | 50 kg                |
| Supplementary load, link arm                    | 0 kg                 | 0 kg                 |
| Supplementary load, rotating column             | 0 kg                 | 0 kg                 |

### Load center of gravity P

For all payloads, the load center of gravity refers to the distance from the face of the mounting flange on axis 6. Refer to the payload diagram for the nominal distance.

### Payload diagram

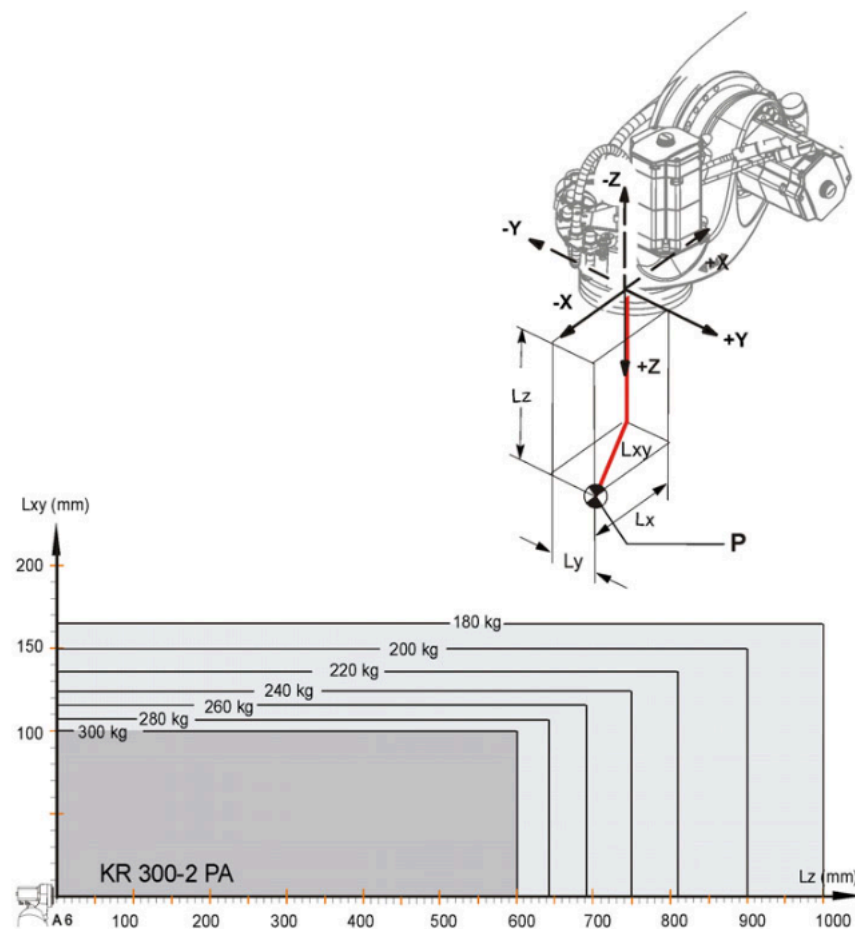


Fig. 4-3: Payload diagram for KR 300-2 PA

**NOTICE**

This loading curve corresponds to the maximum load capacity. Both values (payload and mass moment of inertia) must be checked in all cases. Exceeding this capacity will reduce the service life of the robot and overload the motors and the gears; in any such case the KUKA Roboter GmbH must be consulted beforehand.

The values determined here are necessary for planning the robot application. For commissioning the robot, additional input data are required in accordance with the operating and programming instructions of the KUKA System Software.

The mass inertia must be verified using KUKA.Load. It is imperative for the load data to be entered in the robot controller!

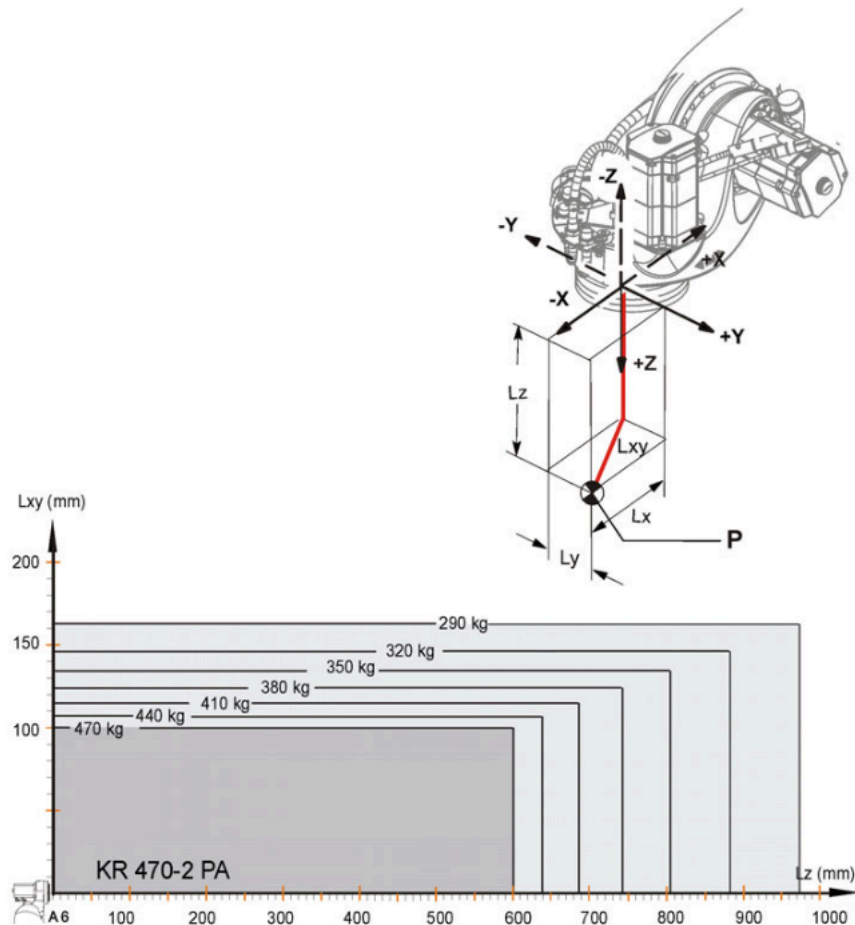


Fig. 4-4: Payload diagram for KR 470-2 PA

#### Mounting flange

|                                       |                                 |
|---------------------------------------|---------------------------------|
| Mounting flange                       | similar to DIN/ISO 9409-1-A160* |
| Screw grade                           | 10.9                            |
| Screw size                            | M12                             |
| Grip length                           | 1.5 x nominal diameter          |
| Depth of engagement                   | min. 15 mm, max. 19.5 mm        |
| Locating element                      | 10 <sup>H7</sup>                |
| Through-hole for energy supply system | ø 60 mm                         |

\*The inner locating diameter is ø 125<sup>H7</sup>. This deviates from the standard.

The mounting flange is depicted (>>> Fig. 4-5 ) with axis 6 in the zero position. The symbol  $X_m$  indicates the position of the locating element (bushing) in the zero position.

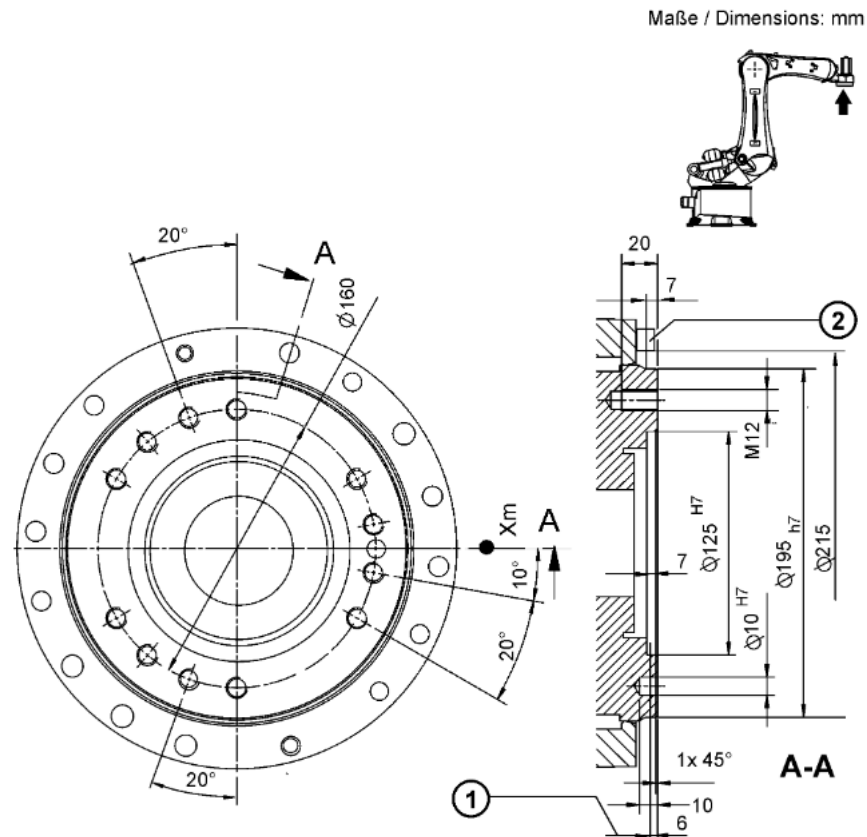


Fig. 4-5: Mounting flange

- 1 Fitting length
- 2 Disruptive contour of screw heads

#### Supplementary load

The robot can carry supplementary loads on the arm. When mounting the supplementary loads, be careful to observe the maximum permissible total load. The dimensions and positions of the installation options can be seen in the diagram. All other threads and holes on the robot are not suitable for attaching additional loads.

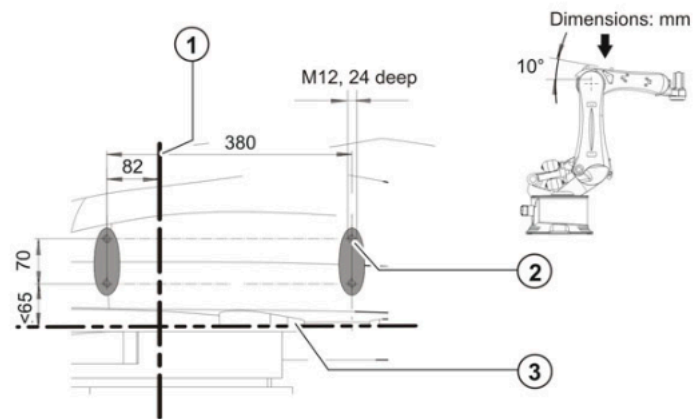


Fig. 4-6: Supplementary load on arm

- |   |                  |   |                               |
|---|------------------|---|-------------------------------|
| 1 | Axis 3           | 3 | Interference contour on right |
| 2 | Fastening thread |   |                               |

#### 4.4 Mounting base data

##### Loads acting on the mounting base

The specified forces and moments already include the payload and the inertia force (weight) of the robot.

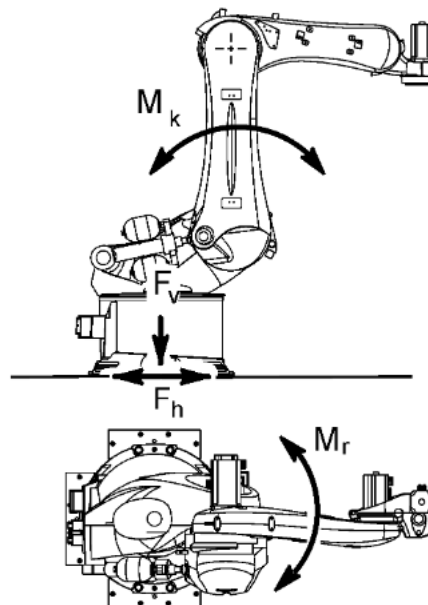


Fig. 4-7: Loads acting on the foundation

| Type of load             | Force/torque/mass                                                                   |
|--------------------------|-------------------------------------------------------------------------------------|
| $F_v$ = vertical force   | $F_{v \text{ normal}} = 36,000 \text{ N}$<br>$F_{v \text{ max}} = 40,500 \text{ N}$ |
| $F_h$ = horizontal force | $F_{h \text{ normal}} = 10,500 \text{ N}$<br>$F_{h \text{ max}} = 23,500 \text{ N}$ |

| Type of load                   | Force/torque/mass                                                                     |
|--------------------------------|---------------------------------------------------------------------------------------|
| $M_k$ = tilting moment         | $M_{k \text{ normal}} = 47,000 \text{ Nm}$<br>$M_{k \text{ max}} = 84,500 \text{ Nm}$ |
| $M_r$ = torque                 | $M_{r \text{ normal}} = 20,500 \text{ Nm}$<br>$M_{r \text{ max}} = 45,500 \text{ Nm}$ |
| Total mass for foundation load | for KR 300 PA: 2,290 kg<br>for KR 470: 2,410 kg                                       |
| Robot                          | 1,940 kg                                                                              |
| Total mass for foundation load | for KR 300 PA: 350 kg<br>for KR 470 PA: 520 kg                                        |



The supplementary loads are not taken into consideration in the calculation of the foundation load. These supplementary loads must be taken into consideration for  $F_v$ .

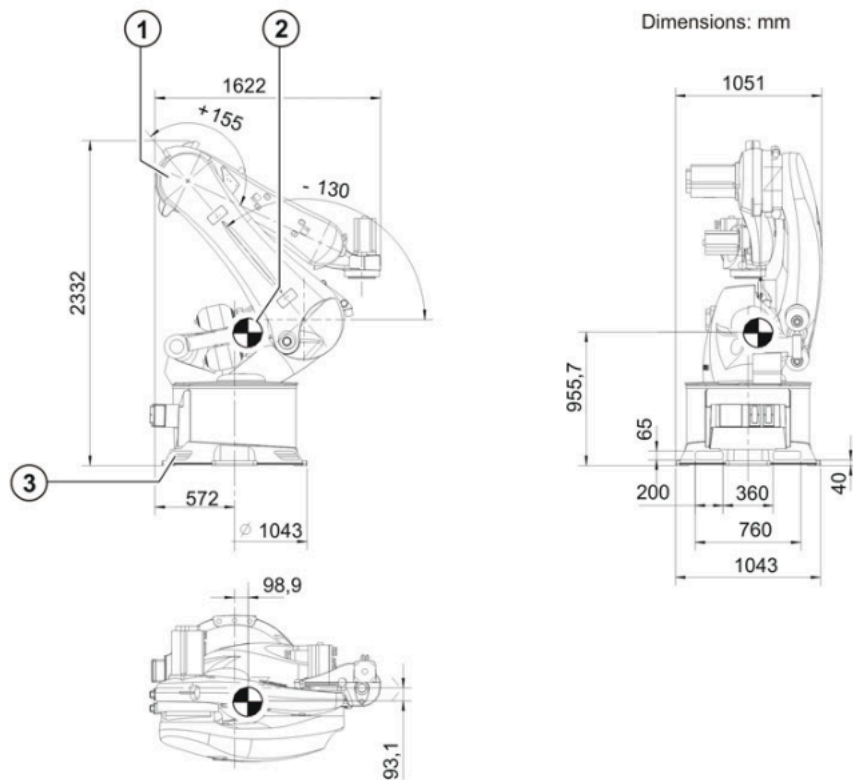
#### Grade of concrete for foundations

When producing foundations from concrete, observe the load-bearing capacity of the ground and the country-specific construction regulations. There must be no layers of insulation or screed between the bedplates and the concrete foundation. The quality of the concrete must meet the requirements of the following standard:

- C20/25 according to DIN EN 206-1:2001/DIN 1045-2:2008

#### 4.5 Transport dimensions

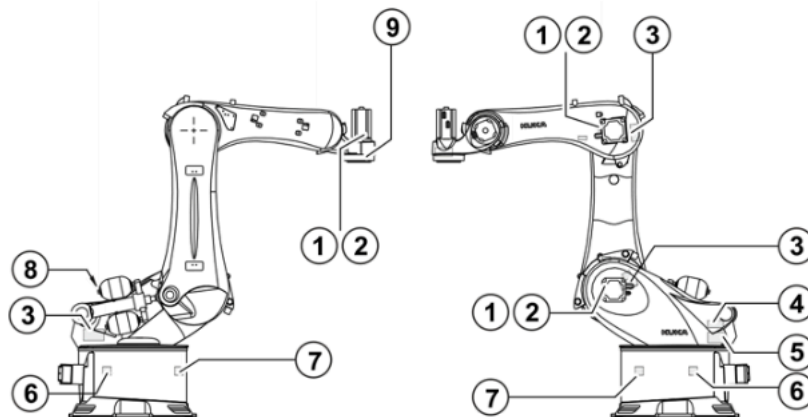
The transport dimensions (>>> Fig. 4-8 ) for the robot can be noted from the following diagram. The position of the center of gravity and the weight vary according to the specific configuration. The specified dimensions refer to the robot without equipment. The following diagram shows the dimensions of the robot when it stands on the floor without wooden transport blocks.

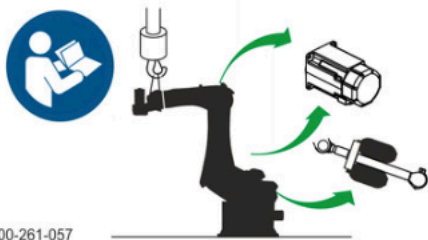
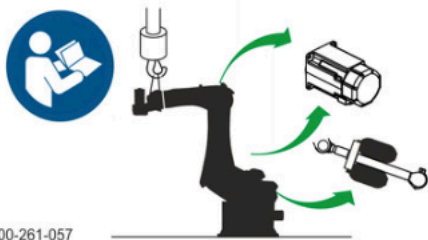
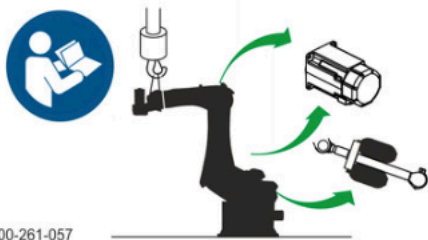
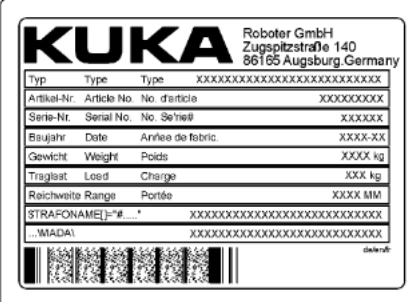
**Fig. 4-8: Transport dimensions**

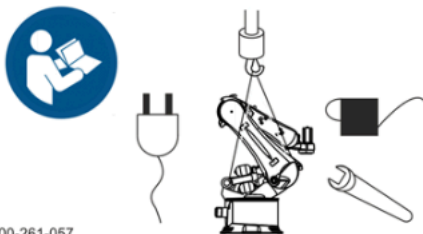
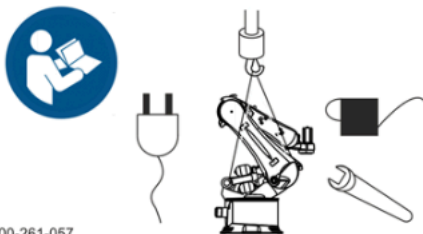
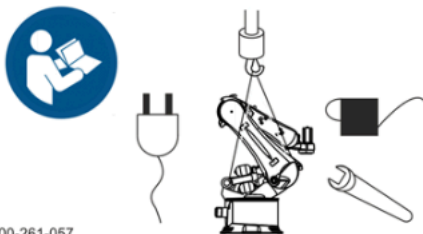
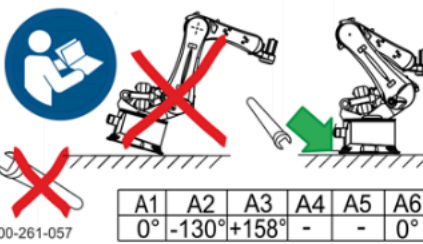
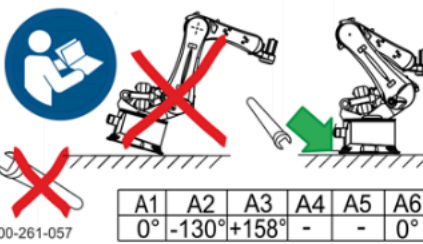
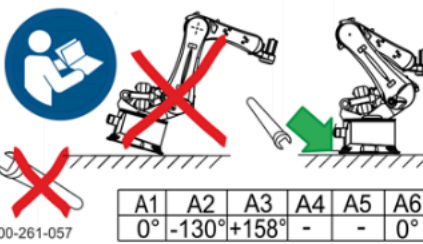
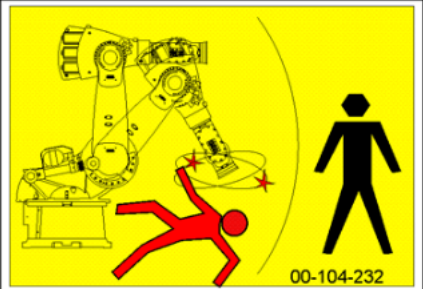
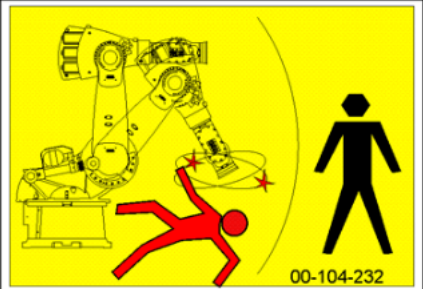
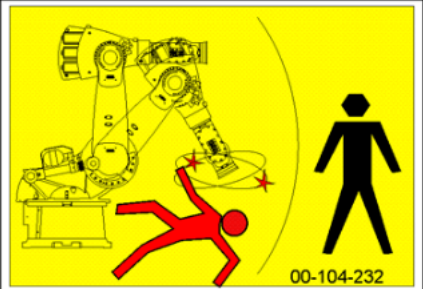
- 1 Robot
- 2 Center of gravity
- 3 Fork slots

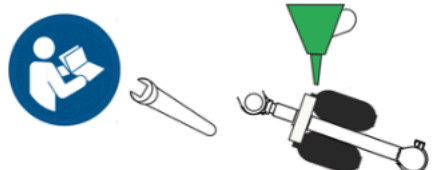
#### 4.6 Plates and labels

**Plates and labels** The following plates and labels are attached to the robot. They must not be removed or rendered illegible. Illegible plates and labels must be replaced.

**Fig. 4-9: Location of plates and labels**

| Item                                                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                   |                                                                                    |                                                                                                   |                                                                                                                  |                                                                                                      |                                                                                                                           |                                                                                                     |                                                                                                            |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1                                                                                                    |  <p><b>High voltage</b><br/>Any improper handling can lead to contact with current-carrying components. Electric shock hazard!</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                   |                                                                                    |                                                                                                   |                                                                                                                  |                                                                                                      |                                                                                                                           |                                                                                                     |                                                                                                            |
| 2                                                                                                    |  <p><b>Hot surface</b><br/>During operation of the robot, surface temperatures may be reached that could result in burn injuries. Protective gloves must be worn!</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                   |                                                                                    |                                                                                                   |                                                                                                                  |                                                                                                      |                                                                                                                           |                                                                                                     |                                                                                                            |
| 3                                                                                                    | <table border="1"> <tr> <td data-bbox="570 688 841 953">  </td><td data-bbox="841 688 1297 953">  </td></tr> <tr> <td data-bbox="570 953 841 1026">  <b>CAUTION</b> </td><td data-bbox="841 953 1297 1026"> <p><b>Before removing the motor or counterbalancing system, secure robot axis to prevent it from moving!</b></p> </td></tr> <tr> <td data-bbox="570 1026 841 1100">  <b>ATTENTION</b> </td><td data-bbox="841 1026 1297 1100"> <p><b>Avant de retirer le moteur ou le système d'équilibrage, protéger l'axe du robot pour éviter tout mouvement!</b></p> </td></tr> <tr> <td data-bbox="570 1100 841 1173">  <b>VORSICHT</b> </td><td data-bbox="841 1100 1297 1173"> <p><b>Vor Entfernen des Motors oder des Gewichtsausgleichs, Roboterachse gegen Bewegungen sichern!</b></p> </td></tr> </table> <p><b>Secure the axes</b><br/>Before exchanging any motor or counterbalancing system, secure the corresponding axis through safeguarding by suitable means/ devices to protect against possible movement. The axis can move. Risk of crushing!</p> |  |  |  <b>CAUTION</b> | <p><b>Before removing the motor or counterbalancing system, secure robot axis to prevent it from moving!</b></p> |  <b>ATTENTION</b> | <p><b>Avant de retirer le moteur ou le système d'équilibrage, protéger l'axe du robot pour éviter tout mouvement!</b></p> |  <b>VORSICHT</b> | <p><b>Vor Entfernen des Motors oder des Gewichtsausgleichs, Roboterachse gegen Bewegungen sichern!</b></p> |
|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                   |                                                                                    |                                                                                                   |                                                                                                                  |                                                                                                      |                                                                                                                           |                                                                                                     |                                                                                                            |
|  <b>CAUTION</b>    | <p><b>Before removing the motor or counterbalancing system, secure robot axis to prevent it from moving!</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                   |                                                                                    |                                                                                                   |                                                                                                                  |                                                                                                      |                                                                                                                           |                                                                                                     |                                                                                                            |
|  <b>ATTENTION</b> | <p><b>Avant de retirer le moteur ou le système d'équilibrage, protéger l'axe du robot pour éviter tout mouvement!</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                   |                                                                                    |                                                                                                   |                                                                                                                  |                                                                                                      |                                                                                                                           |                                                                                                     |                                                                                                            |
|  <b>VORSICHT</b>  | <p><b>Vor Entfernen des Motors oder des Gewichtsausgleichs, Roboterachse gegen Bewegungen sichern!</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                   |                                                                                    |                                                                                                   |                                                                                                                  |                                                                                                      |                                                                                                                           |                                                                                                     |                                                                                                            |
| 4                                                                                                    |  <p><b>Identification plate</b><br/>Content according to Machinery Directive.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                   |                                                                                    |                                                                                                   |                                                                                                                  |                                                                                                      |                                                                                                                           |                                                                                                     |                                                                                                            |

| Item                                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------|
| 5                                                                                                  | <table><tr><td></td><td><br/>00-261-057</td></tr><tr><td><b>CAUTION</b></td><td>Secure the system before beginning work on the robot. Read and observe the safety instructions!</td></tr><tr><td><b>ATTENTION</b></td><td>Bloquer le système avant d'effectuer des travaux sur le robot. Lire et respecter les remarques relatives à la sécurité!</td></tr><tr><td><b>VORSICHT</b></td><td>Vor Arbeiten am Roboter, System sichern. Sicherheitshinweise lesen und beachten!</td></tr></table> <p><b>Work on the robot</b><br/>Before start-up, transportation or maintenance, read and follow the assembly and operating instructions.</p> |                   | <br>00-261-057  | <b>CAUTION</b> | Secure the system before beginning work on the robot. Read and observe the safety instructions! | <b>ATTENTION</b> | Bloquer le système avant d'effectuer des travaux sur le robot. Lire et respecter les remarques relatives à la sécurité! | <b>VORSICHT</b> | Vor Arbeiten am Roboter, System sichern. Sicherheitshinweise lesen und beachten! |
|                   | <br>00-261-057                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
| <b>CAUTION</b>                                                                                     | Secure the system before beginning work on the robot. Read and observe the safety instructions!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
| <b>ATTENTION</b>                                                                                   | Bloquer le système avant d'effectuer des travaux sur le robot. Lire et respecter les remarques relatives à la sécurité!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
| <b>VORSICHT</b>                                                                                    | Vor Arbeiten am Roboter, System sichern. Sicherheitshinweise lesen und beachten!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
| 6                                                                                                  | <table><tr><td></td><td><br/>00-261-057</td></tr><tr><td><b>CAUTION</b></td><td>Move the robot into its transport position before removing the mounting base!</td></tr><tr><td><b>ATTENTION</b></td><td>Amener le robot en position de transport avant de défaire la fixation aux fondations!</td></tr><tr><td><b>VORSICHT</b></td><td>Roboter vor Lösen der Fundamentbefestigung in Tranportstellung bringen!</td></tr></table> <p><b>Transport position</b><br/>Before loosening the bolts of the mounting base, the robot must be in the transport position as indicated in the table. Risk of toppling!</p>                           |                   | <br>00-261-057 | <b>CAUTION</b> | Move the robot into its transport position before removing the mounting base!                   | <b>ATTENTION</b> | Amener le robot en position de transport avant de défaire la fixation aux fondations!                                   | <b>VORSICHT</b> | Roboter vor Lösen der Fundamentbefestigung in Tranportstellung bringen!          |
|                   | <br>00-261-057                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
| <b>CAUTION</b>                                                                                     | Move the robot into its transport position before removing the mounting base!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
| <b>ATTENTION</b>                                                                                   | Amener le robot en position de transport avant de défaire la fixation aux fondations!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
| <b>VORSICHT</b>                                                                                    | Roboter vor Lösen der Fundamentbefestigung in Tranportstellung bringen!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
| 7                                                                                                  | <table><tr><td><br/>00-104-232</td></tr></table> <p><b>Danger zone</b><br/>Entering the danger zone of the robot is prohibited if the robot is in operation or ready for operation. Risk of injury!</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <br>00-104-232 |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |
| <br>00-104-232 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                    |                                                                                                   |                |                                                                                                 |                  |                                                                                                                         |                 |                                                                                  |

| Item                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------|-------------------------|-------------------------|--------------------------|------------------------------|-------------------|------------------|------------|-------------------------|------------------------|------------------------------|----------------------------|------------------|
| 8                          | <div style="display: flex; justify-content: space-around; align-items: center;">  <div style="text-align: center;">  <p><b>p &gt; 14 MPa (140 bar)</b></p> <p>00-261-057</p> </div> </div> <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="background-color: yellow; text-align: center;"><b>⚠ CAUTION</b></td><td>Counterbalancing system pressurized – read and observe safety instructions before beginning work!</td></tr> <tr> <td style="background-color: yellow; text-align: center;"><b>⚠ ATTENTION</b></td><td>Le système d'équilibrage est sous pression, lire et respecter les remarques relatives à la sécurité avant d'effectuer des travaux!</td></tr> <tr> <td style="background-color: yellow; text-align: center;"><b>⚠ VORSICHT</b></td><td>Gewichtsausgleich unter Druck, vor Arbeiten Sicherheitshinweise lesen und beachten!</td></tr> </table> <p><b>Counterbalancing system</b></p> <p>The system is pressurized with oil and nitrogen. Read and follow the assembly and operating instructions before commencing work on the counterbalancing system. Risk of injury!</p> | <b>⚠ CAUTION</b> | Counterbalancing system pressurized – read and observe safety instructions before beginning work! | <b>⚠ ATTENTION</b>     | Le système d'équilibrage est sous pression, lire et respecter les remarques relatives à la sécurité avant d'effectuer des travaux! | <b>⚠ VORSICHT</b> | Gewichtsausgleich unter Druck, vor Arbeiten Sicherheitshinweise lesen und beachten! |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>⚠ CAUTION</b>           | Counterbalancing system pressurized – read and observe safety instructions before beginning work!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>⚠ ATTENTION</b>         | Le système d'équilibrage est sous pression, lire et respecter les remarques relatives à la sécurité avant d'effectuer des travaux!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>⚠ VORSICHT</b>          | Gewichtsausgleich unter Druck, vor Arbeiten Sicherheitshinweise lesen und beachten!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| 9                          | <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td><b>Schrauben</b></td><td><b>M12 Qualität 10.9</b></td></tr> <tr> <td><b>Einschraubtiefe</b></td><td><b>min. 15mm max. 19,5mm</b></td></tr> <tr> <td><b>Klemmlänge</b></td><td><b>min. 15mm</b></td></tr> <tr> <td><b>Fastening screws</b></td><td><b>M12 quality 10.9</b></td></tr> <tr> <td><b>Engagement length</b></td><td><b>min. 15mm max. 19,5mm</b></td></tr> <tr> <td><b>Screw grip</b></td><td><b>min. 15mm</b></td></tr> <tr> <td><b>Vis</b></td><td><b>M12 qualité 10.9</b></td></tr> <tr> <td><b>Longueur vissée</b></td><td><b>min. 15mm max. 19,5mm</b></td></tr> <tr> <td><b>Longueur de serrage</b></td><td><b>min. 15mm</b></td></tr> </table> <p style="text-align: right; font-size: small;">Art.Nr. 00-180-008</p> <p><b>Mounting flange on in-line wrist</b></p> <p>The values specified on this plate apply for the installation of tools on the mounting flange of the wrist and must be observed.</p>                                                                                                                                                                                                                                                                                                | <b>Schrauben</b> | <b>M12 Qualität 10.9</b>                                                                          | <b>Einschraubtiefe</b> | <b>min. 15mm max. 19,5mm</b>                                                                                                       | <b>Klemmlänge</b> | <b>min. 15mm</b>                                                                    | <b>Fastening screws</b> | <b>M12 quality 10.9</b> | <b>Engagement length</b> | <b>min. 15mm max. 19,5mm</b> | <b>Screw grip</b> | <b>min. 15mm</b> | <b>Vis</b> | <b>M12 qualité 10.9</b> | <b>Longueur vissée</b> | <b>min. 15mm max. 19,5mm</b> | <b>Longueur de serrage</b> | <b>min. 15mm</b> |
| <b>Schrauben</b>           | <b>M12 Qualität 10.9</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>Einschraubtiefe</b>     | <b>min. 15mm max. 19,5mm</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>Klemmlänge</b>          | <b>min. 15mm</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>Fastening screws</b>    | <b>M12 quality 10.9</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>Engagement length</b>   | <b>min. 15mm max. 19,5mm</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>Screw grip</b>          | <b>min. 15mm</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>Vis</b>                 | <b>M12 qualité 10.9</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>Longueur vissée</b>     | <b>min. 15mm max. 19,5mm</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |
| <b>Longueur de serrage</b> | <b>min. 15mm</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                                                                                                   |                        |                                                                                                                                    |                   |                                                                                     |                         |                         |                          |                              |                   |                  |            |                         |                        |                              |                            |                  |

## 4.7 Stopping distances and times

### 4.7.1 General information

Information concerning the data:

- The stopping distance is the angle traveled by the robot from the moment the stop signal is triggered until the robot comes to a complete standstill.
- The stopping time is the time that elapses from the moment the stop signal is triggered until the robot comes to a complete standstill.
- The data are given for the main axes A1, A2 and A3. The main axes are the axes with the greatest deflection.
- Superposed axis motions can result in longer stopping distances.
- Stopping distances and stopping times in accordance with DIN EN ISO 10218-1, Annex B.
- Stop categories:
  - Stop category 0 » STOP 0
  - Stop category 1 » STOP 1
 according to IEC 60204-1

- The values specified for Stop 0 are guide values determined by means of tests and simulation. They are average values which conform to the requirements of DIN EN ISO 10218-1. The actual stopping distances and stopping times may differ due to internal and external influences on the braking torque. It is therefore advisable to determine the exact stopping distances and stopping times where necessary under the real conditions of the actual robot application.
- Measuring technique  
The stopping distances were measured using the robot-internal measuring technique.
- The wear on the brakes varies depending on the operating mode, robot application and the number of STOP 0 triggered. It is therefore advisable to check the stopping distance at least once a year.

#### 4.7.2 Terms used

| Term      | Description                                                                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| m         | Mass of the rated load and the supplementary load on the arm.                                                                                                                                          |
| Phi       | Angle of rotation (°) about the corresponding axis. This value can be entered in the controller via the KCP/smartPAD and can be displayed on the KCP/smartPAD.                                         |
| POV       | Program override (%) = velocity of the robot motion. This value can be entered in the controller via the KCP/smartPAD and can be displayed on the KCP/smartPAD.                                        |
| Extension | Distance (l in %) (>>> Fig. 4-10 ) between axis 1 and the intersection of axes 4 and 5. With parallelogram robots, the distance between axis 1 and the intersection of axis 6 and the mounting flange. |
| KCP       | KUKA Control Panel<br><br>Teach pendant for the KR C2/KR C2 edition2005<br><br>The KCP has all the operator control and display functions required for operating and programming the industrial robot. |
| smartPAD  | Teach pendant for the KR C4<br><br>The smartPAD has all the operator control and display functions required for operating and programming the industrial robot.                                        |

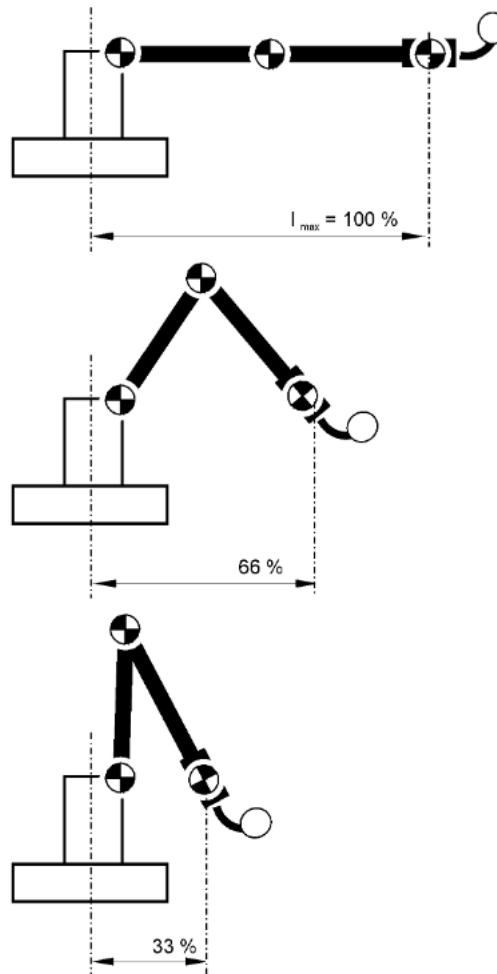


Fig. 4-10: Extension

#### 4.7.3 Stopping distance and stopping times for KR 300 PA

##### 4.7.3.1 Stopping distances and stopping times for STOP 0, axis 1 to axis 3

The table shows the stopping distances and stopping times after a STOP 0 (category 0 stop) is triggered. The values refer to the following configuration:

- Extension  $l = 100\%$
- Program override POV = 100%
- Mass  $m$  = maximum load (rated load + supplementary load on arm)

|        | Stopping distance (°) | Stopping time (s) |
|--------|-----------------------|-------------------|
| Axis 1 |                       |                   |
| Axis 2 |                       |                   |
| Axis 3 |                       |                   |

No data are currently available for STOP 0 for axes 1-3.

## 4.7.3.2 Stopping distances and stopping times for STOP 1, axis 1

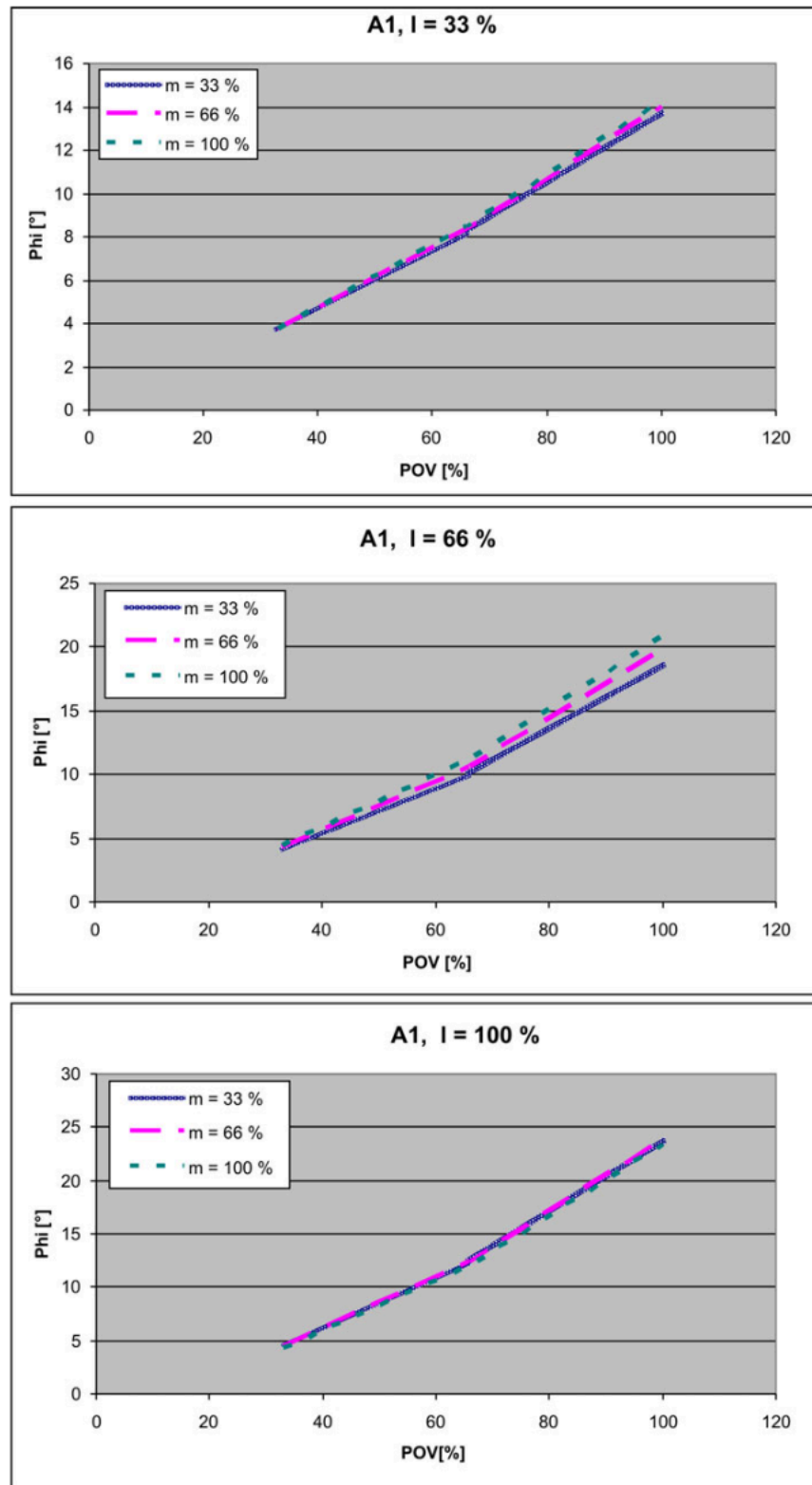


Fig. 4-11: Stopping distances for STOP 1, axis 1

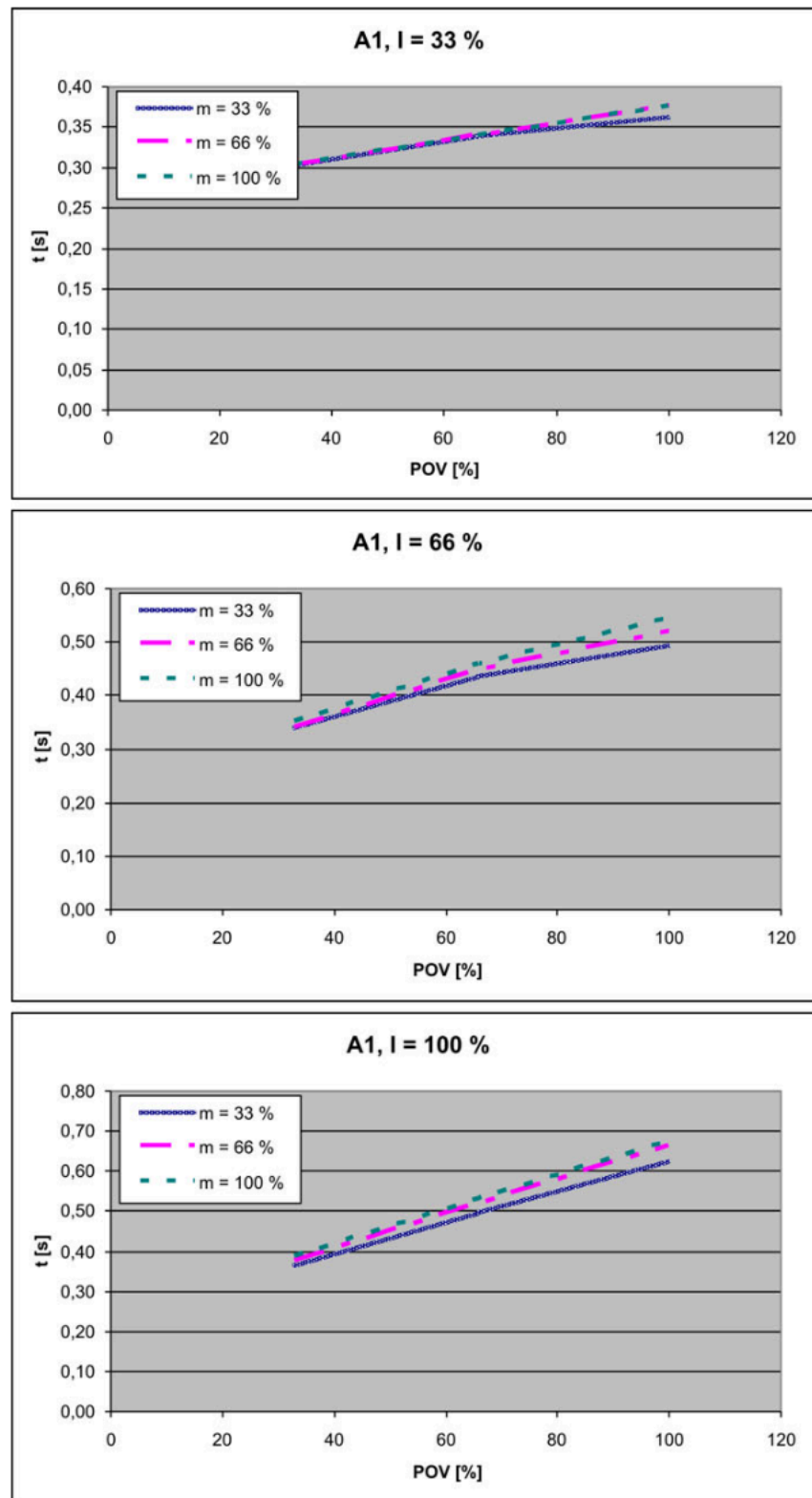


Fig. 4-12: Stopping times for STOP 1, axis 1

## 4.7.3.3 Stopping distances and stopping times for STOP 1, axis 2

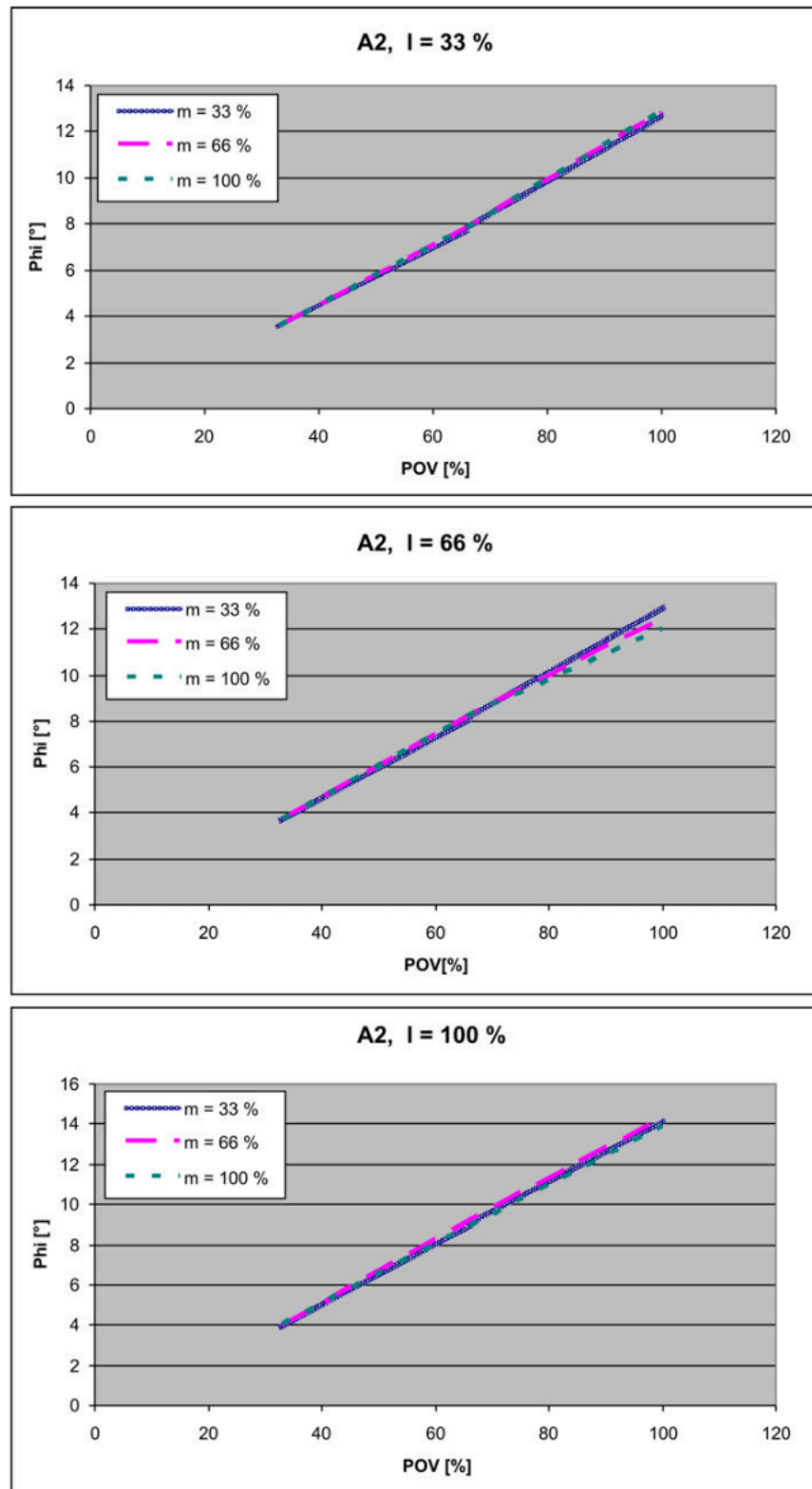


Fig. 4-13: Stopping distances for STOP 1, axis 2

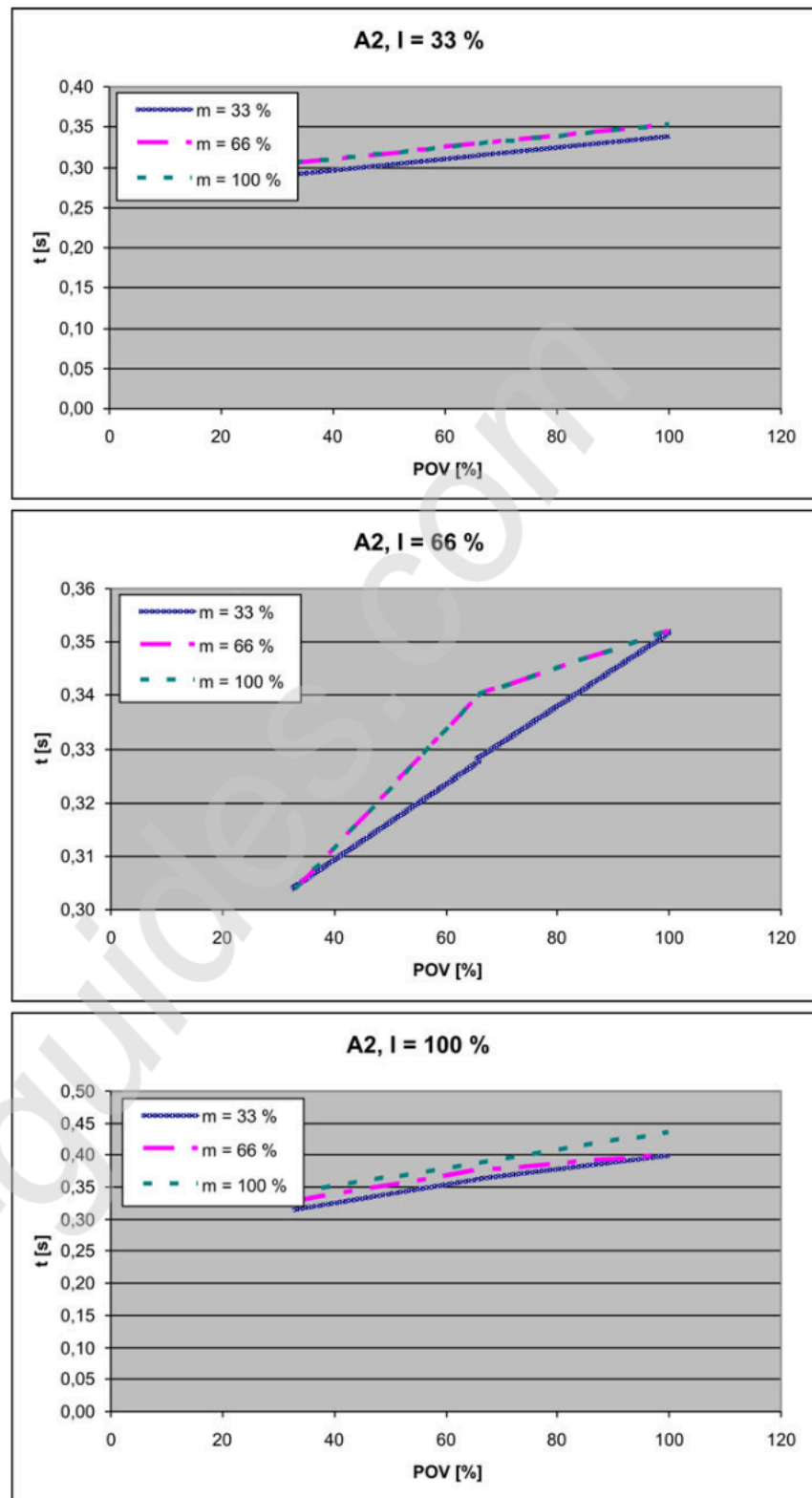


Fig. 4-14: Stopping times for STOP 1, axis 2

## 4.7.3.4 Stopping distances and stopping times for STOP 1, axis 3

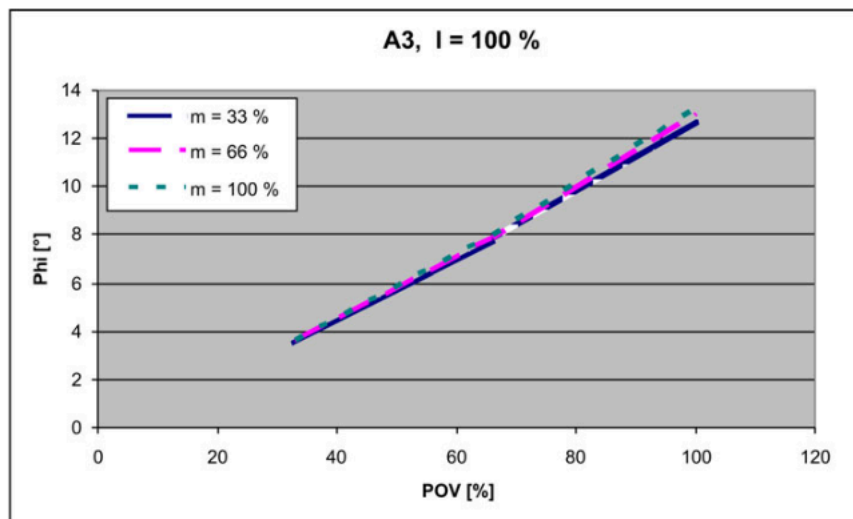


Fig. 4-15: Stopping distances for STOP 1, axis 3

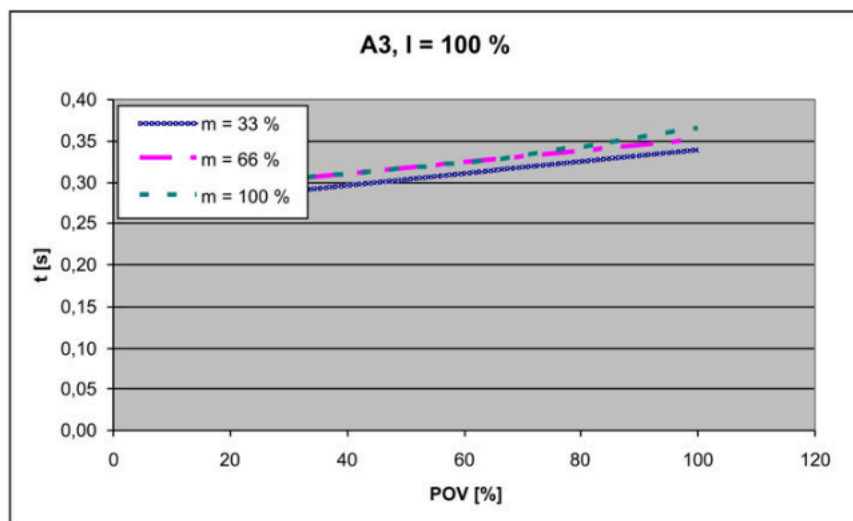


Fig. 4-16: Stopping times for STOP 1, axis 3

## 4.7.4 Stopping distance and stopping times for KR 470 PA

## 4.7.4.1 Stopping distances and stopping times for STOP 0, axis 1 to axis 3

The table shows the stopping distances and stopping times after a STOP 0 (category 0 stop) is triggered. The values refer to the following configuration:

- Extension  $l = 100\%$
- Program override  $POV = 100\%$
- Mass  $m = \text{maximum load (rated load + supplementary load on arm)}$

|        | Stopping distance (°) | Stopping time (s) |
|--------|-----------------------|-------------------|
| Axis 1 |                       |                   |
| Axis 2 |                       |                   |
| Axis 3 |                       |                   |

No data are currently available for STOP 0 for axes 1-3.

## 4.7.4.2 Stopping distances and stopping times for STOP 1, axis 1

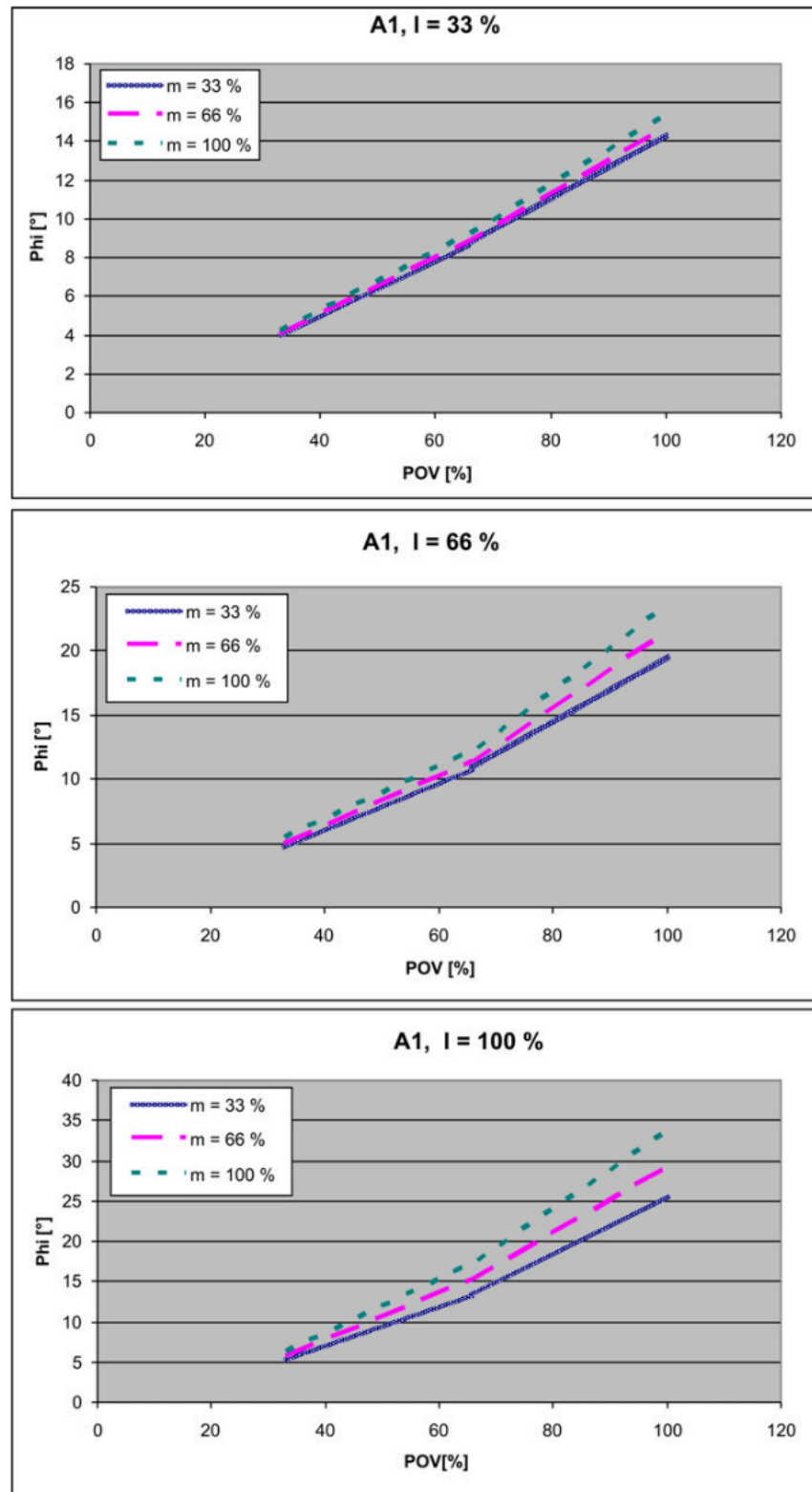


Fig. 4-17: Stopping distances for STOP 1, axis 1

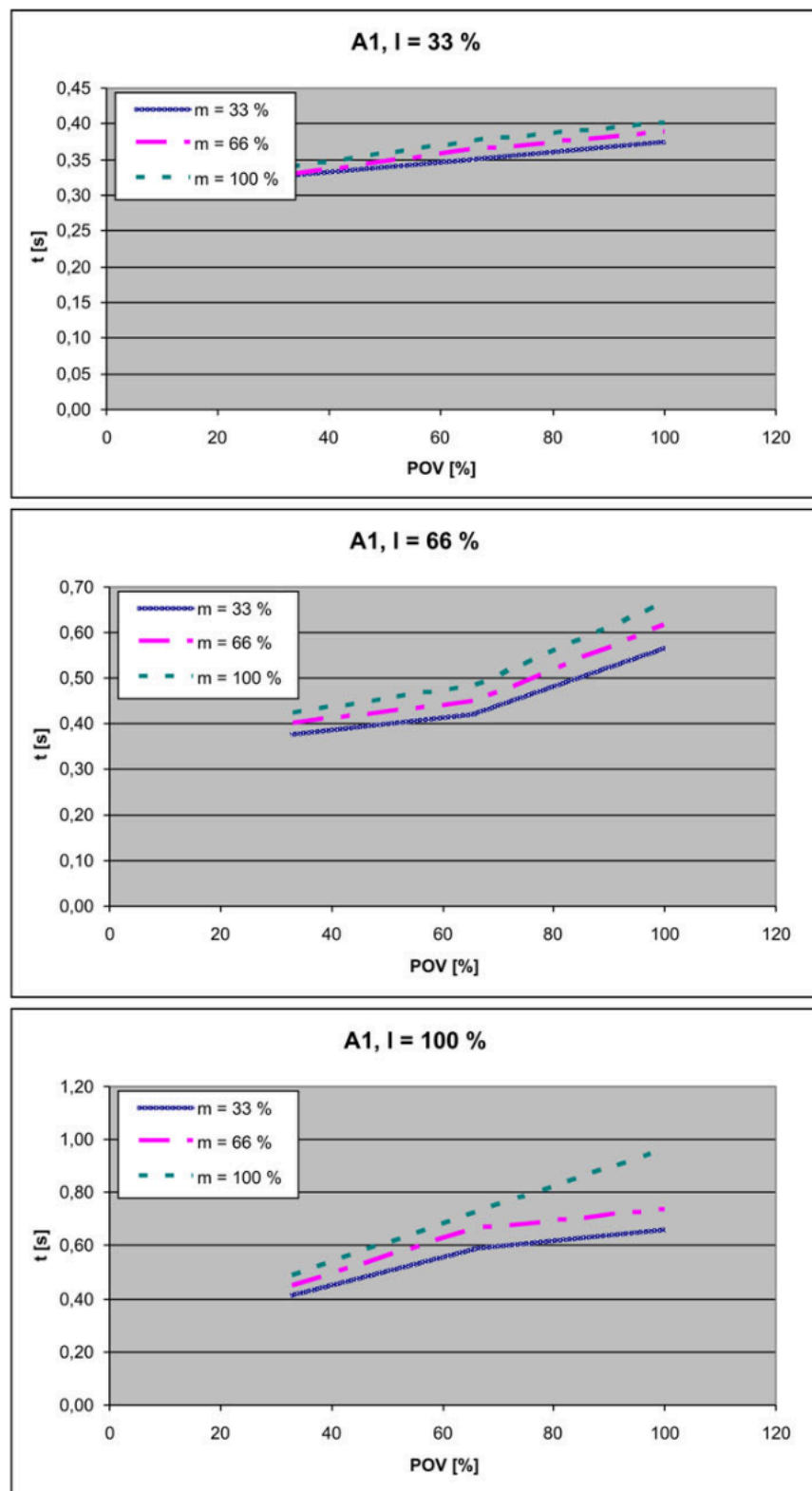


Fig. 4-18: Stopping times for STOP 1, axis 1

## 4.7.4.3 Stopping distances and stopping times for STOP 1, axis 3

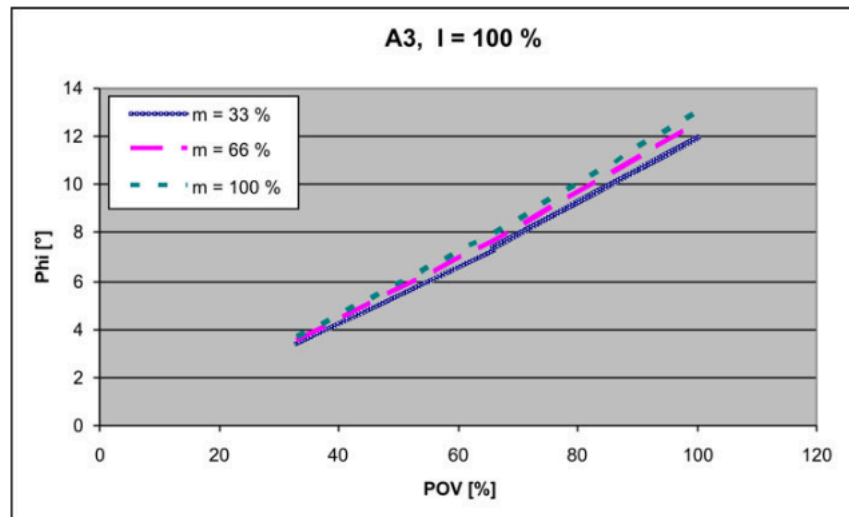


Fig. 4-19: Stopping distances for STOP 1, axis 3

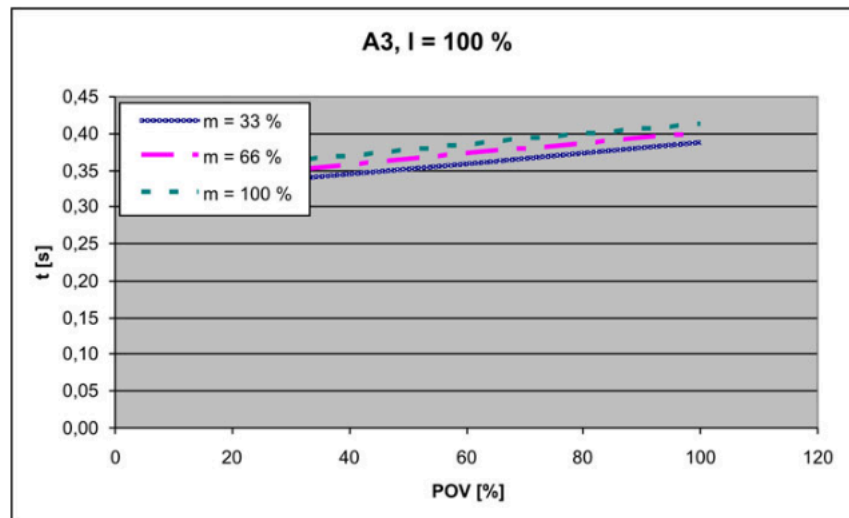


Fig. 4-20: Stopping times for STOP 1, axis 3

## 4.7.4.4 Stopping distances and stopping times for STOP 1, axis 2

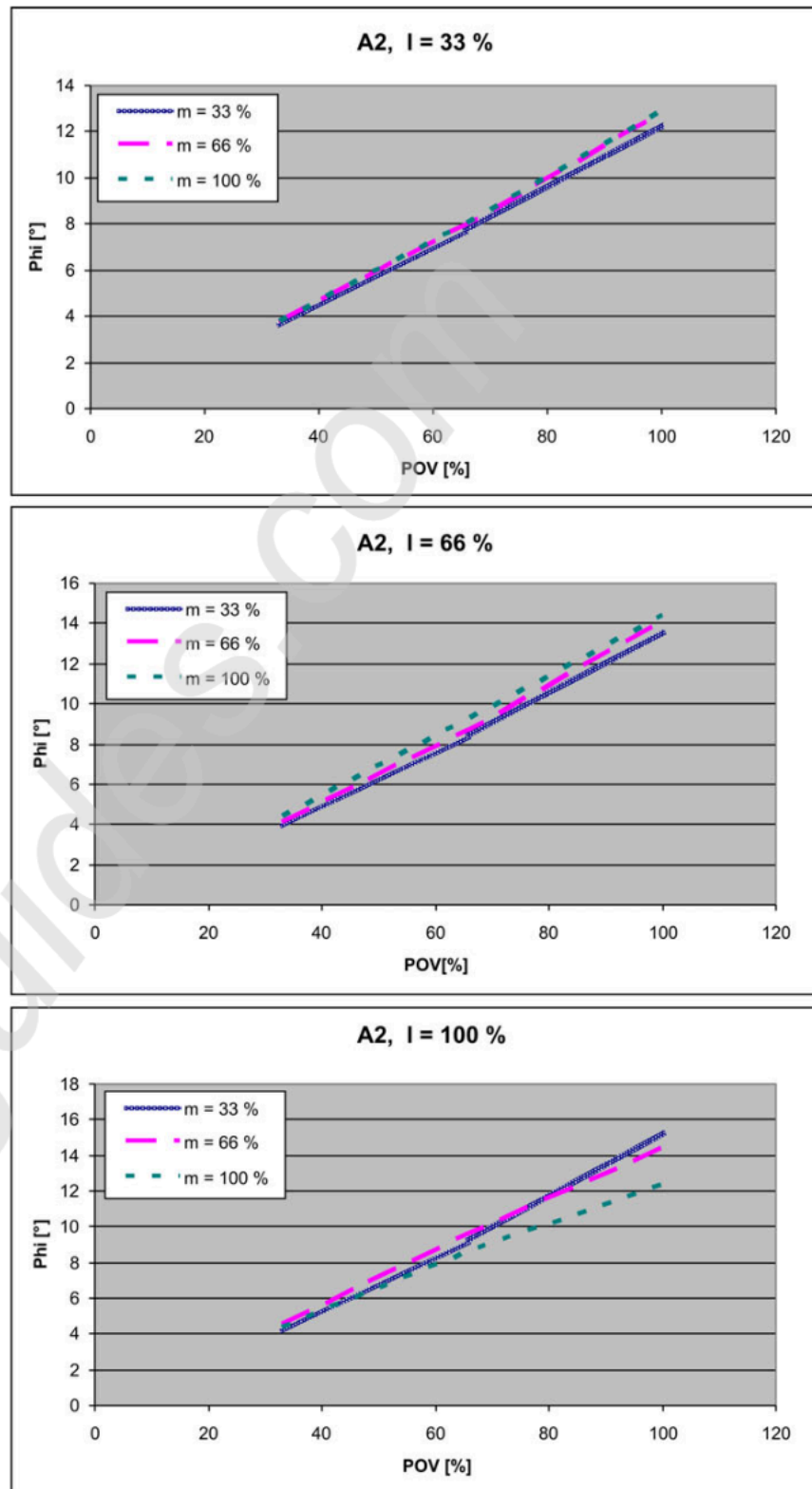


Fig. 4-21: Stopping distances for STOP 1, axis 2

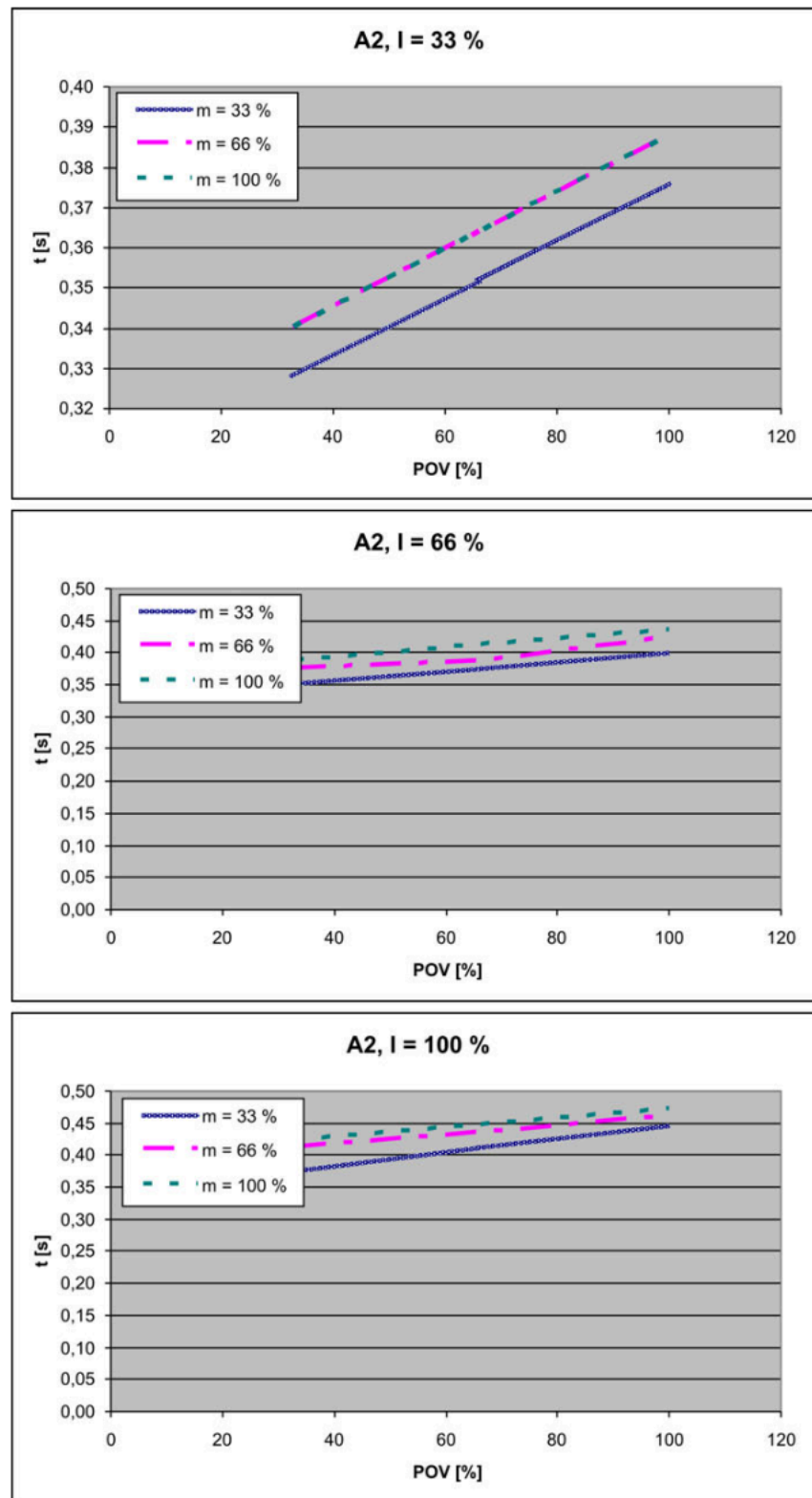


Fig. 4-22: Stopping times for STOP 1, axis 2



## 5 Safety

### 5.1 General



■ This "Safety" chapter refers to a mechanical component of an industrial robot.

■ If the mechanical component is used together with a KUKA robot controller, the "Safety" chapter of the operating instructions or assembly instructions of the robot controller must be used!

This contains all the information provided in this "Safety" chapter. It also contains additional safety information relating to the robot controller which must be observed.

■ Where this "Safety" chapter uses the term "industrial robot", this also refers to the individual mechanical component if applicable.

#### 5.1.1 Liability

The device described in this document is either an industrial robot or a component thereof.

Components of the industrial robot:

- Manipulator
- Robot controller
- Teach pendant
- Connecting cables
- External axes (optional)  
e.g. linear unit, turn-tilt table, positioner
- Software
- Options, accessories

The industrial robot is built using state-of-the-art technology and in accordance with the recognized safety rules. Nevertheless, misuse of the industrial robot may constitute a risk to life and limb or cause damage to the industrial robot and to other material property.

The industrial robot may only be used in perfect technical condition in accordance with its designated use and only by safety-conscious persons who are fully aware of the risks involved in its operation. Use of the industrial robot is subject to compliance with this document and with the declaration of incorporation supplied together with the industrial robot. Any functional disorders affecting safety must be rectified immediately.

#### Safety information

Safety information cannot be held against KUKA Roboter GmbH. Even if all safety instructions are followed, this is not a guarantee that the industrial robot will not cause personal injuries or material damage.

No modifications may be carried out to the industrial robot without the authorization of KUKA Roboter GmbH. Additional components (tools, software, etc.), not supplied by KUKA Roboter GmbH, may be integrated into the industrial robot. The user is liable for any damage these components may cause to the industrial robot or to other material property.

In addition to the Safety chapter, this document contains further safety instructions. These must also be observed.

**5.1.2 Intended use of the industrial robot**

The industrial robot is intended exclusively for the use designated in the "Purpose" chapter of the operating instructions or assembly instructions.

Any use or application deviating from the intended use is deemed to be misuse and is not allowed. The manufacturer is not liable for any damage resulting from such misuse. The risk lies entirely with the user.

Operation of the industrial robot in accordance with its intended use also requires compliance with the operating and assembly instructions for the individual components, with particular reference to the maintenance specifications.

**Misuse**

Any use or application deviating from the intended use is deemed to be misuse and is not allowed. This includes e.g.:

- Transportation of persons and animals
- Use as a climbing aid
- Operation outside the specified operating parameters
- Use in potentially explosive environments
- Operation without additional safeguards
- Outdoor operation
- Underground operation

**5.1.3 EC declaration of conformity and declaration of incorporation**

The industrial robot constitutes partly completed machinery as defined by the EC Machinery Directive. The industrial robot may only be put into operation if the following preconditions are met:

- The industrial robot is integrated into a complete system.  
or: The industrial robot, together with other machinery, constitutes a complete system.  
or: All safety functions and safeguards required for operation in the complete machine as defined by the EC Machinery Directive have been added to the industrial robot.
- The complete system complies with the EC Machinery Directive. This has been confirmed by means of an assessment of conformity.

**Declaration of conformity**

The system integrator must issue a declaration of conformity for the complete system in accordance with the Machinery Directive. The declaration of conformity forms the basis for the CE mark for the system. The industrial robot must always be operated in accordance with the applicable national laws, regulations and standards.

The robot controller is CE certified under the EMC Directive and the Low Voltage Directive.

**Declaration of incorporation**

The partly completed machinery is supplied with a declaration of incorporation in accordance with Annex II B of the EC Machinery Directive 2006/42/EC. The assembly instructions and a list of essential requirements complied with in accordance with Annex I are integral parts of this declaration of incorporation.

The declaration of incorporation declares that the start-up of the partly completed machinery is not allowed until the partly completed machinery has been incorporated into machinery, or has been assembled with other parts to form machinery, and this machinery complies with the terms of the EC Machinery Directive, and the EC declaration of conformity is present in accordance with Annex II A.

### 5.1.4 Terms used

| Term                                 | Description                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Axis range                           | Range of each axis, in degrees or millimeters, within which it may move. The axis range must be defined for each axis.                                                                                                                                                                                       |
| Stopping distance                    | Stopping distance = reaction distance + braking distance<br>The stopping distance is part of the danger zone.                                                                                                                                                                                                |
| Workspace                            | The manipulator is allowed to move within its workspace. The workspace is derived from the individual axis ranges.                                                                                                                                                                                           |
| Operator (User)                      | The user of the industrial robot can be the management, employer or delegated person responsible for use of the industrial robot.                                                                                                                                                                            |
| Danger zone                          | The danger zone consists of the workspace and the stopping distances.                                                                                                                                                                                                                                        |
| Service life                         | The service life of a safety-relevant component begins at the time of delivery of the component to the customer.<br><br>The service life is not affected by whether the component is used in a robot controller or elsewhere or not, as safety-relevant components are also subject to aging during storage. |
| KCP                                  | KUKA Control Panel<br><br>Teach pendant for the KR C2/KR C2 edition2005<br><br>The KCP has all the operator control and display functions required for operating and programming the industrial robot.                                                                                                       |
| KUKA smartPAD                        | see "smartPAD"                                                                                                                                                                                                                                                                                               |
| Manipulator                          | The robot arm and the associated electrical installations                                                                                                                                                                                                                                                    |
| Safety zone                          | The safety zone is situated outside the danger zone.                                                                                                                                                                                                                                                         |
| smartPAD                             | Teach pendant for the KR C4<br><br>The smartPAD has all the operator control and display functions required for operating and programming the industrial robot.                                                                                                                                              |
| Stop category 0                      | The drives are deactivated immediately and the brakes are applied. The manipulator and any external axes (optional) perform path-oriented braking.<br><br><b>Note:</b> This stop category is called STOP 0 in this document.                                                                                 |
| Stop category 1                      | The manipulator and any external axes (optional) perform path-maintaining braking. The drives are deactivated after 1 s and the brakes are applied.<br><br><b>Note:</b> This stop category is called STOP 1 in this document.                                                                                |
| Stop category 2                      | The drives are not deactivated and the brakes are not applied. The manipulator and any external axes (optional) are braked with a normal braking ramp.<br><br><b>Note:</b> This stop category is called STOP 2 in this document.                                                                             |
| System integrator (plant integrator) | System integrators are people who safely integrate the industrial robot into a complete system and commission it.                                                                                                                                                                                            |
| T1                                   | Test mode, Manual Reduced Velocity ( $\leq 250$ mm/s)                                                                                                                                                                                                                                                        |
| T2                                   | Test mode, Manual High Velocity ( $> 250$ mm/s permissible)                                                                                                                                                                                                                                                  |
| External axis                        | Motion axis which is not part of the manipulator but which is controlled using the robot controller, e.g. KUKA linear unit, turn-tilt table, Posiflex.                                                                                                                                                       |

## 5.2 Personnel

The following persons or groups of persons are defined for the industrial robot:

- User

- Personnel



All persons working with the industrial robot must have read and understood the industrial robot documentation, including the safety chapter.

**User**

The user must observe the labor laws and regulations. This includes e.g.:

- The user must comply with his monitoring obligations.
- The user must carry out briefing at defined intervals.

**Personnel**

Personnel must be instructed, before any work is commenced, in the type of work involved and what exactly it entails as well as any hazards which may exist. Instruction must be carried out regularly. Instruction is also required after particular incidents or technical modifications.

Personnel includes:

- System integrator
- Operators, subdivided into:
  - Start-up, maintenance and service personnel
  - Operating personnel
  - Cleaning personnel



Installation, exchange, adjustment, operation, maintenance and repair must be performed only as specified in the operating or assembly instructions for the relevant component of the industrial robot and only by personnel specially trained for this purpose.

**System integrator**

The industrial robot is safely integrated into a complete system by the system integrator.

The system integrator is responsible for the following tasks:

- Installing the industrial robot
- Connecting the industrial robot
- Performing risk assessment
- Implementing the required safety functions and safeguards
- Issuing the declaration of conformity
- Attaching the CE mark
- Creating the operating instructions for the complete system

**Operator**

The operator must meet the following preconditions:

- The operator must be trained for the work to be carried out.
- Work on the industrial robot must only be carried out by qualified personnel. These are people who, due to their specialist training, knowledge and experience, and their familiarization with the relevant standards, are able to assess the work to be carried out and detect any potential hazards.



Work on the electrical and mechanical equipment of the industrial robot may only be carried out by specially trained personnel.

**5.3 Workspace, safety zone and danger zone**

Workspaces are to be restricted to the necessary minimum size. A workspace must be safeguarded using appropriate safeguards.

The safeguards (e.g. safety gate) must be situated inside the safety zone. In the case of a stop, the manipulator and external axes (optional) are braked and come to a stop within the danger zone.

The danger zone consists of the workspace and the stopping distances of the manipulator and external axes (optional). It must be safeguarded by means of physical safeguards to prevent danger to persons or the risk of material damage.

## 5.4 Overview of protective equipment

The protective equipment of the mechanical component may include:

- Mechanical end stops
- Mechanical axis range limitation (optional)
- Axis range monitoring (optional)
- Release device (optional)
- Labeling of danger areas

Not all equipment is relevant for every mechanical component.

### 5.4.1 Mechanical end stops

Depending on the robot variant, the axis ranges of the main and wrist axes of the manipulator are partially limited by mechanical end stops.

Additional mechanical end stops can be installed on the external axes.

 **WARNING** If the manipulator or an external axis hits an obstruction or a mechanical end stop or axis range limitation, the manipulator can no longer be operated safely. The manipulator must be taken out of operation and KUKA Roboter GmbH must be consulted before it is put back into operation.

### 5.4.2 Mechanical axis range limitation (optional)

Some manipulators can be fitted with mechanical axis range limitation in axes A1 to A3. The adjustable axis range limitation systems restrict the working range to the required minimum. This increases personal safety and protection of the system.

In the case of manipulators that are not designed to be fitted with mechanical axis range limitation, the workspace must be laid out in such a way that there is no danger to persons or material property, even in the absence of mechanical axis range limitation.

If this is not possible, the workspace must be limited by means of photoelectric barriers, photoelectric curtains or obstacles on the system side. There must be no shearing or crushing hazards at the loading and transfer areas.

 This option is not available for all robot models. Information on specific robot models can be obtained from KUKA Roboter GmbH.

### 5.4.3 Axis range monitoring (optional)

Some manipulators can be fitted with dual-channel axis range monitoring systems in main axes A1 to A3. The positioner axes may be fitted with additional axis range monitoring systems. The safety zone for an axis can be adjusted

and monitored using an axis range monitoring system. This increases personal safety and protection of the system.



This option is not available for the KR C4. This option is not available for all robot models. Information on specific robot models can be obtained from KUKA Roboter GmbH.

#### 5.4.4 Options for moving the manipulator without drive energy



The system user is responsible for ensuring that the training of personnel with regard to the response to emergencies or exceptional situations also includes how the manipulator can be moved without drive energy.

##### Description

The following options are available for moving the manipulator without drive energy after an accident or malfunction:

- Release device (optional)  
The release device can be used for the main axis drive motors and, depending on the robot variant, also for the wrist axis drive motors.
- Brake release device (option)  
The brake release device is designed for robot variants whose motors are not freely accessible.
- Moving the wrist axes directly by hand  
There is no release device available for the wrist axes of variants in the low payload category. This is not necessary because the wrist axes can be moved directly by hand.



Information about the options available for the various robot models and about how to use them can be found in the assembly and operating instructions for the robot or requested from KUKA Roboter GmbH.

##### **NOTICE**

Moving the manipulator without drive energy can damage the motor brakes of the axes concerned. The motor must be replaced if the brake has been damaged. The manipulator may therefore be moved without drive energy only in emergencies, e.g. for rescuing persons.

#### 5.4.5 Labeling on the industrial robot

All plates, labels, symbols and marks constitute safety-relevant parts of the industrial robot. They must not be modified or removed.

Labeling on the industrial robot consists of:

- Identification plates
- Warning signs
- Safety symbols
- Designation labels
- Cable markings
- Rating plates



Further information is contained in the technical data of the operating instructions or assembly instructions of the components of the industrial robot.

## 5.5 Safety measures

### 5.5.1 General safety measures

The industrial robot may only be used in perfect technical condition in accordance with its intended use and only by safety-conscious persons. Operator errors can result in personal injury and damage to property.

It is important to be prepared for possible movements of the industrial robot even after the robot controller has been switched off and locked out. Incorrect installation (e.g. overload) or mechanical defects (e.g. brake defect) can cause the manipulator or external axes to sag. If work is to be carried out on a switched-off industrial robot, the manipulator and external axes must first be moved into a position in which they are unable to move on their own, whether the payload is mounted or not. If this is not possible, the manipulator and external axes must be secured by appropriate means.

**⚠ DANGER** In the absence of operational safety functions and safeguards, the industrial robot can cause personal injury or material damage. If safety functions or safeguards are dismantled or deactivated, the industrial robot may not be operated.

**⚠ DANGER** Standing underneath the robot arm can cause death or injuries. For this reason, standing underneath the robot arm is prohibited!

**⚠ CAUTION** The motors reach temperatures during operation which can cause burns to the skin. Contact must be avoided. Appropriate safety precautions must be taken, e.g. protective gloves must be worn.

#### KCP/smartPAD

The user must ensure that the industrial robot is only operated with the KCP/smartPAD by authorized persons.

If more than one KCP/smartPAD is used in the overall system, it must be ensured that each device is unambiguously assigned to the corresponding industrial robot. They must not be interchanged.

**⚠ WARNING** The operator must ensure that decoupled KCPs/smartPADs are immediately removed from the system and stored out of sight and reach of personnel working on the industrial robot. This serves to prevent operational and non-operational EMERGENCY STOP devices from becoming interchanged. Failure to observe this precaution may result in death, severe injuries or considerable damage to property.

#### External keyboard, external mouse

An external keyboard and/or external mouse may only be used if the following conditions are met:

- Start-up or maintenance work is being carried out.
- The drives are switched off.
- There are no persons in the danger zone.

The KCP/smartPAD must not be used as long as an external keyboard and/or external mouse are connected to the control cabinet.

The external keyboard and/or external mouse must be removed from the control cabinet as soon as the start-up or maintenance work is completed or the KCP/smartPAD is connected.

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Modifications</b> | <p>After modifications to the industrial robot, checks must be carried out to ensure the required safety level. The valid national or regional work safety regulations must be observed for this check. The correct functioning of all safety functions must also be tested.</p> <p>New or modified programs must always be tested first in Manual Reduced Velocity mode (T1).</p> <p>After modifications to the industrial robot, existing programs must always be tested first in Manual Reduced Velocity mode (T1). This applies to all components of the industrial robot and includes modifications to the software and configuration settings.</p> |
| <b>Faults</b>        | <p>The following tasks must be carried out in the case of faults in the industrial robot:</p> <ul style="list-style-type: none"> <li>■ Switch off the robot controller and secure it (e.g. with a padlock) to prevent unauthorized persons from switching it on again.</li> <li>■ Indicate the fault by means of a label with a corresponding warning (tag-out).</li> <li>■ Keep a record of the faults.</li> <li>■ Eliminate the fault and carry out a function test.</li> </ul>                                                                                                                                                                        |

### 5.5.2 Transportation

|                                 |                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Manipulator</b>              | <p>The prescribed transport position of the manipulator must be observed. Transportation must be carried out in accordance with the operating instructions or assembly instructions of the robot.</p> <p>Avoid vibrations and impacts during transportation in order to prevent damage to the manipulator.</p>                      |
| <b>Robot controller</b>         | <p>The prescribed transport position of the robot controller must be observed. Transportation must be carried out in accordance with the operating instructions or assembly instructions of the robot controller.</p> <p>Avoid vibrations and impacts during transportation in order to prevent damage to the robot controller.</p> |
| <b>External axis (optional)</b> | <p>The prescribed transport position of the external axis (e.g. KUKA linear unit, turn-tilt table, positioner) must be observed. Transportation must be carried out in accordance with the operating instructions or assembly instructions of the external axis.</p>                                                                |

### 5.5.3 Start-up and recommissioning

Before starting up systems and devices for the first time, a check must be carried out to ensure that the systems and devices are complete and operational, that they can be operated safely and that any damage is detected.

The valid national or regional work safety regulations must be observed for this check. The correct functioning of all safety circuits must also be tested.



The passwords for logging onto the KUKA System Software as "Expert" and "Administrator" must be changed before start-up and must only be communicated to authorized personnel.

**WARNING** The robot controller is preconfigured for the specific industrial robot. If cables are interchanged, the manipulator and the external axes (optional) may receive incorrect data and can thus cause personal injury or material damage. If a system consists of more than one manipulator, always connect the connecting cables to the manipulators and their corresponding robot controllers.

**!** If additional components (e.g. cables), which are not part of the scope of supply of KUKA Roboter GmbH, are integrated into the industrial robot, the user is responsible for ensuring that these components do not adversely affect or disable safety functions.

**NOTICE** If the internal cabinet temperature of the robot controller differs greatly from the ambient temperature, condensation can form, which may cause damage to the electrical components. Do not put the robot controller into operation until the internal temperature of the cabinet has adjusted to the ambient temperature.

## Function test

The following tests must be carried out before start-up and recommissioning:  
It must be ensured that:

- The industrial robot is correctly installed and fastened in accordance with the specifications in the documentation.
- There is no damage to the robot that could be attributed to external forces.  
Example: Dents or abrasion that could be caused by an impact or collision.

**WARNING** In the case of such damage, the affected components must be exchanged. In particular, the motor and counterbalancing system must be checked carefully. External forces can cause non-visible damage. For example, it can lead to a gradual loss of drive power from the motor, resulting in unintended movements of the manipulator. Death, injuries or considerable damage to property may otherwise result.

- There are no foreign bodies or loose parts on the industrial robot.
- All required safety equipment is correctly installed and operational.
- The power supply ratings of the industrial robot correspond to the local supply voltage and mains type.
- The ground conductor and the equipotential bonding cable are sufficiently rated and correctly connected.
- The connecting cables are correctly connected and the connectors are locked.

## Machine data

It must be ensured that the rating plate on the robot controller has the same machine data as those entered in the declaration of incorporation. The machine data on the rating plate of the manipulator and the external axes (optional) must be entered during start-up.

**WARNING** The industrial robot must not be moved if incorrect machine data are loaded. Death, severe injuries or considerable damage to property may otherwise result. The correct machine data must be loaded.

### 5.5.4 Manual mode

Manual mode is the mode for setup work. Setup work is all the tasks that have to be carried out on the industrial robot to enable automatic operation. Setup work includes:

- Jog mode
- Teaching
- Programming
- Program verification

The following must be taken into consideration in manual mode:

- If the drives are not required, they must be switched off to prevent the manipulator or the external axes (optional) from being moved unintentionally.
- New or modified programs must always be tested first in Manual Reduced Velocity mode (T1).
- The manipulator, tooling or external axes (optional) must never touch or project beyond the safety fence.
- Workpieces, tooling and other objects must not become jammed as a result of the industrial robot motion, nor must they lead to short-circuits or be liable to fall off.
- All setup work must be carried out, where possible, from outside the safeguarded area.

If the setup work has to be carried out inside the safeguarded area, the following must be taken into consideration:

#### In Manual Reduced Velocity mode (T1):

- If it can be avoided, there must be no other persons inside the safeguarded area.  
If it is necessary for there to be several persons inside the safeguarded area, the following must be observed:
  - Each person must have an enabling device.
  - All persons must have an unimpeded view of the industrial robot.
  - Eye-contact between all persons must be possible at all times.
- The operator must be so positioned that he can see into the danger area and get out of harm's way.

#### In Manual High Velocity mode (T2):

- This mode may only be used if the application requires a test at a velocity higher than possible in T1 mode.
- Teaching and programming are not permissible in this operating mode.
- Before commencing the test, the operator must ensure that the enabling devices are operational.
- The operator must be positioned outside the danger zone.
- There must be no other persons inside the safeguarded area. It is the responsibility of the operator to ensure this.

### 5.5.5 Automatic mode

Automatic mode is only permissible in compliance with the following safety measures:

- All safety equipment and safeguards are present and operational.
- There are no persons in the system.
- The defined working procedures are adhered to.

If the manipulator or an external axis (optional) comes to a standstill for no apparent reason, the danger zone must not be entered until an EMERGENCY STOP has been triggered.

### 5.5.6 Maintenance and repair

After maintenance and repair work, checks must be carried out to ensure the required safety level. The valid national or regional work safety regulations must be observed for this check. The correct functioning of all safety functions must also be tested.

The purpose of maintenance and repair work is to ensure that the system is kept operational or, in the event of a fault, to return the system to an operational state. Repair work includes troubleshooting in addition to the actual repair itself.

The following safety measures must be carried out when working on the industrial robot:

- Carry out work outside the danger zone. If work inside the danger zone is necessary, the user must define additional safety measures to ensure the safe protection of personnel.
- Switch off the industrial robot and secure it (e.g. with a padlock) to prevent it from being switched on again. If it is necessary to carry out work with the robot controller switched on, the user must define additional safety measures to ensure the safe protection of personnel.
- If it is necessary to carry out work with the robot controller switched on, this may only be done in operating mode T1.
- Label the system with a sign indicating that work is in progress. This sign must remain in place, even during temporary interruptions to the work.
- The EMERGENCY STOP devices must remain active. If safety functions or safeguards are deactivated during maintenance or repair work, they must be reactivated immediately after the work is completed.



#### **DANGER**

Before work is commenced on live parts of the robot system, the main switch must be turned off and secured against being switched on again. The system must then be checked to ensure that it is deenergized.

It is not sufficient, before commencing work on live parts, to execute an EMERGENCY STOP or a safety stop, or to switch off the drives, as this does not disconnect the robot system from the mains power supply. Parts remain energized. Death or severe injuries may result.

Faulty components must be replaced using new components with the same article numbers or equivalent components approved by KUKA Roboter GmbH for this purpose.

Cleaning and preventive maintenance work is to be carried out in accordance with the operating instructions.

#### **Robot controller**

Even when the robot controller is switched off, parts connected to peripheral devices may still carry voltage. The external power sources must therefore be switched off if work is to be carried out on the robot controller.

The ESD regulations must be adhered to when working on components in the robot controller.

Voltages in excess of 50 V (up to 600 V) can be present in various components for several minutes after the robot controller has been switched off! To prevent life-threatening injuries, no work may be carried out on the industrial robot in this time.

Water and dust must be prevented from entering the robot controller.

**Counterbalancing system**

Some robot variants are equipped with a hydropneumatic, spring or gas cylinder counterbalancing system.

The hydropneumatic and gas cylinder counterbalancing systems are pressure equipment and, as such, are subject to obligatory equipment monitoring and the provisions of the Pressure Equipment Directive.

The user must comply with the applicable national laws, regulations and standards pertaining to pressure equipment.

Inspection intervals in Germany in accordance with Industrial Safety Order, Sections 14 and 15. Inspection by the user before commissioning at the installation site.

The following safety measures must be carried out when working on the counterbalancing system:

- The manipulator assemblies supported by the counterbalancing systems must be secured.
- Work on the counterbalancing systems must only be carried out by qualified personnel.

**Hazardous substances**

The following safety measures must be carried out when handling hazardous substances:

- Avoid prolonged and repeated intensive contact with the skin.
- Avoid breathing in oil spray or vapors.
- Clean skin and apply skin cream.



To ensure safe use of our products, we recommend that our customers regularly request up-to-date safety data sheets from the manufacturers of hazardous substances.

**5.5.7 Decommissioning, storage and disposal**

The industrial robot must be decommissioned, stored and disposed of in accordance with the applicable national laws, regulations and standards.

**5.6 Applied norms and regulations**

| Name       | Definition                                                                                                                                                                                                                                                            | Edition |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 2006/42/EC | <b>Machinery Directive:</b><br>Directive 2006/42/EC of the European Parliament and of the Council of 17 May 2006 on machinery, and amending Directive 95/16/EC (recast)                                                                                               | 2006    |
| 2014/30/EC | <b>EMC Directive:</b><br>Directive 2014/30/EC of the European Parliament and of the Council of 26 February 2014 on the approximation of the laws of the Member States concerning electromagnetic compatibility<br><br>This directive is valid from the 20/04/2016 on. | 2014    |

|                |                                                                                                                                                                                                                                                                                                                                                           |      |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 2004/108/EC    | <b>EMC Directive:</b><br><br>Directive 2004/108/EC of the European Parliament and of the Council of 15 December 2004 on the approximation of the laws of the Member States concerning electromagnetic compatibility<br><br>This directive is valid until 19/04/2016.                                                                                      | 2004 |
| 2014/68/EC     | <b>Pressure Equipment Directive:</b><br><br>Directive 2014/68/EC of the European Parliament and of the Council of 15 May 2014 on the approximation of the laws of the Member States concerning pressure equipment<br><br>(Only applicable for robots with hydropneumatic counterbalancing system.)<br><br>This directive is valid from the 19/07/2016 on. | 2014 |
| 97/23/EC       | <b>Pressure Equipment Directive:</b><br><br>Directive 97/23/EC of the European Parliament and of the Council of 29 May 1997 on the approximation of the laws of the Member States concerning pressure equipment<br><br>(Only applicable for robots with hydropneumatic counterbalancing system.)<br><br>This directive is valid until 18/07/2016.         | 1997 |
| EN ISO 13850   | <b>Safety of machinery:</b><br><br>Emergency stop - Principles for design                                                                                                                                                                                                                                                                                 | 2008 |
| EN ISO 13849-1 | <b>Safety of machinery:</b><br><br>Safety-related parts of control systems - Part 1: General principles of design                                                                                                                                                                                                                                         | 2008 |
| EN ISO 13849-2 | <b>Safety of machinery:</b><br><br>Safety-related parts of control systems - Part 2: Validation                                                                                                                                                                                                                                                           | 2012 |
| EN ISO 12100   | <b>Safety of machinery:</b><br><br>General principles of design, risk assessment and risk reduction                                                                                                                                                                                                                                                       | 2010 |
| EN ISO 10218-1 | <b>Industrial robots – Safety requirements</b><br><br>Part 1: Robot<br><br><b>Note:</b> Content equivalent to <b>ANSI/RIA R.15.06-2012, Part 1</b>                                                                                                                                                                                                        | 2011 |
| EN 614-1 + A1  | <b>Safety of machinery:</b><br><br>Ergonomic design principles - Part 1: Terms and general principles                                                                                                                                                                                                                                                     | 2009 |
| EN 61000-6-2   | <b>Electromagnetic compatibility (EMC):</b><br><br>Part 6-2: Generic standards; Immunity for industrial environments                                                                                                                                                                                                                                      | 2005 |

|                          |                                                                                                                           |      |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------|------|
| <b>EN 61000-6-4 + A1</b> | <b>Electromagnetic compatibility (EMC):</b><br>Part 6-4: Generic standards; Emission standard for industrial environments | 2011 |
| <b>EN 60204-1 + A1</b>   | <b>Safety of machinery:</b><br>Electrical equipment of machines - Part 1: General requirements                            | 2009 |

## 6 Planning

### 6.1 Mounting base for 175 mm concrete thickness

#### Description

The mounting base (>>> Fig. 6-1 ) with centering is used when the robot is fastened to the floor, i.e. directly on a concrete foundation.

The mounting base consists of:

- Plate
- Chemical anchors (resin-bonded anchors) with Dynamic Set
- Fasteners

This mounting variant requires a level and smooth surface on a concrete foundation with adequate load bearing capacity. The concrete foundation must be able to accommodate the forces occurring during operation. The minimum dimensions must be observed.

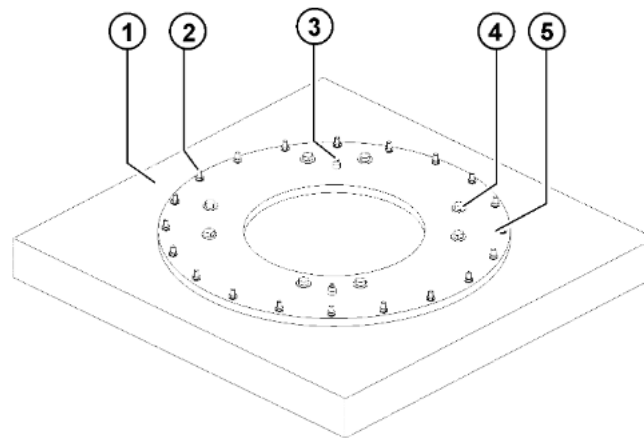


Fig. 6-1: Mounting base 175 mm

- |                                         |                |
|-----------------------------------------|----------------|
| 1 Concrete foundation                   | 4 Hexagon bolt |
| 2 Chemical anchor (resin-bonded anchor) | 5 Plate        |
| 3 Pin                                   |                |

#### Grade of concrete for foundations

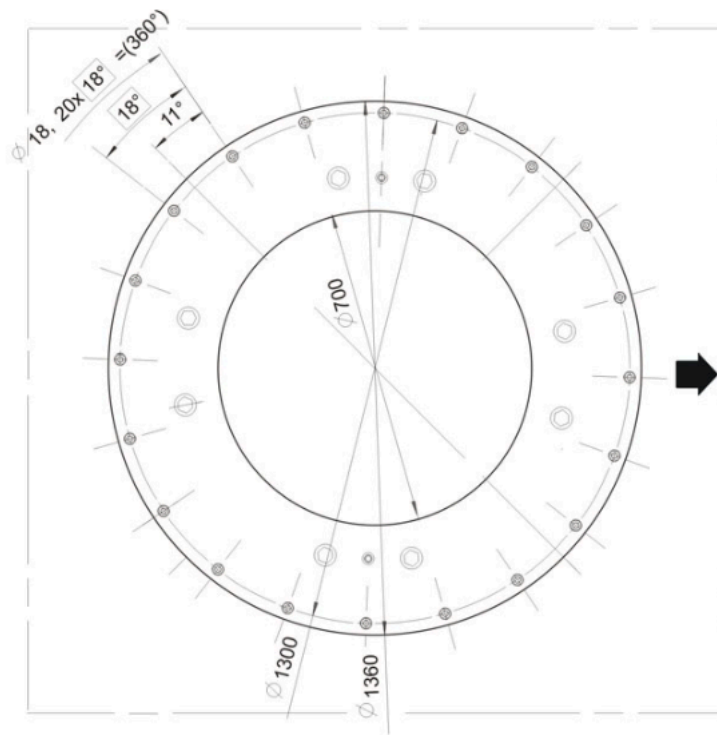
When producing foundations from concrete, observe the load-bearing capacity of the ground and the country-specific construction regulations. There must be no layers of insulation or screed between the bedplates and the concrete foundation. The quality of the concrete must meet the requirements of the following standard:

- C20/25 according to DIN EN 206-1:2001/DIN 1045-2:2008

#### Dimensioned drawing

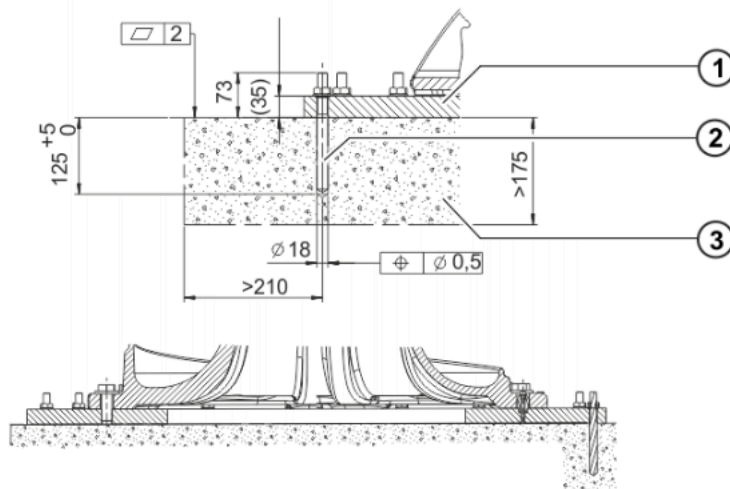
The following illustration provides all the necessary information on the mounting base, together with the required foundation data (>>> Fig. 6-2 ).

Dimensions: mm

**Fig. 6-2: Mounting base, dimensioned drawing**

To ensure that the anchor forces are safely transmitted to the foundation, observe the dimensions for concrete foundations specified in the following illustration (>>> Fig. 6-3).

Dimensions: mm

**Fig. 6-3: Cross-section of foundation 175 mm**

- 1 Bedplate
- 2 Chemical anchors (resin-bonded anchors) with Dynamic Set
- 3 Concrete foundation

## 6.2 Mounting base for 200 mm concrete thickness

### Description

The mounting base for 200 mm concrete thickness (>>> Fig. 6-4 ) is used when the robot is fastened to the floor, i.e. directly on a concrete foundation with a thickness of at least 200 mm.

The mounting base with centering consists of:

- Bedplates
- Chemical anchors (resin-bonded anchors) with Dynamic Set
- Fastening elements

This mounting variant requires a level and smooth surface on a concrete foundation with adequate load bearing capacity. The concrete foundation must be able to accommodate the forces occurring during operation. There must be no layers of insulation or screed between the bedplates and the concrete foundation.

The minimum dimensions must be observed.

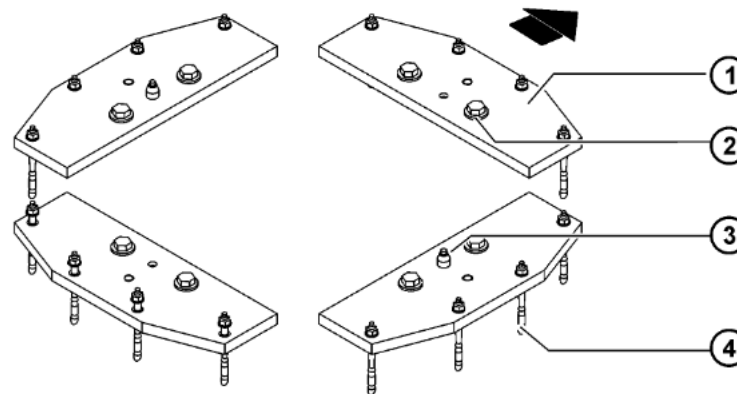


Fig. 6-4: Mounting base 200 mm

- |                |                                     |
|----------------|-------------------------------------|
| 1 Bedplate     | 3 Pin with Allen screw              |
| 2 Hexagon bolt | 4 Chemical anchors with Dynamic Set |

### Grade of concrete for foundations

When producing foundations from concrete, observe the load-bearing capacity of the ground and the country-specific construction regulations. There must be no layers of insulation or screed between the bedplates and the concrete foundation. The quality of the concrete must meet the requirements of the following standard:

- C20/25 according to DIN EN 206-1:2001/DIN 1045-2:2008

### Dimensioned drawing

The following illustrations provide all the necessary information on the mounting base, together with the required foundation data.

Dimensions: mm

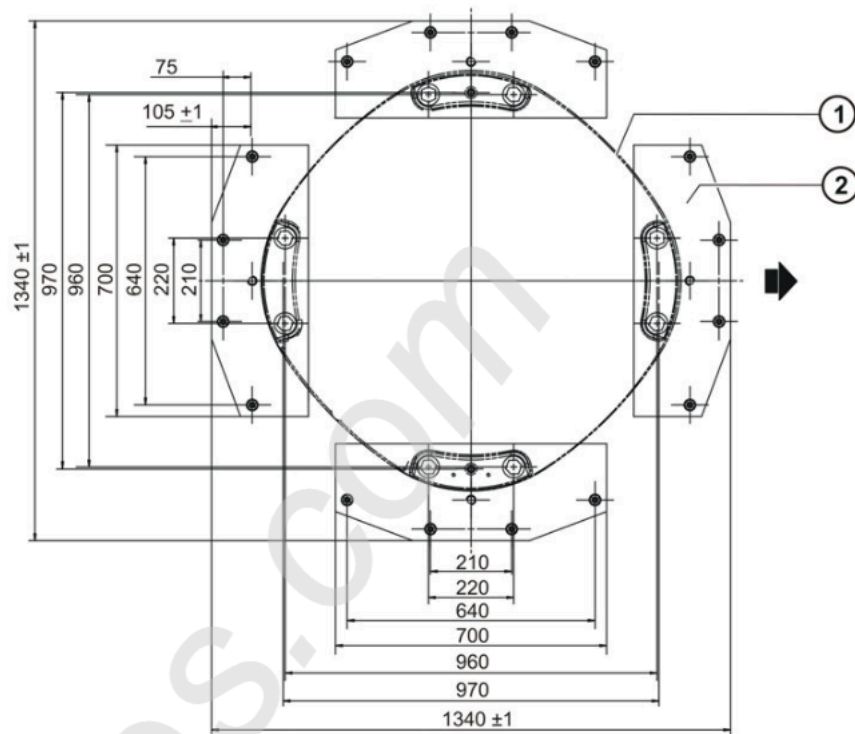


Fig. 6-5: Mounting base 200 mm, dimensioned drawing

- 1 Robot contour                      2 Bedplates

To ensure that the anchor forces are safely transmitted to the foundation, observe the dimensions for concrete foundations specified in the following illustration.

Dimensions: mm

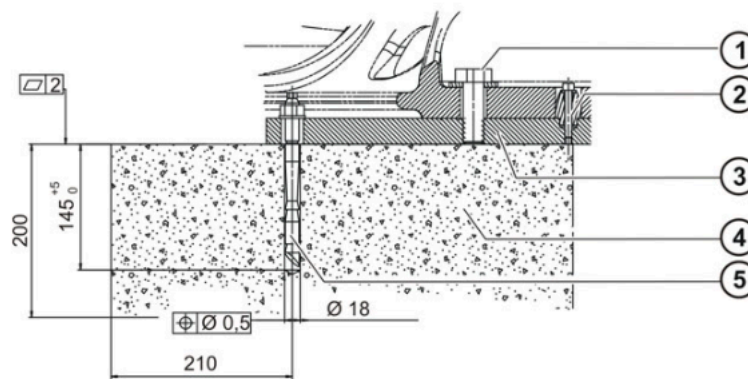


Fig. 6-6: Cross-section of foundation 200 mm

- 1 Hexagon bolt                      4 Concrete foundation  
2 Pin                                    5 Chemical anchor  
3 Bedplate

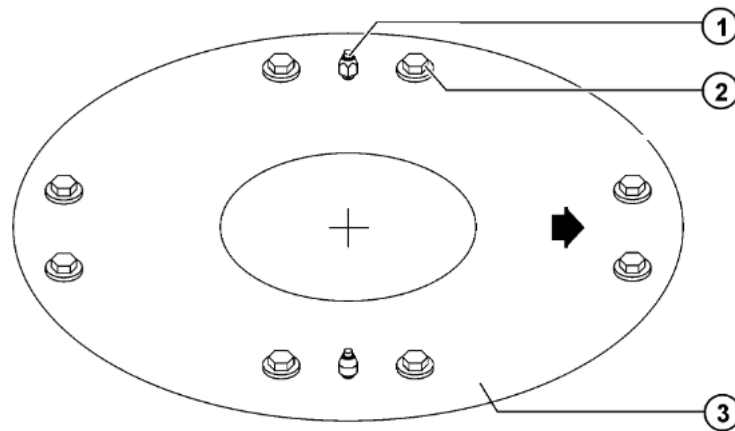
### 6.3 Machine frame mounting

#### Description

The machine frame mounting assembly (>>> Fig. 6-7 ) with centering is used when the robot is fastened on a steel structure, a booster frame (pedestal) or a KUKA linear unit. It must be ensured that the substructure is able to withstand safely the forces occurring during operation (foundation loads). The following diagram contains all the necessary information that must be observed when preparing the mounting surface.

The machine frame mounting assembly consists of (>>> Fig. 6-8 ):

- Pin with fasteners
- Sword pin with fasteners
- Hexagon bolts with conical spring washers



**Fig. 6-7: Machine frame mounting**

- |                     |                    |
|---------------------|--------------------|
| 1 Pin, sword pin    | 3 Mounting surface |
| 2 Hexagon bolt (8x) |                    |

|                     |                    |
|---------------------|--------------------|
| 1 Hexagon bolt (8x) | 3 Pin              |
| 2 Sword pin         | 4 Mounting surface |

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The ground conductors are connected via ring cable lugs.

Wiring diagrams, connector pin allocations and connector designations can be found in the section "Description of the connecting cables".

The following points must be observed when planning and routing the connecting cables:

- The bending radius for fixed routing – 150 mm for motor cables and 60 mm for control cables – must not be exceeded.
- Protect cables against exposure to mechanical stress.
- Route the cables without mechanical stress – no tensile forces on the connectors.
- Cables are only to be installed indoors.
- Observe permissible temperature range (fixed installation) of 263 K (-10 °C) to 343 K (+70 °C).
- Route the motor cables and the control cables separately in metal ducts; if necessary, additional measures must be taken to ensure electromagnetic compatibility (EMC).

#### Interface for energy supply systems

The robot can be equipped with an energy supply system between axis 1 and axis 6 and a second energy supply system between axis 6 and the tool. The A1 interface required for this is located on the rear, and the A6 interface is located on the swing frame and on the tool of the robot. The interfaces are equipped with connections for cables and hoses depending on the application. Detailed information on the connector pin allocation, threaded unions, etc. is given in separate documentation.

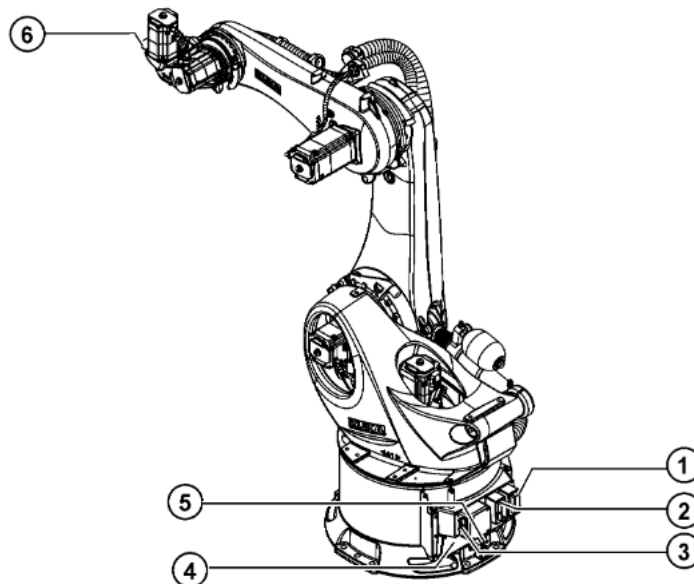


Fig. 6-9: Interfaces A1 to A6

- |                                       |                                                 |
|---------------------------------------|-------------------------------------------------|
| 1 Multi-function housing (MFH), X30.1 | 4 Energy supply system, interface A1            |
| 2 Multi-function housing (MFH), X30.4 | 5 Ground conductor connection M8 ring cable lug |
| 3 Control cable X31                   | 6 Energy supply system, interface A6            |



## 7 Transportation

### 7.1 Transporting the robot

Before transporting the robot, always move the robot into its transport position (>>> Fig. 7-1 ). It must be ensured that the robot is stable while it is being transported. The robot must remain in its transport position until it has been fastened in position. Before the robot is lifted, it must be ensured that it is free from obstructions. Remove all transport safeguards, such as nails and screws, in advance. First remove any rust or glue on contact surfaces.

#### Transport position

The robot must be in the transport position (>>> Fig. 7-1 ) before it can be transported. The robot is in the transport position when the axes are in the following positions:

| Axis               | A1 | A2    | A3    | A6 |
|--------------------|----|-------|-------|----|
| Transport position | 0° | -130° | +155° | 0° |

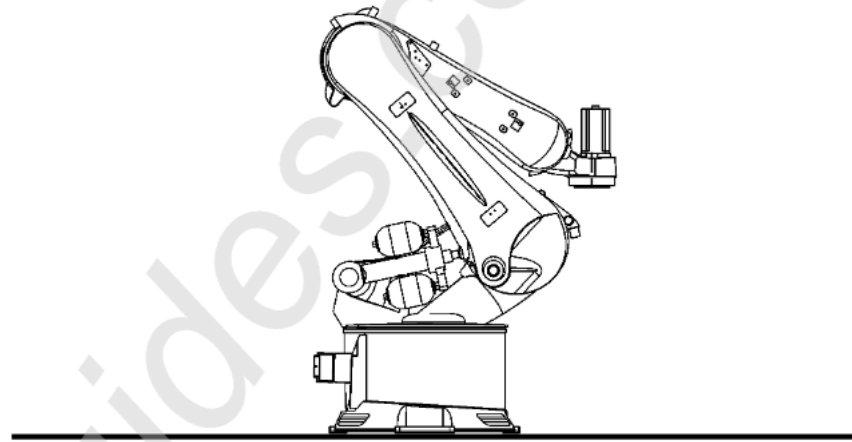
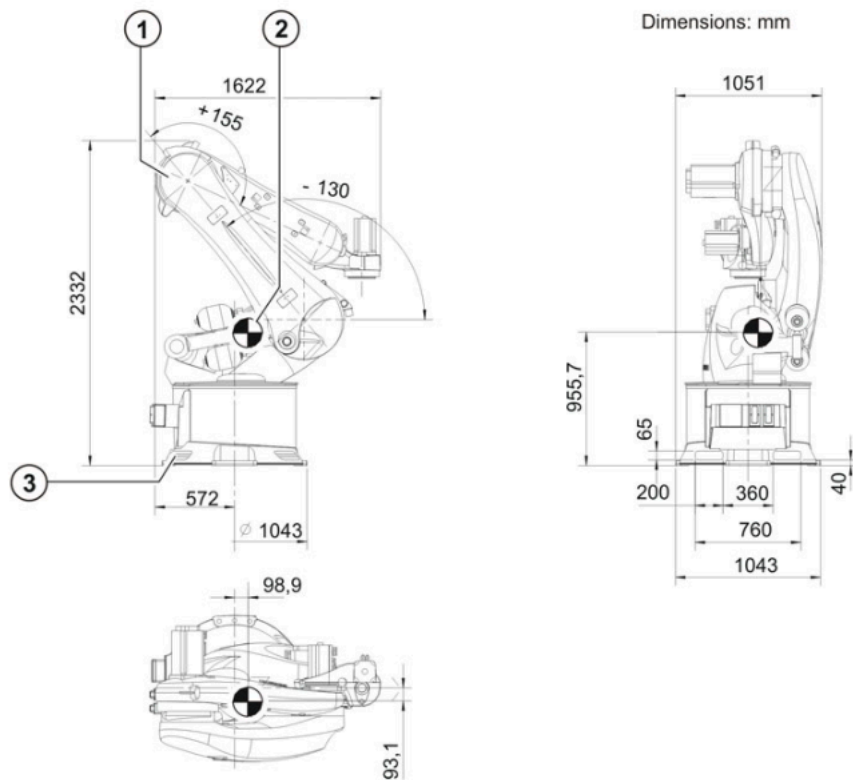


Fig. 7-1: Transport position

#### Transport dimensions

The transport dimensions (>>> Fig. 7-2 ) for the robot can be noted from the following diagram. The position of the center of gravity and the weight vary according to the specific configuration and the position of axes 2 and 3. The specified dimensions refer to the robot without equipment.

**Fig. 7-2: Transport dimensions**

- 1 Robot
- 2 Center of gravity
- 3 Fork slots

**Transportation**

The robot can be transported by fork lift truck or using lifting tackle.



**WARNING** Use of unsuitable handling equipment may result in damage to the robot or injury to persons. Only use authorized handling equipment with a sufficient load-bearing capacity. Only transport the robot in the manner specified here.

**Transportation by fork lift truck**

For transport by fork lift truck (>>> Fig. 7-3 ), two fork slots are provided in the base frame. The robot can be picked up by the fork lift truck from the front and rear. The fork lift truck must have a minimum payload capacity of 3.5 t.



Avoid excessive loading of the fork slots through undue inward or outward movement of hydraulically adjustable forks of the fork lift truck.

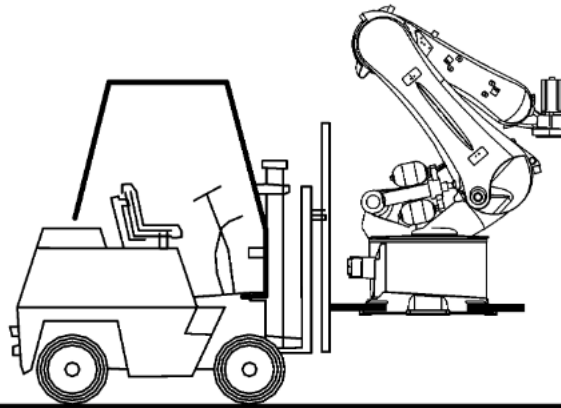


Fig. 7-3: Transportation by fork lift truck

### Transportation using lifting tackle

The robot can also be transported using lifting tackle (>>> Fig. 7-4 ). The robot must be in the transport position. The lifting tackle is attached at 3 points to M20 DIN 580 eyebolts. All ropes must be long enough and must be routed in such a way that the robot is not damaged. Installed tools and pieces of equipment can cause undesirable shifts in the center of gravity.



#### **WARNING**

The robot may tip during transportation. Risk of personal injury and damage to property.

If the robot is being transported using lifting tackle, special care must be exercised to prevent it from tipping. Additional safeguarding measures must be taken. It is forbidden to pick up the robot in any other way using a crane!

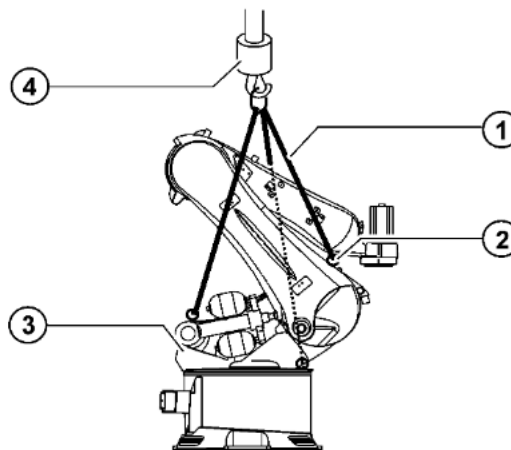


Fig. 7-4: Transportation using lifting tackle

- |                  |                   |
|------------------|-------------------|
| 1 Lifting tackle | 3 Rotating column |
| 2 M20 eyebolt    | 4 Crane           |



## 8 Start-up and recommissioning

### 8.1 Mounting base for 175 mm concrete thickness

#### Description

The mounting base (>>> Fig. 8-1 ) with centering is used when the robot is fastened to the floor, i.e. directly on a concrete foundation.

The mounting base consists of:

- Plate
- Chemical anchors (resin-bonded anchors) with Dynamic Set
- Fasteners

This mounting variant requires a level and smooth surface on a concrete foundation with adequate load bearing capacity. The concrete foundation must be able to accommodate the forces occurring during operation. The minimum dimensions must be observed.

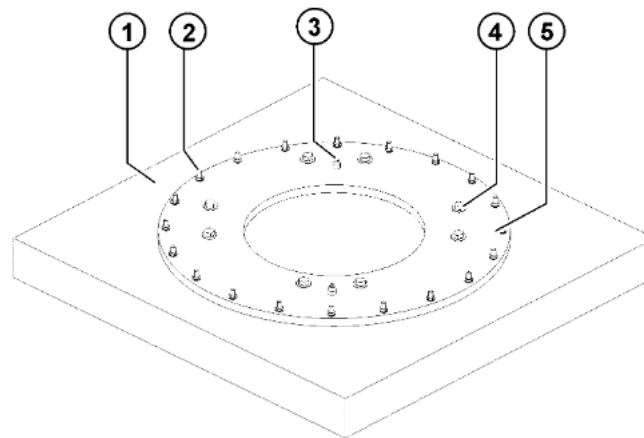


Fig. 8-1: Mounting base 175 mm

- |   |                                       |   |              |
|---|---------------------------------------|---|--------------|
| 1 | Concrete foundation                   | 4 | Hexagon bolt |
| 2 | Chemical anchor (resin-bonded anchor) | 5 | Plate        |
| 3 | Pin                                   |   |              |

#### Grade of concrete for foundations

When producing foundations from concrete, observe the load-bearing capacity of the ground and the country-specific construction regulations. There must be no layers of insulation or screed between the bedplates and the concrete foundation. The quality of the concrete must meet the requirements of the following standard:

- C20/25 according to DIN EN 206-1:2001/DIN 1045-2:2008

#### Dimensioned drawing

The following illustration provides all the necessary information on the mounting base, together with the required foundation data (>>> Fig. 8-2 ).

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## 8.2 Mounting base for 200 mm concrete thickness

### Description

The mounting base for 200 mm concrete thickness (>>> Fig. 8-4 ) is used when the robot is fastened to the floor, i.e. directly on a concrete foundation with a thickness of at least 200 mm.

The mounting base with centering consists of:

- Bedplates
- Chemical anchors (resin-bonded anchors) with Dynamic Set
- Fastening elements

This mounting variant requires a level and smooth surface on a concrete foundation with adequate load bearing capacity. The concrete foundation must be able to accommodate the forces occurring during operation. There must be no layers of insulation or screed between the bedplates and the concrete foundation.

The minimum dimensions must be observed.

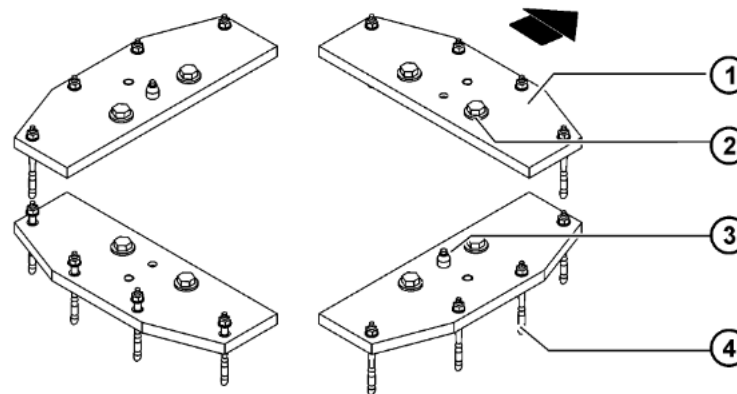


Fig. 8-4: Mounting base 200 mm

- |                |                                     |
|----------------|-------------------------------------|
| 1 Bedplate     | 3 Pin with Allen screw              |
| 2 Hexagon bolt | 4 Chemical anchors with Dynamic Set |

### Grade of concrete for foundations

When producing foundations from concrete, observe the load-bearing capacity of the ground and the country-specific construction regulations. There must be no layers of insulation or screed between the bedplates and the concrete foundation. The quality of the concrete must meet the requirements of the following standard:

- C20/25 according to DIN EN 206-1:2001/DIN 1045-2:2008

### Dimensioned drawing

The following illustrations provide all the necessary information on the mounting base, together with the required foundation data.

Dimensions: mm

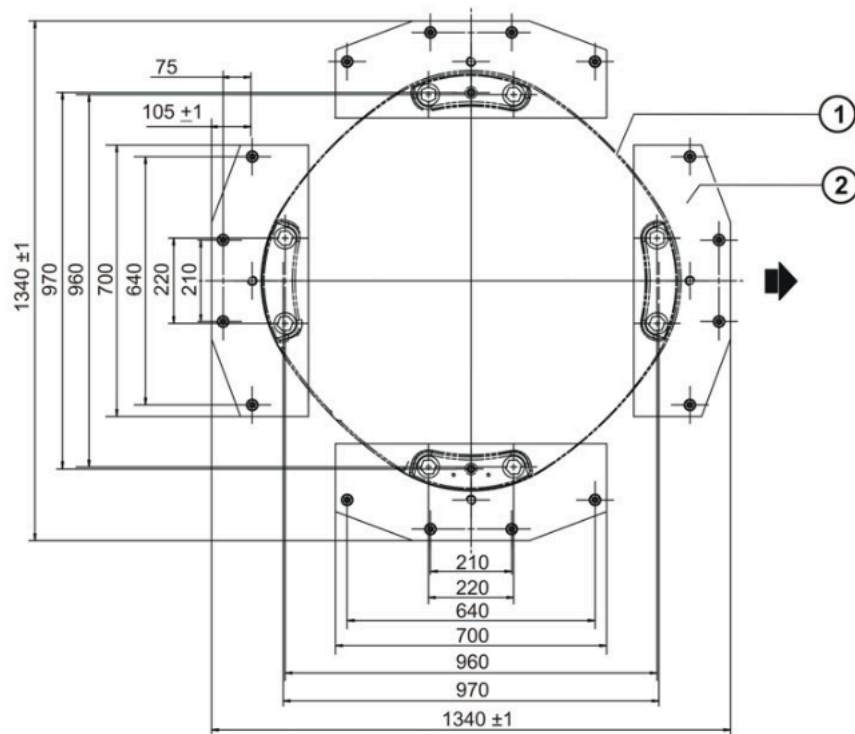


Fig. 8-5: Mounting base 200 mm, dimensioned drawing

1 Robot contour

2 Bedplates

To ensure that the anchor forces are safely transmitted to the foundation, observe the dimensions for concrete foundations specified in the following illustration.

Dimensions: mm

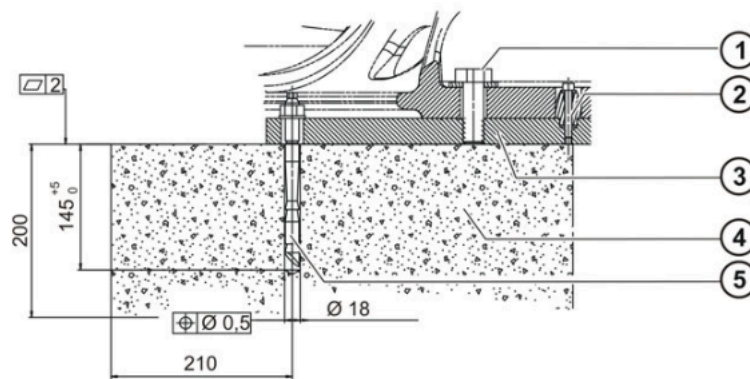


Fig. 8-6: Cross-section of foundation 200 mm

1 Hexagon bolt

4 Concrete foundation

2 Pin

5 Chemical anchor

3 Bedplate

### 8.3 Installing the machine frame mounting assembly

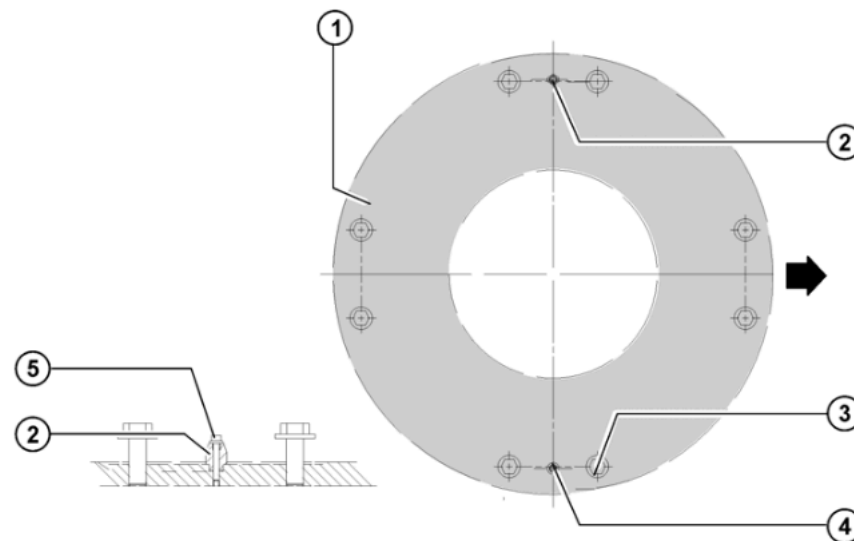
#### Description

The machine frame mounting assembly with centering (>>> Fig. 6-8 ) is used when the robot is fastened on a steel structure, a booster frame (pedestal) or a KUKA linear unit. It must be ensured that the substructure is able to withstand safely the forces occurring during operation (foundation loads). The following diagram contains all the necessary information that must be observed when preparing the mounting surface.

For further data concerning the machine frame mounting assembly, see (>>> Fig. 8-7 ).

The machine frame mounting assembly consists of:

- Pin with fasteners
- Sword pin with fasteners
- Hexagon bolts with conical spring washers



**Fig. 8-7: Installing the mounting base**

- |                    |                   |
|--------------------|-------------------|
| 1 Mounting surface | 4 Sword pin       |
| 2 Pin              | 5 Allen screw, 2x |
| 3 Allen screw, 8x  |                   |

#### Preconditions

- Substructure and mounting surface are prepared as described in Section (>>> 6.3 "Machine frame mounting" Page 57).

#### Procedure for installation

1. Insert the pin on the left and fasten it with an M8x55-8.8 Allen screw and conical spring washer.
2. Insert the sword pin on the right and fasten it with an M8x55-8.8 Allen screw and conical spring washer.
3. Tighten both M8x55-8.8 Allen screws with a torque wrench ( $M_A = 23.0$  Nm).

### 8.4 Installing the robot

#### Description

This description is valid for the installation of floor-mounted robots with the assemblies "Mounting base for 175 mm, 200 mm concrete thickness" and the "Machine frame mounting" assembly. 8 hexagon bolts with conical spring

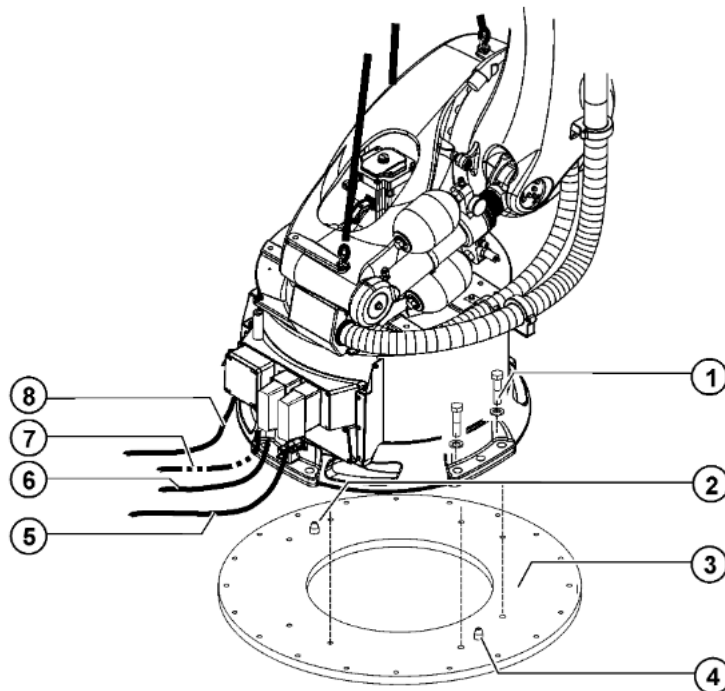
washers are used for fastening the robot to the bedplates or to a machine frame. Pins and sword pins are provided to ensure correct positioning.

### Preconditions

- The mounting surface has been prepared for the robot.
- The connecting cables are installed on the system side.
- The site of installation must be accessible with a fork lift truck or with a crane. Minimum payload capacity 2,500 kg

### Procedure

1. Check that the pins are undamaged and fitted securely (>>> Fig. 8-8 ).
2. Bring the robot to the installation site by crane or fork lift truck.
3. Lower the robot vertically onto the mounting face. Ensure that an entirely vertical position is maintained in order to prevent damage to the pins.
4. Remove the fork lift truck or lifting tackle.
5. Insert 8 M24x70-8.8-A2K hexagon bolts together with conical spring washers.
6. Tighten 8 hexagon bolts with a torque wrench in diagonally opposite sequence. Gradually increase the tightening torque to 640 Nm.
7. Connect motor cables X30 and X30.2.
8. Connect control cable X31.
9. Establish the equipotential bonding between the robot and the robot controller and between the robot and the system potential via the threaded bolts.
10. Check the equipotential bonding in accordance with VDE 0100 and EN 60204-1.
11. If present, mount the tool and connect the energy supply systems.



**Fig. 8-8: Installing the robot**

- |                    |                                        |
|--------------------|----------------------------------------|
| 1 Hexagon bolts    | 5 Motor cable                          |
| 2 Pin              | 6 Motor cable                          |
| 3 Mounting surface | 7 Equipotential bonding, threaded bolt |
| 4 Pin, sword pin   | 8 Control cable                        |

**WARNING** The holding-down bolts of the robot on the foundation must be retightened once after 100 operating hours with the specified tightening torque.

Put the robot system into operation in accordance with the "Start-up" chapter of the operating and programming instructions for the KUKA System Software (KSS) and the "Start-up" chapter of the operating instructions for the robot controller.

## 8.5 Description of the connecting cables

**Configuration** The connecting cables are used to transfer power and signals between the robot controller and the robot.

The connecting cables comprise:

- Motor cable (2x)
- Control cable
- Ground conductor, optional

**Interface** For connection of the connecting cables between the robot controller and the robot, the following connectors are available at the interfaces:

| Cable designation                                                      | Connector designation robot controller - robot | Connection            |
|------------------------------------------------------------------------|------------------------------------------------|-----------------------|
| Motor cable                                                            | X20.1 - X30.1                                  | HAN size 24           |
| Motor cable                                                            | X20.4 - X30.4                                  | HAN size 24           |
| Data cable                                                             | X21 - X31                                      | Rectangular connector |
| Ground conductor / equipotential bonding 16 mm <sup>2</sup> (optional) |                                                | M8 ring cable lug     |



For the connecting cables, a ground conductor is always required to provide a low-resistance connection between the robot and the control cabinet in accordance with DIN EN 60204. The ground conductor is not part of the scope of supply and can be ordered as an option. The connection must be made by the customer. The tapped holes for connecting the ground conductor are located on the base frame of the robot.

**Standard connecting cable**

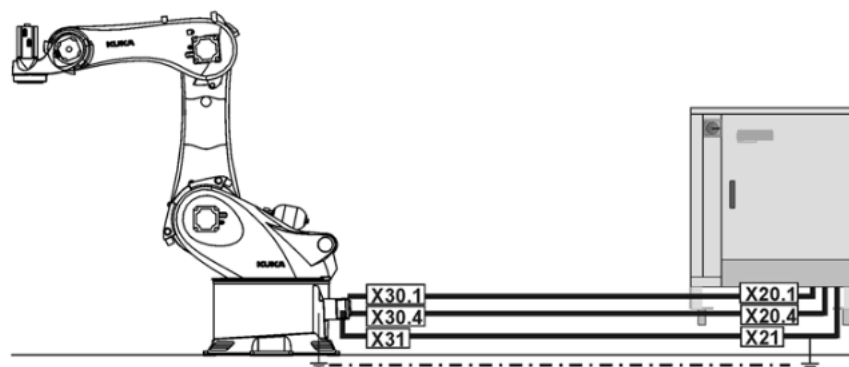
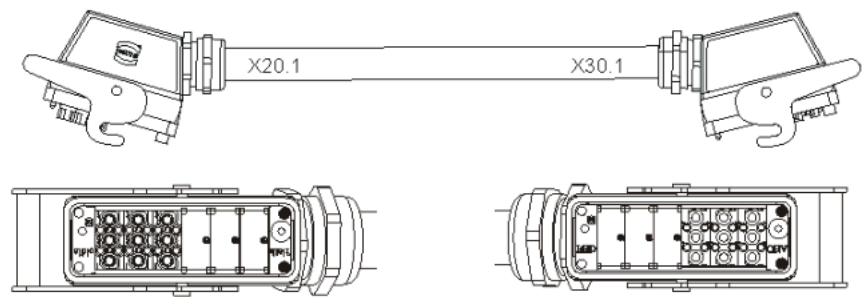
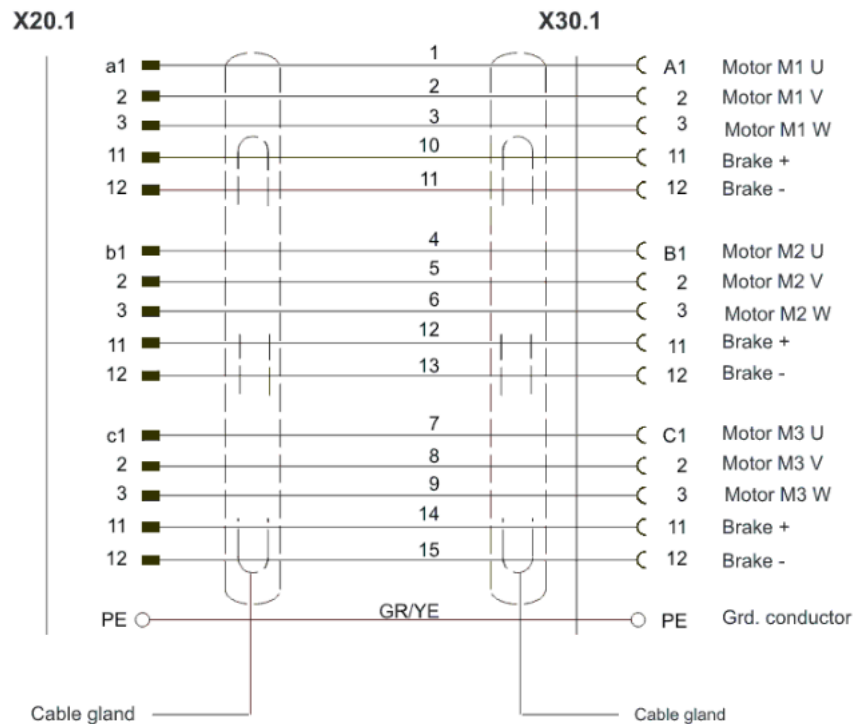
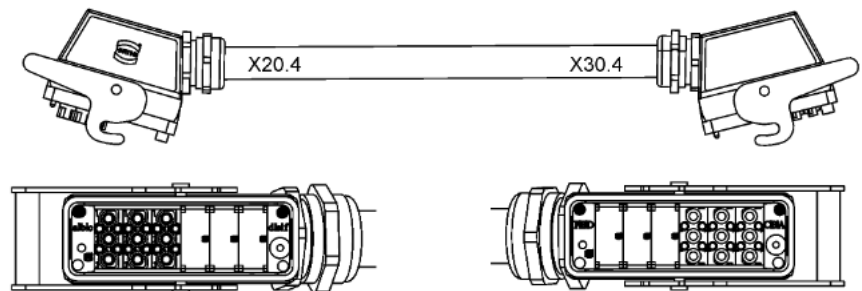


Fig. 8-9: Connecting cables, overview

**Fig. 8-10: Connecting cable, motor cable, X20.1 - X30.1****Fig. 8-11: Wiring diagram, motor cable X20.1 - X30.1****Fig. 8-12: Connecting cable, motor cable, X20.4 - X30.4**

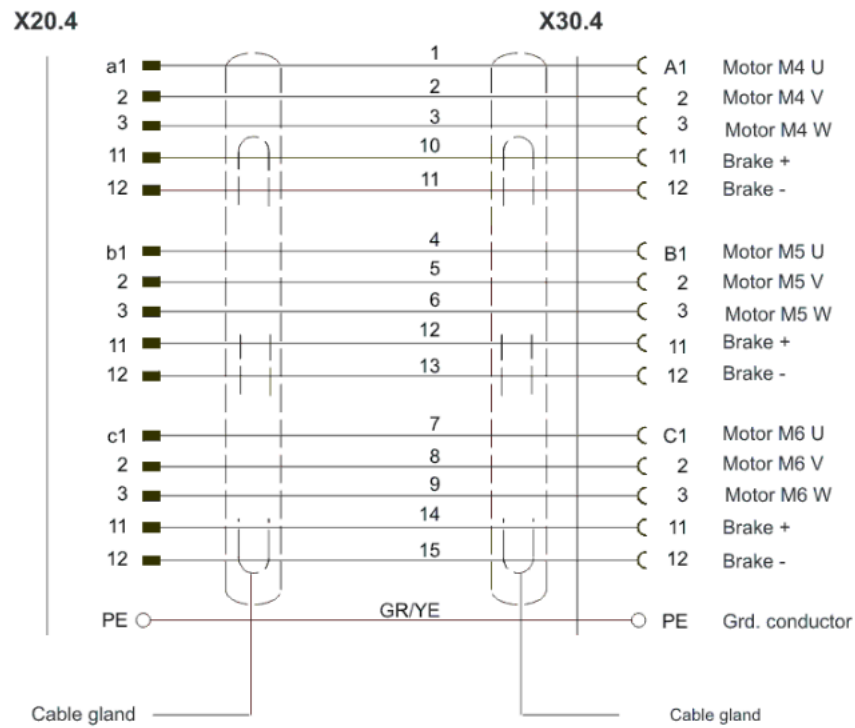


Fig. 8-13: Wiring diagram, motor cable X20.4-X30.4

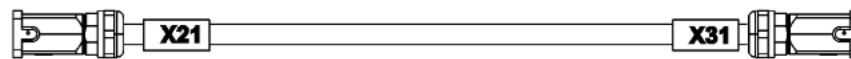


Fig. 8-14: Connecting cable, data cable X21 - X31

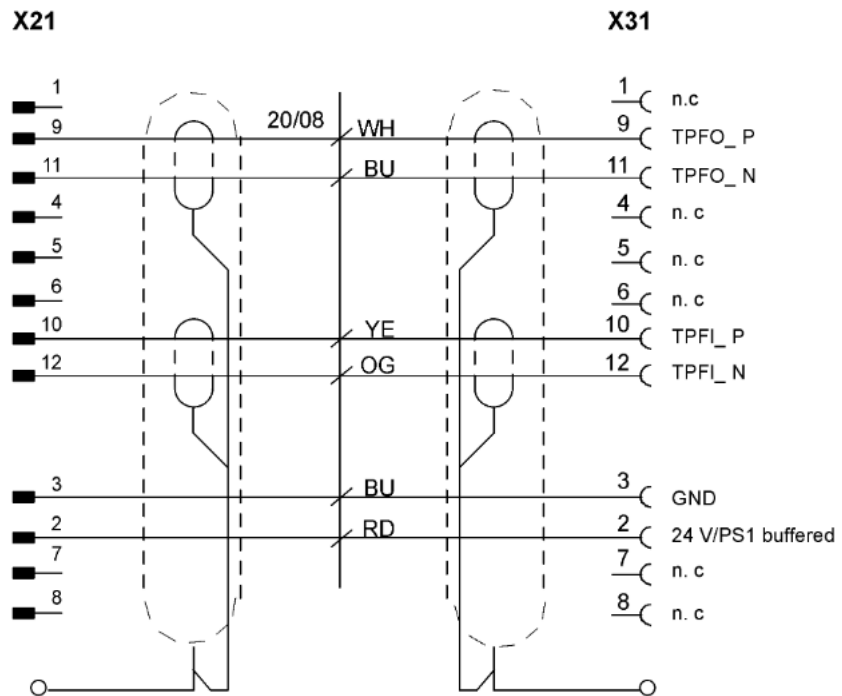


Fig. 8-15: Connecting cable, wiring diagram, data cable X21 - X31

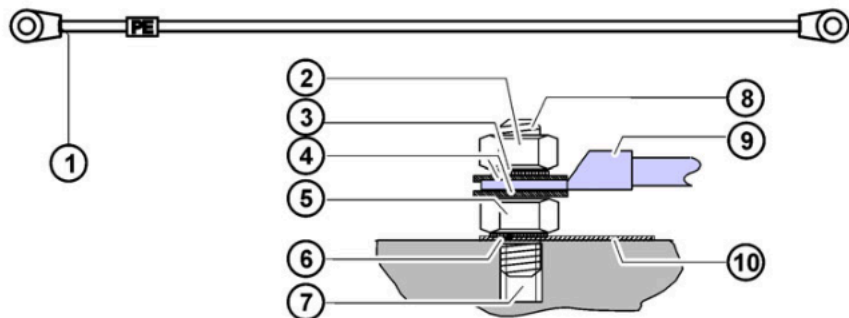


Fig. 8-16: Connecting cable, ground conductor

- |   |                          |    |                                                |
|---|--------------------------|----|------------------------------------------------|
| 1 | Ground conductor         | 6  | Conical spring washer                          |
| 2 | Hexagon nut              | 7  | Robot                                          |
| 3 | Conical spring washer 2x | 8  | Setscrew                                       |
| 4 | Plain washer 2x          | 9  | Ground conductor connection, M8 ring cable lug |
| 5 | Hexagon nut              | 10 | Ground plate                                   |

## 9 Options

### 9.1 Release device (optional)

#### Description

The release device can be used to move the manipulator manually after an accident or malfunction. The release device can be used for all the motors. It is only for use in exceptional circumstances and emergencies (e.g. for freeing people).

The release device is mounted on the base frame of the manipulator. This assembly also includes a ratchet and a set of plates with one plate for each motor. The plate specifies the direction of rotation for the ratchet and shows the corresponding direction of motion of the manipulator.

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## 10 Appendix

### 10.1 Tightening torques

#### Tightening torques

The following tightening torques (Nm) are valid for screws and nuts where no other specifications are given.

The specified values apply to lightly oiled black (e.g. phosphated) and coated (e.g. mech.galv., zinc flake coatings) screws and nuts.

| Screw thread | Strength class |           |           |
|--------------|----------------|-----------|-----------|
|              | 8.8            | 10.9      | 12.9      |
| M1.6         | 0.17 Nm        | 0.24 Nm   | 0.28 Nm   |
| M2           | 0.35 Nm        | 0.48 Nm   | 0.56 Nm   |
| M2.5         | 0.68 Nm        | 0.93 Nm   | 1.10 Nm   |
| M3           | 1.2 Nm         | 1.6 Nm    | 2.0 Nm    |
| M4           | 2.8 Nm         | 3.8 Nm    | 4.4 Nm    |
| M5           | 5.6 Nm         | 7.5 Nm    | 9.0 Nm    |
| M6           | 9.5 Nm         | 12.5 Nm   | 15.0 Nm   |
| M8           | 23.0 Nm        | 31.0 Nm   | 36.0 Nm   |
| M10          | 45.0 Nm        | 60.0 Nm   | 70.0 Nm   |
| M12          | 78.0 Nm        | 104.0 Nm  | 125.0 Nm  |
| M14          | 125.0 Nm       | 165.0 Nm  | 195.0 Nm  |
| M16          | 195.0 Nm       | 250.0 Nm  | 305.0 Nm  |
| M20          | 370.0 Nm       | 500.0 Nm  | 600.0 Nm  |
| M24          | 640.0 Nm       | 860.0 Nm  | 1030.0 Nm |
| M30          | 1330.0 Nm      | 1700.0 Nm | 2000.0 Nm |

| Screw thread | Strength class                   |                                                                               |
|--------------|----------------------------------|-------------------------------------------------------------------------------|
|              | 8.8<br>ISO7991<br>Hexagon socket | 10.9<br>ISO7380, ISO07381<br>Fillister head<br>screw with hexa-<br>gon socket |
| M3           | 0.8 Nm                           | 0.8 Nm                                                                        |
| M4           | 1.9 Nm                           | 1.9 Nm                                                                        |
| M5           | 3.8 Nm                           | 3.8 Nm                                                                        |

Tighten M5 domed cap nuts with a torque of 4.2 Nm.

### 10.2 Safety data sheets

#### 10.2.1 Safety data sheet for Microlube GL 261 lubricating grease

The following extract from the safety data sheet according to 91/155/EEC must be observed.

##### Section 1 Designation of substance/formulation and manufacturer

| Name of substance/preparation |                  |
|-------------------------------|------------------|
| Trade name:                   | Microlube GL 261 |

| Name of substance/preparation |                    |
|-------------------------------|--------------------|
| Article no.:                  | 020195             |
| Use:                          | Lubricating grease |

| Manufacturer designation |                                |
|--------------------------|--------------------------------|
| Company:                 | Klüber Lubrication München AG  |
| Address:                 | Geisenhausenerstr. 7           |
| Postal code:             | D-81379 Munich                 |
| Country:                 | Germany                        |
| Phone:                   | +49 (0)89 7876 0               |
| Fax:                     | +49 (0)89 7876 333             |
| Information center:      | Material Compliance Management |
| Emergency hotline:       | +49 (0)89 7876 700             |

## Section 2 Composition / Information about the components

|                            |                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------|
| Chemical characterization: | Mineral oil, lithium special soap, UV indicator                                       |
| Hazardous components:      | This product contains no dangerous components above the legally defined limit values. |

## Section 3 Possible hazards

The preparation is **NOT** classified as hazardous in accordance with Directive 1999/45/EC in its altered and adapted version.

|                              |                                              |
|------------------------------|----------------------------------------------|
| Physical / chemical hazards: | Not classified as dangerous.                 |
| Human health hazards:        | May irritate eyes and skin.                  |
| Environmental hazards:       | Unlikely to be harmful to aquatic organisms. |

|                              |                                              |
|------------------------------|----------------------------------------------|
| Physical / chemical hazards: | Not classified as dangerous.                 |
| Human health hazards:        | May irritate eyes and skin.                  |
| Environmental hazards:       | Unlikely to be harmful to aquatic organisms. |

| Effects and symptoms |                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------|
| Eyes:                | May cause mild eye irritation.                                                                                    |
| Skin:                | Can dry out the skin and lead to irritation and/or dermatitis. Allergic reactions are possible in the worst case. |
| Inhalation:          | Inhalation of oil mist or vapors at elevated temperatures may cause respiratory irritation.                       |
| Ingestion:           | Ingestion may cause gastrointestinal irritation (e.g. diarrhea).                                                  |

## Section 4 First aid measures

|                    |                                                                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contact with eyes: | Rinse eyes immediately with plenty of water for several minutes. If irritation occurs, consult a doctor.                                                                                                                                                     |
| Skin contact:      | Wash affected areas of skin with soap and water, or use suitable cleaning agent. Change clothing and shoes if they become contaminated with product. Wash clothing before reuse. Clean shoes thoroughly before reuse. If irritation occurs consult a doctor. |
| Inhalation:        | Take affected person into fresh air. Consult a doctor if symptoms persist.                                                                                                                                                                                   |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ingestion:</b>          | Do <b>not</b> induce vomiting. If the person is unconscious, do not give anything by mouth. Consult a physician immediately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Notes to physician:</b> | <p>Treatment should in general be symptomatic and directed at relieving any effects.</p> <p>Note regarding high-pressure applications.</p> <p>Injection into the skin due to contact with a product under high pressure constitutes a major medical emergency. Within a few hours the tissue swells up and becomes discolored and extremely painful, with severe subcutaneous necrosis.</p> <p>Surgical treatment is absolutely imperative. Comprehensive opening of the wound and the tissue beneath it is necessary in order to reduce tissue loss and to prevent or limit lasting damage. The high pressure can cause the product to penetrate extensive areas of tissue layers.</p> |

## Section 5 Fire-fighting measures

|                                                     |                                                                                                                |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Suitable extinguishing agents:</b>               | In the event of a fire: use water spray (mist), foam, dry chemicals or CO <sub>2</sub> .                       |
| <b>Unsuitable extinguishing agents:</b>             | <b>Do not</b> use water jets.                                                                                  |
| <b>Special instructions for extinguishing work:</b> | Contaminated extinguishing water must be collected separately and prevented from entering the drainage system. |
| <b>Hazardous decomposition products:</b>            | Decomposition products include various oxides (e.g. carbon oxides).                                            |
| <b>Unusual fire/explosion hazards:</b>              | This product is not inherently explosive in accordance with the applicable rules.                              |
| <b>Special fire-fighting measures:</b>              | -                                                                                                              |
| <b>Protection of fire-fighters:</b>                 | Fire-fighters must wear self-contained positive pressure breathing apparatus (SCBA) and full protective gear.  |

## Section 6 Measures after unintended release

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal safety precautions:</b>       | In case of spillage, isolate the hazard area. Avoid contact with the hazardous substance and avoid inhaling vapors. Ensure adequate ventilation. Where there is insufficient ventilation, wear suitable respiratory equipment. Use suitable protective equipment.                                                                                                                                                     |
| <b>Environmental protection measures:</b> | Prevent the product from entering the drainage system, surface waters and soil. Notify the competent authorities, if required.                                                                                                                                                                                                                                                                                        |
| <b>Large spills:</b>                      | Stop the leak if you can do so without risk. Remove container from spill area. Flush spilled material into a wastewater treatment plant. Contain spilled material using a non-combustible absorbent (e.g. sand, soil, vermiculite, diatomaceous earth) and collect in the containers provided. Disposal should be entrusted to a recognized waste disposal company. Contaminated absorbents can be just as dangerous. |
| <b>Small spills:</b>                      | Stop the leak if you can do so without risk. Remove container from spill area. Dilute with water and mop up, or add absorbent material and place the substance in a suitable container for disposal. Disposal should be entrusted to a recognized waste disposal company.                                                                                                                                             |

## Section 7 Handling and storage

|                    |                                                                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling:</b>   | Prevent contact with skin or clothing. Spilled and leaked product must be prevented from coming into contact with soil and surface waters. Wash thoroughly after handling. |
| <b>Storage:</b>    | Keep containers tightly sealed. Keep containers in a cool, well-ventilated area.                                                                                           |
| <b>Unsuitable:</b> | Avoid prolonged exposure to increased temperatures.                                                                                                                        |

|                           |    |
|---------------------------|----|
| <b>VCI storage class:</b> | 11 |
|---------------------------|----|

## Section 8 Exposure limits and personal protective equipment

|                                      |                                                                                      |
|--------------------------------------|--------------------------------------------------------------------------------------|
| <b>Threshold limit values (TLV):</b> | No occupational exposure threshold limit values have been assigned for this product. |
|--------------------------------------|--------------------------------------------------------------------------------------|

| Protective measures                                     |                                                                                                                      |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Limitation and monitoring of exposure in the workplace: | Provide exhaust ventilation or other engineering controls.                                                           |
| Hygiene measures:                                       | Wash hands, forearms and face thoroughly after handling chemical products.                                           |
| Respiratory protection:                                 | Sufficient ventilation is recommended in industry. If ventilation is insufficient, respiratory protection is useful. |
| Hand protection:                                        | Wear chemical-resistant gloves (recommended: nitrile gloves) if prolonged or repeated contact is likely.             |
| Eye protection:                                         | Protective goggles with side shields to guard against splashing.                                                     |
| Skin and body:                                          | Wear appropriate clothing to avoid prolonged skin contact.                                                           |

## Section 9 Physical and chemical properties

|                        |                                                           |
|------------------------|-----------------------------------------------------------|
| <b>Physical state:</b> | Paste                                                     |
| <b>Color:</b>          | Tawny                                                     |
| <b>Odor:</b>           | Specific                                                  |
| <b>Flash point:</b>    | -                                                         |
| <b>Drop point:</b>     | >220 °C (DIN ISO 2176)                                    |
| <b>Density:</b>        | <1,000 kg/m <sup>3</sup> (<1 g/cm <sup>3</sup> ) at 20 °C |
| <b>Solubility:</b>     | Insoluble in water                                        |

## Section 10 Stability and reactivity

|                                              |                                                                                                                                                                          |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Incompatible with various substances:</b> | Reactive or incompatible with oxidizing materials.                                                                                                                       |
| <b>Hazardous polymerization:</b>             | The product is stable. No hazardous polymerization occurs under normal conditions.                                                                                       |
| <b>Hazardous decomposition products:</b>     | Decomposition products may include various oxides (e.g. carbon oxides). No hazardous decomposition products should be formed under normal conditions of storage and use. |

## Section 11 Toxicological information

|                         |                                                                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chronic effects:</b> | No particular effects or risks known.                                                                                                           |
| <b>Effects on eyes:</b> | May cause mild eye irritation.                                                                                                                  |
| <b>Effects on skin:</b> | Prolonged or repeated contact can dry out the skin and lead to irritation and/or dermatitis. Allergic reactions are possible in the worst case. |

|                     |                                                                                          |
|---------------------|------------------------------------------------------------------------------------------|
| Effect if inhaled:  | Vapors and aerosols may cause irritation of the mucous membranes of the nose and throat. |
| Effect if ingested: | May cause nausea, vomiting and diarrhea.                                                 |

## Section 12 Ecological information

|                              |                                                   |
|------------------------------|---------------------------------------------------|
| Persistence / degradability: | The product can be separated by mechanical means. |
| Mobility:                    | The product is insoluble in water.                |
| Environmental hazards:       | Prevent from entering wastewater and soil.        |

## Section 13 Disposal information

| Disposal information                         |                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disposal information / waste specifications: | Spilled and leaked product must be prevented from coming into contact with soil and surface waters. Observe local, regional and national regulations. Use only approved transporters, recyclers, treatment, storage or disposal facilities. If necessary, disposal should be entrusted to a recognized waste disposal company. |

| Hazardous waste                                                                                                                                    |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--|
| On the basis of the supplier's current state of knowledge, this product is not considered as hazardous waste as defined by EU directive 91/689/EC. |  |

## Section 14 Transport information

|                 |                                                                                        |
|-----------------|----------------------------------------------------------------------------------------|
| Classification: | Not hazardous as defined by the transport regulations (ADR/RID, ADN, IMDG, ICAO/IATA). |
|-----------------|----------------------------------------------------------------------------------------|

## Section 15 Regulations

|                                                                                                                                                       |                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| EU regulations:                                                                                                                                       | EC Directives 1999/45/EC and 67/548/EEC |
| Classification acc. to the German Administrative Regulation on the Classification of Substances Hazardous to Water into Water Hazard Classes (VwVwS): | 1, low hazard to waters, Annex 4        |

## Section 16 Other information

| Notes for the reader                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>All information is based on the current state of our knowledge. It is intended only to describe our product with regard to the safety data. It is not intended to provide assurance of particular properties.</p> <p>The product may only be used for the scope of work specified above; any other use requires prior consultation with KUKA. Using the product for any purpose other than for its designated use could lead to risks which are not described in this document.</p> <p>Further information on the use of the product may be found in the relevant technical specifications.</p> |  |

### 10.2.2 Safety data sheet for Castrol Hyspin ZZ 46 hydraulic oil

The following extract from the safety data sheet according to 91/155/EEC must be observed.

**Section 1 Designation of substance/formulation and manufacturer**

| Name of substance/preparation           |                                                                                                                                     |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product name:</b>                    | <b>Castrol Hyspin ZZ 46</b>                                                                                                         |
| <b>SDS no.:</b>                         | 450828                                                                                                                              |
| <b>Use of substance or formulation:</b> | Hydraulic oil<br>For specific instructions for use, see the corresponding technical data sheet or contact a company representative. |

| Manufacturer designation           |                                                                           |
|------------------------------------|---------------------------------------------------------------------------|
| <b>Company:</b>                    | <b>Deutsche Castrol Vertriebsgesellschaft mbH</b>                         |
| <b>Address:</b>                    | Max-Born-Str. 2, D-22761 Hamburg                                          |
| <b>Country:</b>                    | Germany                                                                   |
| <b>Phone:</b>                      | Central customer service/environment/product safety:<br>+49 (0)40 3594-01 |
| <b>Emergency telephone number:</b> | Carechem: +44 (0)208 762 8322 (24 hours)                                  |

**Section 2 Composition / Information about the components**

Highly refined base oil (IP 346 DMSO extract < 3%). Additives.

This product contains no dangerous components above the legally defined limit values.

**Section 3 Possible hazards**

The preparation is not classified as hazardous in accordance with Directive 1999/45/EC in its altered and adapted version.

|                                     |                                              |
|-------------------------------------|----------------------------------------------|
| <b>Physical / chemical hazards:</b> | Not classified as dangerous.                 |
| <b>Human health hazards:</b>        | Not classified as dangerous.                 |
| <b>Environmental hazards:</b>       | Unlikely to be harmful to aquatic organisms. |

| Effects and symptoms |                                                                                                                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eyes:</b>         | No significant health hazards identified.                                                                                                                                                                                                                                                |
| <b>Skin:</b>         | No significant health hazards identified.<br>Note regarding high-pressure applications.<br>Injection into the skin due to contact with a product under high pressure constitutes a major medical emergency. See Notes to physician under "First aid measures" in this safety data sheet. |
| <b>Inhalation:</b>   | No significant health hazards identified.                                                                                                                                                                                                                                                |
| <b>Ingestion:</b>    | No significant health hazards identified.                                                                                                                                                                                                                                                |

**Section 4 First aid measures**

|                           |                                                                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contact with eyes:</b> | In case of contact, rinse eyes immediately with plenty of water for at least 15 minutes. If irritation occurs, consult a doctor.                                                                                                             |
| <b>Skin contact:</b>      | In case of contact, rinse skin immediately with plenty of water. Change clothing and shoes if they become contaminated with product. Wash clothing before reuse. Clean shoes thoroughly before reuse. If irritation occurs consult a doctor. |
| <b>Inhalation:</b>        | Take affected person into fresh air. Consult a doctor if symptoms persist.                                                                                                                                                                   |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ingestion:</b>          | Do NOT induce vomiting unless explicitly directed to do so by medical personnel. Never give anything by mouth to an unconscious person. If large quantities of this product have been swallowed, call a physician immediately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Notes to physician:</b> | <p>Treatment should in general be symptomatic and directed at relieving any effects.</p> <p>Note regarding high-pressure applications.</p> <p>Injection into the skin due to contact with a product under high pressure constitutes a major medical emergency. The injuries appear minor at first, but within a few hours the tissue swells up and becomes discolored and extremely painful, with severe subcutaneous necrosis.</p> <p>Surgical treatment is absolutely imperative. Comprehensive opening of the wound and the tissue beneath it is necessary in order to reduce tissue loss and to prevent or limit lasting damage. The high pressure can cause the product to penetrate extensive areas of tissue layers.</p> |

## Section 5 Fire-fighting measures

|                                          |                                                                                                               |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Suitable extinguishing agents:</b>    | In the event of a fire, use foam, dry-chemical or carbon dioxide extinguisher or spray.                       |
| <b>Unsuitable extinguishing agents:</b>  | Do NOT use water jets.                                                                                        |
| <b>Hazardous decomposition products:</b> | These products are carbon oxides (CO, CO <sub>2</sub> ).                                                      |
| <b>Unusual fire/explosion hazards:</b>   | Not specified.                                                                                                |
| <b>Special fire-fighting measures:</b>   | Not specified.                                                                                                |
| <b>Protection of fire-fighters:</b>      | Fire-fighters must wear self-contained positive pressure breathing apparatus (SCBA) and full protective gear. |

## Section 6 Measures after unintended release

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal safety precautions:</b>                       | Contact emergency personnel immediately. Keep unnecessary personnel away. Use suitable protective equipment (see Section: "Exposure limits and personal protective equipment"). Follow all fire-fighting procedures (see Section: "Fire-fighting measures").                                                                                                                                                                                                                                                                                                                    |
| <b>Environmental precautions and clean-up methods:</b>    | If no emergency personnel are available, contain spilled material. For small spills add an absorbent (soil may be used in the absence of other suitable materials) and scoop up material. Place it in a sealed watertight container for subsequent disposal. For larger spills, take steps to contain the material to ensure that runoff cannot reach a waterway. Place spilled material in an appropriate container for disposal. Prevent spilled material from coming into contact with soil and surface waters. See Section "Disposal information" for disposal information. |
| <b>Personal protection in the event of a large spill:</b> | Splash goggles, full suit, boots, gloves.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## Section 7 Handling and storage

|                    |                                                                                  |
|--------------------|----------------------------------------------------------------------------------|
| <b>Handling:</b>   | Wash thoroughly after handling.                                                  |
| <b>Storage:</b>    | Keep containers tightly sealed. Keep containers in a cool, well-ventilated area. |
| <b>Unsuitable:</b> | Prolonged exposure to increased temperatures.                                    |

## Section 8 Exposure limits and personal protective equipment

| <b>Ingredient name:</b> | <b>Threshold limit values (TLV) acc. to ACGIH (USA)</b>                                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Base oil – unspecified  | STEL: 10 mg/m <sup>3</sup> 15 minute(s). Form: Mineral oil mist<br>TWA: 5 mg/m <sup>3</sup> 8 hour(s). Form: Mineral oil mist |

Where no exposure limits exist in law, the ACGIH values are included for information and as reference. For further information, please consult your supplier. It must be noted that all mists, vapors and dusts also contain other components of this preparation. For this reason, the specific limit values given in the safety data sheet for particular components may not be applicable to the product and are given here only as a guideline.

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Protective measures</b> | <p>Provide exhaust ventilation or other engineering controls to keep the relevant airborne concentrations below their respective occupational exposure limits.</p> <p>All chemicals should be assessed with regard to their risks to health and appropriate control measures put in place to prevent or adequately control exposure. A hierarchy of control measures exists (e.g. elimination, substitution, general ventilation, containment, systems of work, changing the process or activity) which must be considered before using personal protective equipment. Personal protective equipment should conform to the applicable standards. For this purpose, please consult your supplier of personal protective equipment. Relevant information can be obtained from the European Committee for Standardization: <a href="http://www.cenorm.be/cenorm/index.htm">http://www.cenorm.be/cenorm/index.htm</a>.</p> <p>The final choice of protective equipment will depend on a risk assessment. It must always be ensured that all items of personal protective equipment are compatible with one another.</p> |
| <b>Hygiene measures:</b>   | Wash hands, forearms and face thoroughly after handling chemical products and before eating, smoking or using the toilet, as well as at the end of the working day.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Personal protective equipment |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Respiratory organs:           | <p>Respiratory protection is not normally required where there is adequate natural or local exhaust ventilation to control exposure. Where there is insufficient ventilation, wear suitable respiratory equipment. Respiratory protection equipment must be checked in order to ensure a correct fit every time it is used.</p> <p>Air-filtering respirators, also called air-purifying respirators, are not adequate under conditions of oxygen deficiency (e.g. low oxygen concentration) and are not suitable where airborne concentrations of chemicals with a significant hazard are present. In these cases, air-supplied breathing apparatus is required.</p> <p>Provided an air-filtering or air-purifying respirator is suitable, a particulate filter for mist or fumes can be used. Use filter type P or comparable standard. A combination filter for particles and organic gases and vapors (boiling point &gt;65 °C) may be necessary if vapor or abnormal odor is also present due to high product temperature. Use filter type AP or comparable standard.</p> |
| Skin and body:                | <p>Use of protective clothing is good industrial practice.</p> <p>Cotton or polyester/cotton overalls only provide protection against light, superficial contamination which will not soak through to the skin. Overalls should be laundered on a regular basis. When the risk of skin exposure is high (e.g. when cleaning up spillages or if there is a risk of splashing), then chemical-resistant aprons and/or impervious chemical suits and boots will be required.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Hands:                        | <p>Wear protective gloves if prolonged or repeated contact is likely. Wear chemical-resistant gloves.</p> <p>Recommended: nitrile gloves.</p> <p>Cold-resistant clothing is required when handling ice-cold products.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Eyes:                         | Protective goggles with side shields to guard against splashing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Section 9 Physical and chemical properties

|                      |                                                                                                               |
|----------------------|---------------------------------------------------------------------------------------------------------------|
| Flash point:         | 220 °C (closed cup)                                                                                           |
| Pour point:          | -21 °C                                                                                                        |
| Color:               | Straw-colored                                                                                                 |
| Odor:                | Oily                                                                                                          |
| Physical state:      | Liquid                                                                                                        |
| Density:             | 876 kg/m <sup>3</sup> (0.876 g/cm <sup>3</sup> ) at 15 °C                                                     |
| Solubility:          | Insoluble in water                                                                                            |
| LogK <sub>OW</sub> : | The product is much more soluble in octanol; log(octanol/water) >3                                            |
| Viscosity:           | Kinematic: 46 mm <sup>2</sup> /s (46 cSt) at 40 °C<br>Kinematic: 6.65 mm <sup>2</sup> /s (6.65 cSt) at 100 °C |

## Section 10 Stability and reactivity

|                                       |                                                          |
|---------------------------------------|----------------------------------------------------------|
| Incompatible with various substances: | Reacts with oxidizing agents.                            |
| Hazardous polymerization:             | Will not occur.                                          |
| Hazardous decomposition products:     | These products are carbon oxides (CO, CO <sub>2</sub> ). |

**Section 11 Toxicological information**

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute toxicity: | <p>Unlikely to cause more than transient stinging or redness if accidental eye contact occurs.</p> <p>Unlikely to cause harm to the skin on brief or occasional contact. Prolonged or repeated exposure may dry out the skin and lead to dermatitis.</p> <p>Unlikely to cause harm if accidentally swallowed in small doses. Larger quantities may cause nausea and diarrhea.</p> <p>At normal ambient temperatures this product is unlikely to present an inhalation hazard because of its low volatility. May be harmful by inhalation if exposure to vapor, mists or fumes resulting from thermal decomposition products occurs.</p> |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Chronic toxicity:**

|                       |                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carcinogenic effects: | No component of this product at levels greater than or equal to 0.1% is identified as a carcinogen by ACGIH, the International Agency for Research on Cancer (IARC) or the European Commission (EC). |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Section 12 Ecological information**

|                               |                                                                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Persistence / degradability:  | Inherently biodegradable.                                                                                                                       |
| Mobility:                     | Spilled substance can enter the ground and contaminate soil and groundwater.                                                                    |
| Bioaccumulative potential:    | This product is not expected to bioaccumulate through food chains in the environment.                                                           |
| Environmental hazards:        | Unlikely to be harmful to aquatic organisms.                                                                                                    |
| Other ecological information: | Spilled product can lead to formation of a film on the surface of water; this can reduce the oxygen exchange, thereby causing organisms to die. |

**Section 13 Disposal information****Disposal information**

|                                              |                                                                          |
|----------------------------------------------|--------------------------------------------------------------------------|
| Disposal information / waste specifications: | Disposal must be carried out by an authorized waste disposal contractor. |
|----------------------------------------------|--------------------------------------------------------------------------|

**Used/contaminated product**

|                               |                                                                   |
|-------------------------------|-------------------------------------------------------------------|
| European Waste Catalog (EWC): | 13 01 10* Non-chlorinated hydraulic oils with a mineral oil base. |
|-------------------------------|-------------------------------------------------------------------|

Use of the product for purposes other than those specified and/or impurities can necessitate the use of a different waste code number for the waste producer.

**Section 14 Transport information**

|                 |                                                                                        |
|-----------------|----------------------------------------------------------------------------------------|
| Classification: | Not hazardous as defined by the transport regulations (ADR/RID, ADN, IMDG, ICAO/IATA). |
|-----------------|----------------------------------------------------------------------------------------|

**Section 15 Regulations**

| Labeling requirements |                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| Risk (R) phrases:     | This product is not classified according to the Dangerous Substances Order / EU regulations.                                 |
| EU regulations:       | Classification and labeling have been performed according to EU directives 1999/45/EC and 67/548/EEC as amended and adapted. |

| Miscellaneous provisions                            |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inventories:                                        | <b>Australian inventory (AICS):</b> In compliance<br><b>Canadian inventory (DSL):</b> In compliance<br><b>Chinese inventory (IECS):</b> Not determined.<br><b>EC inventory (EINECS/ELINCS):</b> In compliance<br><b>Japanese inventory (ENCS):</b> In compliance<br><b>Korean inventory (ECL):</b> Not listed<br><b>Philippine inventory (PICCS):</b> Not determined.<br><b>US inventory (TSCA):</b> In compliance |
| Water hazard classification (WGK):                  | 1 (low hazard to waters), classification acc. to the German Administrative Regulation on the Classification of Substances Hazardous to Waters into Water Hazard Classes (VwVwS)                                                                                                                                                                                                                                    |
| Statutory Order on Hazardous Incidents (StörfallV): | 12th ordinance of the German Immission Control Act (StörfallV): Not listed                                                                                                                                                                                                                                                                                                                                         |

## Section 16 Other information

| History                 |                           |
|-------------------------|---------------------------|
| Date of issue:          | 29/03/2006                |
| Date of previous issue: | No previous validation    |
| Created by:             | Product Stewardship Group |

| Notes for the reader                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>(#) Indicates information altered since the last version.</p> <p>All reasonably practicable steps have been taken to ensure this data sheet and the health, safety and environmental information contained in it is accurate as at the date of issue specified below. No warranty or representation, express or implied, is made as to the accuracy or completeness of the data and information in this data sheet.</p> <p>The data and advice given are valid if the product is sold for the application(s) specified. The product should not be used for purposes other than the applications specified without prior consultation with us. It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. The BP Group accepts no responsibility for any damage or injury resulting from uses other than the stated product use of the material, from any failure to adhere to recommendations, or from hazards inherent in the nature of the material. Those purchasing the product for supply to third parties for use at work have a duty to take all necessary steps to ensure that any person handling or using the product is provided with the information on this data sheet. Employers have a duty to tell employees and others who may be affected of any hazards described in this sheet and of any precautions that should be taken.</p> |  |

### 10.2.3 Safety data sheet for RV OIL SB150 lubricating oil

The following extract from the safety data sheet according to 1907/2006/EEC, Article 31 must be observed.

## Section 1 Identification of the substance or mixture and of the company

### 1.1 Product identifier

|                 |              |
|-----------------|--------------|
| Trade name:     | RV OIL SB150 |
| Article number: | 36567        |

### 1.2 Relevant identified uses of the substance or mixture and uses advised against

No further relevant information available.

|                              |                 |
|------------------------------|-----------------|
| Use of substance or mixture: | Lubricating oil |
|------------------------------|-----------------|

### 1.3 Details of the supplier of the safety data sheet

| Manufacturer designation |                                                                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manufacturer/supplier:   | Kyodo Yushi Co., Ltd.<br>2-2-30 Tsujido Kandai, Fujisawa-shi<br>Kanagawa 251-8588<br>JAPAN<br>Nabtesco Corporation SERVICE CENTER<br>TSU PLANT 594 Katada-cho, Tsu-shi, Mie-ken 514-8533 |

| Department for information: |                                |
|-----------------------------|--------------------------------|
| Phone:                      | +81-59-237-4672                |
| Fax:                        | +81-59-237-4697                |
| e-mail address:             | PR Service Center@nabtesco.com |

### 1.4 Emergency telephone number

|                   |                 |
|-------------------|-----------------|
| Emergency hotline | +81-59-237-4672 |
|-------------------|-----------------|

## Section 2 Possible hazards

### 2.1 Classification of the substance or mixture

|                                                       |                                                                 |
|-------------------------------------------------------|-----------------------------------------------------------------|
| Classification acc. to Regulation (EC) No. 1272/2008: | This product is not classified according to the CLP Regulation. |
|-------------------------------------------------------|-----------------------------------------------------------------|

### 2.2 Label elements

|                                                       |                                                                                                          |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Classification acc. to Regulation (EC) No. 1272/2008: | Not applicable                                                                                           |
| Hazard symbols:                                       | Not applicable                                                                                           |
| Signal word:                                          | Not applicable                                                                                           |
| Hazard statements:                                    | Not applicable                                                                                           |
| Safety instructions:                                  | P501 Dispose of contents/container in accordance with local/regional/national/international regulations. |

### 2.3 Other hazards

| Results of PBT and vPvB assessment |                |
|------------------------------------|----------------|
| PBT:                               | Not applicable |
| vPvB:                              | Not applicable |

## Section 3 Composition / Information about the components

|                                   |                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Chemical characterization:</b> | Mixture                                                                                             |
| <b>Description:</b>               | Mixture of the substances listed below with non-hazardous admixtures.                               |
| <b>Hazardous components:</b>      | Not applicable                                                                                      |
| <b>Additional information:</b>    | The exact wording of the hazard statements listed above can be found in (>>> "Section 16" Page 94). |

## Section 4 First aid measures

### ■ 4.1 Description of first aid measures

|                                 |                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>General information:</b>     | No special measures required.                                                                                           |
| <b>After inhalation:</b>        | Remove to fresh air; consult a doctor if symptoms persist.                                                              |
| <b>After skin contact:</b>      | Dab away from affected skin using cotton or cellulose material and then thoroughly wash with water and a mild cleanser. |
| <b>After contact with eyes:</b> | Rinse eyes under running water for several minutes with the eyelids open                                                |
| <b>After swallowing:</b>        | Consult a doctor if symptoms persist.                                                                                   |

- 4.2 Most important symptoms and effects, both acute and delayed  
No further relevant information available.
- 4.3 Indication of any immediate medical attention and special treatment needed  
No further relevant information available.

## Section 5 Fire-fighting measures

### ■ 5.1 Extinguishing agents

|                                       |                                                                                                                                                               |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable extinguishing agents:</b> | CO <sub>2</sub> , dry powder or water spray. Tackle larger fires with alcohol-resistant foam. Adjust the fire-extinguishing measures to suit the environment. |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

- 5.2 Special hazards arising from the substance or mixture  
Toxic gases may evolve when heated or in the event of fire.
- 5.3 Advice for fire-fighters

|                                      |                                          |
|--------------------------------------|------------------------------------------|
| <b>Special protective equipment:</b> | Wear self-contained breathing apparatus. |
|--------------------------------------|------------------------------------------|

## Section 6 Measures after unintended release

- 6.1 Personal precautions, protective equipment and emergency procedures  
High risk of slipping due to leakage/spillage of product.
- 6.2 Environmental protection measures  
Do not discharge into the drainage system, surface waters or groundwater.
- 6.3 Methods and material for containment and cleaning up  
Absorb with liquid-binding material (sand, diatomaceous earth, acid binders, universal binders, sawdust).
- 6.4 Reference to other sections  
No hazardous materials are released.  
For information on safe handling, see (>>> "Section 7" Page 90).  
For information on personal protective equipment, see (>>> "Section 8" Page 90).  
For information on disposal, see (>>> "Section 13" Page 93).

**Section 7 Handling and storage**

## ■ 7.1 Precautions for safe handling

|                                                                |                                                                              |
|----------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Safety measures:</b>                                        | Avoid exposure to heat and direct sunlight.<br>No special measures required. |
| <b>Instructions relating to fire and explosion protection:</b> | The product forms flammable fumes when heated.                               |

## ■ 7.2 Conditions for safe storage, including any incompatibilities

| Storage                                                      |                                        |
|--------------------------------------------------------------|----------------------------------------|
| <b>Requirements regarding storage rooms and containers:</b>  | No special requirements.               |
| <b>Information on storage with other products:</b>           | Keep separately from oxidizing agents. |
| <b>Further specifications concerning storage conditions:</b> | Protect from dirt.                     |

| Storage class                                                                                    |   |
|--------------------------------------------------------------------------------------------------|---|
| <b>Classification according to German Ordinance on Industrial Safety and Health (BetrSichV):</b> | - |

## ■ 7.3 Specific end use(s)

No further relevant information available.

**Section 8 Limitation and monitoring of exposure / Personal protective equipment**

Additional information for the design of technical systems: No further information, see (&gt;&gt;&gt; "Section 7" Page 90).

## ■ 8.1 Control parameters

|                                                                       |                                                                                                                        |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Components with applicable occupational exposure limit values:</b> | The product contains no relevant amounts of materials that are subject to monitoring of limit values in the workplace. |
| <b>Additional information:</b>                                        | These values are based on the lists valid at the time of issue.                                                        |

## ■ 8.2 Limitation and monitoring of exposure

| Personal protective equipment                   |                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General protective and hygiene measures:</b> | The usual precautions when handling chemicals must be observed.                                                                                                                                                                                                                                                                                                                        |
| <b>Respiratory protection:</b>                  | Not necessary if room is well-ventilated.                                                                                                                                                                                                                                                                                                                                              |
| <b>Hand protection:</b>                         | Oil-resistant gloves.<br><br>The glove material must be impermeable and resistant to the product / substance / preparation.<br><br>Due to lack of testing, no recommendation can be provided as to the glove material for the product / preparation / chemical mixture.<br><br>The selection of the glove material must consider breakthrough times, permeation rates and degradation. |

| Personal protective equipment                   |                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Glove material:</b>                          | The selection of a suitable glove depends not only on the material but also on further quality characteristics. These will vary from manufacturer to manufacturer. Since the product is a preparation composed of several substances, the resistance of glove materials cannot be determined beforehand and must therefore be tested before use. |
| <b>Breakthrough time of the glove material:</b> | The precise breakthrough time must be obtained from the glove manufacturer and observed.                                                                                                                                                                                                                                                         |
| <b>Eye protection:</b>                          | Safety goggles recommended when pouring the product.                                                                                                                                                                                                                                                                                             |

## Section 9

## Physical and chemical properties

## ■ 9.1 Information on basic physical and chemical properties

## General information

|                                                 |                        |
|-------------------------------------------------|------------------------|
| <b>Appearance:</b>                              |                        |
| <b>Physical state</b>                           | Liquid                 |
| <b>Color</b>                                    | Green                  |
| <b>Odor:</b>                                    | Perceptible            |
| <b>Odor threshold:</b>                          | Not determined         |
| <b>pH value:</b>                                | Not determined         |
| <b>Change in state:</b>                         | Not determined         |
| <b>Melting point / range:</b>                   |                        |
| <b>Change in state:</b>                         | Not determined         |
| <b>Boiling point / range:</b>                   |                        |
| <b>Flash point:</b>                             | 260 °C (COC)           |
| <b>Ignition temperature:</b>                    | Not determined         |
| <b>Decomposition temperature:</b>               | Not determined         |
| <b>Risk of explosion:</b>                       | Not determined         |
| <b>Lower explosion limit:</b>                   | Not determined         |
| <b>Upper explosion limit:</b>                   | Not determined         |
| <b>Vapor pressure:</b>                          | Not determined         |
| <b>Density at 15 °C:</b>                        | 0.87 g/cm <sup>3</sup> |
| <b>Relative density:</b>                        | Not determined         |
| <b>Vapor density:</b>                           | Not determined         |
| <b>Evaporation rate:</b>                        | Not determined         |
| <b>Solubility / miscibility in water:</b>       | Insoluble              |
| <b>Partition coefficient (n-octanol/water):</b> | Not determined         |
| <b>Viscosity:</b>                               |                        |
| <b>Dynamic</b>                                  | Not determined         |
| <b>Kinematic</b>                                | Not determined         |

## ■ 9.2 Other information

No further relevant information available.

## Section 10

## Stability and reactivity

- 10.1 Reactivity  
No further relevant information available.
- 10.2 Chemical stability  
**Thermal decomposition / conditions to be avoided:**  
To avoid thermal decomposition, do not overheat.
- 10.3 Possibility of hazardous reactions  
No hazardous reactions known.
- 10.4 Conditions to be avoided  
No further relevant information available.
- 10.5 Incompatible materials  
No further relevant information available.
- 10.6 Hazardous decomposition products:  
Carbon monoxide and carbon dioxide  
Toxic metal oxide fumes

## Section 11 Toxicological information

- 11.1 Information on toxicological effects

| Acute toxicity                                      |                                                                       |
|-----------------------------------------------------|-----------------------------------------------------------------------|
| <b>Primary irritation effect</b>                    |                                                                       |
| <b>Caustic/irritant for the skin:</b>               | Based on the available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation:</b>           | Based on the available data, the classification criteria are not met. |
| <b>Sensitization of the respiratory tract/skin:</b> | Based on the available data, the classification criteria are not met. |

### Additional toxicological information:

The product is not subject to labeling obligations due to the method of calculation in the latest version of the "EC General Classification Directive for Preparations".

With correct handling and use for the designated purpose, and on the basis of our experience and the information available to us, the product does not produce any effects harmful to health.

| CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction). |                                                                       |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Germ cell mutagenicity:</b>                                             | Based on the available data, the classification criteria are not met. |
| <b>Carcinogenicity:</b>                                                    | Based on the available data, the classification criteria are not met. |
| <b>Reproductive toxicity:</b>                                              | Based on the available data, the classification criteria are not met. |
| <b>Specific target organ toxicity - single exposure:</b>                   | Based on the available data, the classification criteria are not met. |
| <b>Specific target organ toxicity - repeated exposure:</b>                 | Based on the available data, the classification criteria are not met. |
| <b>Aspiration hazard:</b>                                                  | Based on the available data, the classification criteria are not met. |

## Section 12 Ecological information

- 12.1 Toxicity  
Aquatic toxicity: No further relevant information available.

- 12.2 Persistence and degradability  
Not easily biodegradable.
- 12.3 Bioaccumulative potential  
No further relevant information available.
- 12.4 Mobility in soil  
No further relevant information available.

**Additional ecological information:****General information:**

Water hazard classification 1 (manufacturer's classification): low hazard to waters.

Prevent from entering groundwater, surface waters or the drainage system in an undiluted form or in larger quantities.

- 12.5 Results of PBT and vPvB assessment  
PBT: Not applicable  
vPvB: Not applicable
- 12.6 Other adverse effects  
No further relevant information available.

**Section 13****Disposal information**

- 13.1 Waste treatment methods

|                                                 |                                                                                                   |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Recommendation:</b>                          | Hand over to a hazardous waste collector or bring it to a problematic substance collection point. |
| <b>Contaminated packaging / recommendation:</b> | Disposal in accordance with the applicable regulations.                                           |

**Section 14****Transport information**

- 14.1 UN number  
ADR, ADN, IMDG, IATA: Not applicable
- 14.2 UN proper shipping name  
ADR, ADN, IMDG, IATA: Not applicable
- 14.3 Transport hazard class(es)  
ADR, ADN, IMDG, IATA class: Not applicable
- 14.4 Packing group  
ADR, IMDG, IATA: Not applicable
- 14.5 Environmental hazards  
Marine pollutant: No
- 14.6 Special precautions for user  
Not applicable
- 14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code  
Not applicable
- Further information:  
UN "Model Regulation": -

**Section 15****Regulatory information**

- 15.1 Safety, health and environmental regulations / legislation specific to the substance or mixture
- **Directive 2012/18/EU**  
**Named dangerous substances - ANNEX I:** None of the ingredients is listed.

- **National regulations**

**Water hazard classification:** Water hazard classification 1 (manufacturer's classification): low hazard to waters.

- **15.2 Chemical safety assessment**

Chemical safety assessment has not been performed.

**Section 16****Other information**

The information given is based on our present knowledge. However, this shall not constitute a guarantee for any specific product characteristics and shall not establish a legally valid contractual relationship.

**Department issuing data sheet:** Product safety department.

**Abbreviations and acronyms:**

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonized System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

**10.2.4 Safety data sheet for LGEP 2 extreme pressure grease**

The following extract from the safety data sheet according to 91/155/EEC must be observed.

**Section 1****Identification of the substance/mixture and of the company**

| Product identifier |        |
|--------------------|--------|
| Trade name:        | LGEP 2 |

| Relevant identified uses of the substance or mixture and uses advised against |                    |
|-------------------------------------------------------------------------------|--------------------|
| Recommended use:                                                              | Lubricating grease |

| Details of the supplier of the safety data sheet |                                                                                    |
|--------------------------------------------------|------------------------------------------------------------------------------------|
| Supplier:                                        | SKF Maintenance Products<br>P.O. Box 1008<br>NL-3430, BA Nieuwegein<br>Netherlands |
| Telephone:                                       | +31 30 6307200                                                                     |
| e-mail:                                          | sebastien.david@skf.com                                                            |
| WWW                                              | www.skf.com                                                                        |

| Emergency hotline |                                                |
|-------------------|------------------------------------------------|
|                   | +49 (0)30 19240 (Berlin poison control center) |

## Section 2 Possible hazards

| Classification of the substance or mixture        |                                                                                                           |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| DPD classification (Directive 1999/45/EC):        | R52/53                                                                                                    |
| CLP classification (Regulation EC No. 1272/2008): | Aquatic Chronic 3; H412<br>For full text of R phrases and H phrases, see Section 16.                      |
| Most serious harmful effects:                     | Harmful to aquatic organisms with long-lasting effects. May cause slight irritation to the skin and eyes. |

| Label elements |                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| H phrases:     | Harmful to aquatic organisms with long-lasting effects. (H412)                                                                              |
| P phrases:     | Avoid release to the environment. (P273) Dispose of content/ container at a waste collection point according to local regulations. (P501-A) |

| Other hazards |                                                         |
|---------------|---------------------------------------------------------|
|               | Assessment to determine PBT and vPvB has not been made. |

## Section 3 Composition / Information about the components

| Registration number: | CAS/EC no. | Substance                                                           | DSD classification (Directive 67/548/(EEC)/ CLP classification (Regulation EC) No. 1272/2008) | w/w %                   | Note |
|----------------------|------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------|------|
|                      | 85940-28-9 | Phosphorodithionic acids, mixed                                     | Xi; R36/38 N; R51/53                                                                          | 2.5 - 5                 |      |
|                      | 288-917-4  | O,O-bis(2-ethylhexyl and isobutyl and isopropyl) esters, zinc salts | Skin Irrit. 2; H315 Eye Irrit. 2; H319 Aquatic Chronic 2; H411                                | Xi; N; R38; R41; R51/53 |      |

For full text of R phrases and H phrases, see Section 16.

**Other information:** The mineral oils in this product contain < 3% DMSO extract (IP 346).

## Section 4 First aid measures

| Description of first aid measures |                                                                                                                                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation:                       | Ensure plenty of fresh air. Seek medical advice in case of persistent discomfort.                                                                                                                                                     |
| Ingestion:                        | Rinse out mouth thoroughly and drink 1-2 glasses of water in small sips. Seek medical advice in case of discomfort.                                                                                                                   |
| Skin:                             | Remove contaminated clothing. Wash skin with soap and water. Seek medical advice in case of persistent discomfort. Remove contaminated clothing. Wash skin with soap and water. Seek medical advice in case of persistent discomfort. |

| Description of first aid measures |                                                                                                                            |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Eyes:                             | Flush with water (preferably using eye wash equipment) until irritation subsides. Seek medical advice if symptoms persist. |
| More information:                 | When obtaining medical advice, show the safety data sheet or label.                                                        |

| Most important symptoms and effects, both acute and delayed: |                                                                                                                                          |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|                                                              | May cause slight irritation to the skin and eyes. Degreases the skin. Long-term exposure may cause irritation and possible inflammation. |

| Indication of any immediate medical attention and special treatment needed |                                                          |
|----------------------------------------------------------------------------|----------------------------------------------------------|
|                                                                            | Treat symptoms. No special immediate treatment required. |

## Section 5 Fire-fighting measures

| Extinguishing agents             |                                                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| Suitable extinguishing agents:   | Extinguish with powder, foam, carbon dioxide or water mist. Use water or water mist to cool non-ignited stock. |
| Unsuitable extinguishing agents: | Do not use a water stream, as it may spread the fire.                                                          |

| Special hazards arising from the substance or mixture |                                                                                                              |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|                                                       | Not flammable, but combustible. In the event of a fire, liable to generate fumes containing carbon monoxide. |

| Advice for firefighters |                                                                                                                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | Remove containers from danger zone if this can be done without risk. Do not inhale vapors and fumes. Ensure plenty of fresh air. Wear self-contained breathing apparatus (SCBA) with chemical resistant gloves. |

## Section 6 Measures after unintended release

| Personal precautions, protective equipment and emergency procedures |                                                                                           |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| For non-emergency personnel:                                        | Wear safety goggles if there is a risk of splashing. Wear gloves.                         |
| For emergency responders:                                           | In addition to the above: Normal protective clothing equivalent to EN 469 is recommended. |

| Environmental protection measures |                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------------------|
|                                   | No large quantities of spilled material or residue may enter the drainage system. |

| Methods and material for containment and cleaning up |                                                                                                                                              |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | Contain and absorb spill with sand or other absorbent material and transfer to suitable waste containers. Wipe up minor spills with a cloth. |

| Reference to other sections |                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------|
|                             | The type of protective equipment can be found in Section 8. Information on disposal: see Section 13. |

## Section 7 Handling and storage

| Precautions for safe handling |                                                                                                                                                                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | The product should be used under well-ventilated conditions and preferably under process ventilation. Running water and eye wash equipment should be available. Wash hands before breaks, before using restroom facilities, and at the end of work. |

| Conditions for safe storage, including any incompatibilities |                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                              | The product should be stored safely, out of reach of children and away from food, animal feeding stuffs, medicines, etc. Keep in tightly closed original packaging. Do not store with the following: Oxidizing agents. Store below 45 °C. Storage class in accordance with TRGS 510: 10 (flammable liquids, insofar as not storage class 3) |

| Specific end uses |     |
|-------------------|-----|
|                   | No. |

## Section 8 Limitation and monitoring of exposure / Personal protective equipment

| Control parameters |                                                                                                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Legal basis:       | Technical regulations for hazardous substances, occupational exposure limits TRGS 900, issued January 2006.<br>Last amended: GMBI 2013.<br>Contains no substances subject to reporting requirements. |

| Limitation and monitoring of exposure                  |                                                                                                                                                                                  |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appropriate engineering controls:                      | Wear the personal protective equipment specified below.                                                                                                                          |
| Personal protective equipment, eye/face protection:    | Wear safety goggles if there is a risk of splashing. Eye protection must conform to EN 166.                                                                                      |
| Personal protective equipment, skin protection:        | In the event of direct skin contact, wear protective gloves. Material type and thickness: nitrile rubber > 0.38 mm. Breakthrough time: > 8 hours. Gloves must conform to EN 374. |
| Personal protective equipment, respiratory protection: | Not required.                                                                                                                                                                    |
| Environmental exposure controls:                       | Ensure compliance with local regulations for emissions.                                                                                                                          |

## Section 9 Physical and chemical properties

| Information on basic physical and chemical properties |                |
|-------------------------------------------------------|----------------|
| State:                                                | Paste/grease   |
| Color:                                                | Tawny          |
| Odor:                                                 | Characteristic |
| Odor threshold:                                       | No data        |
| pH (solution for use):                                | No data        |
| pH (concentrate):                                     | No data        |
| Melting point/freezing point:                         | No data        |
| Initial boiling point and boiling range:              | No data        |
| Flash point:                                          | >150 °C        |
| Evaporation rate:                                     | No data        |
| Flammability (solid, gas):                            | No data        |

| Information on basic physical and chemical properties |                         |
|-------------------------------------------------------|-------------------------|
| Upper/lower flammability limits:                      | No data                 |
| Upper/lower explosive limits:                         | No data                 |
| Vapor pressure:                                       | No data                 |
| Vapor density:                                        | No data                 |
| Relative density:                                     | < 1.0 g/cm <sup>3</sup> |
| Solubility:                                           | No data                 |
| Partition coefficient (n-octanol/water):              | No data                 |
| Auto-ignition temperature:                            | No data                 |
| Decomposition temperature:                            | No data                 |
| Viscosity:                                            | No data                 |
| Explosive properties:                                 | No data                 |
| Oxidizing properties:                                 | No data                 |

| Other information |    |
|-------------------|----|
|                   | No |

## Section 10 Stability and reactivity

| Reactivity |                                                     |
|------------|-----------------------------------------------------|
|            | Reacts with the following: Strong oxidizing agents. |

| Chemical stability |                                                                                     |
|--------------------|-------------------------------------------------------------------------------------|
|                    | The product is stable when used in accordance with the manufacturer's instructions. |

| Possibility of hazardous reactions: |            |
|-------------------------------------|------------|
|                                     | Not known. |

| Conditions to be avoided |            |
|--------------------------|------------|
|                          | Not known. |

| Incompatible materials |                          |
|------------------------|--------------------------|
|                        | Strong oxidizing agents. |

| Hazardous decomposition products |                                                                                                                           |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|                                  | Product decomposes in fire conditions or when heated to high temperatures, and flammable and toxic gases may be released. |

## Section 11 Toxicological information

| Information on toxicological effects |                                                                                                                                                                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute toxicity - oral:               | The product does not have to be classified. Based on existing data, the classification criteria appear not to be met. Ingestion of large quantities may cause discomfort.<br>LGEP 2: LD50 = 5122 mg/kg<br>85940-28-9: LD50 (rat) = 3000 mg/kg |
| Acute toxicity - dermal:             | The product does not have to be classified. Based on existing data, the classification criteria appear not to be met.<br>LGEP 2: LD50 = 3289 mg/kg                                                                                            |

| Information on toxicological effects             |                                                                                                                                                                 |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute toxicity - inhalation:                     | The product does not have to be classified. Test data are not available.                                                                                        |
| Skin corrosion/irritation                        | The product does not have to be classified. Test data are not available. Degreases the skin. Long-term exposure may cause irritation and possible inflammation. |
| Serious eye damage/eye irritation:               | The product does not have to be classified. According to the assessment findings, the product does not have to be classified.                                   |
| Sensitization of the respiratory tracts or skin: | The product does not have to be classified. Test data are not available.                                                                                        |
| Germ cell mutagenicity:                          | The product does not have to be classified. Test data are not available.                                                                                        |
| Carcinogenic properties:                         | The product does not have to be classified. Test data are not available.                                                                                        |
| Reproductive toxicity:                           | The product does not have to be classified. Test data are not available.                                                                                        |
| Single STOT exposure:                            | The product does not have to be classified. Test data are not available.                                                                                        |
| Repeated STOT exposure:                          | The product does not have to be classified. Test data are not available.                                                                                        |
| Aspiration hazard:                               | The product does not have to be classified. Test data are not available.                                                                                        |
| Other toxicological effects:                     | Not known.                                                                                                                                                      |

## Section 12 Ecological information

| Toxicity                           |                                                                                                                                                                 |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.<br>85940-28-9:<br>Fish: 96h LC50 = 2.2 mg/L (Cyprinodon variegatus) |
| Persistence and degradability      |                                                                                                                                                                 |
|                                    | Test data are not available.                                                                                                                                    |
| Bioaccumulative potential          |                                                                                                                                                                 |
|                                    | No bioaccumulation expected.                                                                                                                                    |
| Mobility in soil                   |                                                                                                                                                                 |
|                                    | Not anticipated to be mobile in soil.                                                                                                                           |
| Results of PBT and vPvB assessment |                                                                                                                                                                 |
|                                    | No assessment has been made.                                                                                                                                    |
| Other adverse effects              |                                                                                                                                                                 |
|                                    | Petroleum products can lead to soil or water contamination.                                                                                                     |

## Section 13 Disposal information

| Waste treatment methods |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>Avoid discharge to drain or surface water. Collect spills and waste in closed, leak-proof containers for disposal at the local hazardous waste site.</p> <p>EWC code: Depends on line of business and use, 13 08 99* oil wastes not otherwise specified</p> <p>Absorbent/cloth contaminated with the product:<br/>EWC code: 15 02 02 absorbents, filter materials (including oil filters not otherwise specified), wiping cloths, protective clothing contaminated by dangerous substances.</p> <p>Contaminated packaging is to be disposed of via the local waste-removal scheme.</p> <p>Empty, cleaned packaging should be disposed of for recycling.</p> |

**Section 14****Transport information**

The product is not subject to the regulations for the transportation of hazardous goods.

| UN number:                                                              |   |
|-------------------------------------------------------------------------|---|
|                                                                         | - |
| UN proper shipping name:                                                |   |
|                                                                         | - |
| Transport hazard classes                                                |   |
|                                                                         | - |
| Dangers to the environment                                              |   |
|                                                                         | - |
| Special precautions for user                                            |   |
|                                                                         | - |
| Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code |   |
|                                                                         | - |

**Section 15****Regulatory information**

| Safety, health and environmental regulations/legislation specific to the substance or mixture |                                                             |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Special provisions:                                                                           | No.<br>Statutory Order on Hazardous Incidents: Not covered. |
| Water hazard classification 1:                                                                | Low hazard to waters.                                       |
| Chemical Safety Assessment                                                                    |                                                             |
|                                                                                               | Chemical safety assessment has not been performed.          |

**Section 16****Other information**

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Changes have been made in the following sections: | 1, 2, 3, 9, 10, 11, 12 & 16                                                                                                                                                                                                                                                                                                                                                                                         |
| Abbreviation explanations:                        | PBT: Persistent, Bioaccumulative and Toxic vPvB: Very Persistent and Very Bioaccumulative<br>STOT: Specific Target Organ Toxicity                                                                                                                                                                                                                                                                                   |
| Method of classification:                         | Calculation based on the risks for the known components.                                                                                                                                                                                                                                                                                                                                                            |
| R phrases:                                        | R36/38 Irritating to eyes and skin.<br><br>R51/53 Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.<br><br>R52/53 Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.                                                                                                                                                           |
| H phrases:                                        | H315 Causes skin irritation.<br><br>H319 Causes severe eye irritation.<br><br>H411 Toxic to aquatic organisms with long-lasting effects.<br><br>H412 Harmful to aquatic organisms with long-lasting effects.                                                                                                                                                                                                        |
| Training:                                         | Thorough knowledge of this safety data sheet is a prerequisite.                                                                                                                                                                                                                                                                                                                                                     |
| More information:                                 | This safety data sheet has been prepared exclusively for this product and only applies to this product. It is based on our current knowledge and information that the supplier was able to make available at the time of its preparation. The safety data sheet complies with the pertinent regulations for the preparation of safety data sheets as per Regulation 907/2006/EC (REACH) with subsequent amendments. |

### 10.2.5 Safety data sheet for Optitemp RB 2 lubricating grease

The following extract from the safety data sheet according to EC Regulation 1907/2006 (REACH), Annex II - "Germany" must be observed.

#### Section 1 Identification of the substance or mixture and of the company

##### ■ 1.1 Product identifier

|               |               |
|---------------|---------------|
| Product name: | Optitemp RB 2 |
| Product code: | 468633-DE03   |
| SDS no.:      | 468633        |
| Product type: | Grease        |

##### ■ 1.2 Relevant identified uses of the substance or mixture and uses advised against

|                              |                                                                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use of substance or mixture: | Lubricating grease for industrial use.<br><br>For specific instructions for use, see the corresponding technical data sheet or contact a company representative. |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

##### ■ 1.3 Details of the supplier of the safety data sheet

| Manufacturer designation |                                                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Supplier:                | BP Europa SE - Geschäftsbereich Industrieschmierstoffe<br>Erkelenzer Strasse 20<br>D-41179 Mönchengladbach<br>Germany |
| Phone:                   | +49 (0)2161 909 30                                                                                                    |

| Manufacturer designation |                     |
|--------------------------|---------------------|
| <b>Fax:</b>              | +49 (0)2161 909 400 |
| <b>e-mail address:</b>   | MSDSadvice@bp.com   |

■ 1.4 Emergency telephone number

|                          |                                                          |
|--------------------------|----------------------------------------------------------|
| <b>Emergency hotline</b> | Carechem: +44 (0)1235 239,670 (available 24 hours a day) |
|--------------------------|----------------------------------------------------------|

## Section 2 Possible hazards

■ 2.1 Classification of the substance or mixture

|                                                                                       |                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product definition:</b>                                                            | Mixture                                                                                                                                                                                                                                                          |
| <b>Classification acc. to Regulation (EC) No. 1272/2008 [CLP/GHS]:</b>                | Not classified.                                                                                                                                                                                                                                                  |
| <b>Classification acc. to Directive 1999/45/EC [Dangerous Preparation Directive]:</b> | The product is not classified as hazardous according to Directive 1999/45/EC and its annexes.<br><br>Sections (>>> "Section 11" Page 110) and (>>> "Section 12" Page 110) contain more detailed information on health hazards, symptoms and environmental risks. |

■ 2.2 Label elements

|                           |                                       |
|---------------------------|---------------------------------------|
| <b>Signal word:</b>       | No signal word                        |
| <b>Hazard statements:</b> | No particular effects or risks known. |

| Safety instructions                 |                |
|-------------------------------------|----------------|
| <b>Prevention:</b>                  | Not applicable |
| <b>Reaction:</b>                    | Not applicable |
| <b>Storage:</b>                     | Not applicable |
| <b>Disposal:</b>                    | Not applicable |
| <b>Supplemental label elements:</b> | Not applicable |

| Special packaging requirements                                  |                |
|-----------------------------------------------------------------|----------------|
| <b>Containers to be fitted with child-resistant fastenings:</b> | Not applicable |
| <b>Tactile warning of danger:</b>                               | Not applicable |

■ 2.3 Other hazards

|                                                             |                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Other hazards which do not result in classification:</b> | Defatting to the skin<br><br>Note regarding high-pressure applications<br><br>Injection into the skin due to contact with a product under high pressure constitutes a major medical emergency. See Notes to physician under "First aid measures" in this safety data sheet. |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Section 3 Composition / Information about the components

|                           |                                                               |
|---------------------------|---------------------------------------------------------------|
| <b>Substance/mixture:</b> | Mixture<br><br>Synthetic lubricant and additives. Thickeners. |
|---------------------------|---------------------------------------------------------------|

This product contains no dangerous components above the legally defined limit values.

## Section 4 First aid measures

■ 4.1 Description of first aid measures

|                                    |                                                                                                                                                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contact with eyes:</b>          | In case of contact, rinse eyes immediately with plenty of water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. Check for and remove any contact lenses. Get medical attention.                     |
| <b>Skin contact:</b>               | Wash skin with soap and water, or use an approved skin cleansing agent. Take off clothing and shoes if they become contaminated with product. Wash clothing before reuse. Clean shoes thoroughly before reuse. If irritation occurs consult a doctor. |
| <b>Inhalation:</b>                 | Take affected person into fresh air. Consult a doctor if symptoms persist.                                                                                                                                                                            |
| <b>Ingestion:</b>                  | Do not induce vomiting unless explicitly directed to do so by medical personnel. Seek medical attention if symptoms occur.                                                                                                                            |
| <b>Protection of first-aiders:</b> | No measures should be taken that involve a risk to personnel or have not been adequately trained.                                                                                                                                                     |

- 4.2 Most important symptoms and effects, both acute and delayed  
See Section (>>> "Section 11" Page 110) for more detailed information on health effects and symptoms.
- 4.3 Indication of any immediate medical attention and special treatment needed

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Notes to physician:</b> | <p>Treatment should in general be symptomatic and directed at relieving any effects.</p> <p>Note regarding high-pressure applications</p> <p>Injection into the skin due to contact with a product under high pressure constitutes a major medical emergency. The injuries appear minor at first, but within a few hours the tissue swells up and becomes discolored and extremely painful, with severe subcutaneous necrosis.</p> <p>Surgical treatment is absolutely imperative. Comprehensive opening of the wound and the tissue beneath it is necessary in order to reduce tissue loss and to prevent or limit lasting damage. The high pressure can cause the product to penetrate extensive areas of tissue layers.</p> |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Section 5 Fire-fighting measures

- 5.1 Extinguishing agents

|                                         |                                                                                                                                    |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable extinguishing agents:</b>   | In the event of a fire, use water spray (mist), alcohol resistant foam, dry chemical or carbon dioxide-based extinguisher or spray |
| <b>Unsuitable extinguishing agents:</b> | Do not use a waterjet.                                                                                                             |

- 5.2 Special hazards arising from the substance or mixture

|                                               |                                                                                                                                          |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazards from the substance or mixture:</b> | In a fire or if heated, a pressure increase will occur and the container may burst.                                                      |
| <b>Hazardous combustion products:</b>         | <p>The combustion products may include the following compounds:</p> <p>Carbon oxides (CO, CO<sub>2</sub>)</p> <p>Metal oxides/oxides</p> |

- 5.3 Advice for fire-fighters

|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special precautions for fire-fighters:</b>          | Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No measures should be taken that involve a risk to personnel or have not been adequately trained. This substance is toxic to aquatic organisms. Extinguishing water contaminated with this product must be contained and prevented from entering surface waters or the sewage or drainage system. |
| <b>Special protective equipment for fire-fighting:</b> | Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Clothing for fire-fighters (including helmets, protective boots and gloves) conforming to European standard EN 469 will provide a basic level of protection for chemical incidents.                                                    |

**Section 6****Measures after unintended release**

- 6.1 Personal precautions, protective equipment and emergency procedures

|                                     |                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>For non-emergency personnel:</b> | No measures should be taken that involve a risk to personnel or have not been adequately trained. Evacuate the environment. Refuse access to personnel who are not required or are unprotected. Do not touch or step on any spilled substance. Floors may be slippery. Use care to avoid falling. Put on appropriate personal protective equipment. |
| <b>For emergency responders:</b>    | If specialized clothing is required to deal with the spillage, take note of any information in (>>> "Section 8" Page 105) on suitable and unsuitable materials. See also the information for non-emergency personnel under "Personal safety precautions".                                                                                           |

- 6.2 Environmental protection measures

Prevent released material from dispersing or flowing away and from coming into contact with soil, surface waters and drainage system. Notify the relevant authorities if the product has caused pollution (sewers, surface waters, ground or air).

- 6.3 Methods and material for containment and cleaning up

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Small spills:</b> | Stop the leak if you can do so without risk. Remove container from spill area. Absorb spill with inert material and place it in a suitable container for disposal. Disposal should be entrusted to a recognized waste disposal company.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Large spills:</b> | Stop the leak if you can do so without risk. Remove container from spill area. Prevent entry into drainage system, surface waters, basements or confined areas. Contain spilled material using a non-combustible absorbent (e.g. sand, soil, vermiculite, diatomaceous earth) and place it in a container provided for the purpose of disposal in accordance with the local regulations. If no emergency personnel are available, contain spilled material. Remove spilled material with a vacuum cleaner or a shovel and deposit it in suitable disposal or recycling containers. Cover the surface contaminated by the spilled product with an oil absorption agent. Disposal should be entrusted to a recognized waste disposal company. |

#### 6.4 Reference to other sections

See Section (>>> "Section 1" Page 101) for emergency contact information.

See Section (>>> "Section 5" Page 103) for fire-fighting measures.

See Section (>>> "Section 8" Page 105) for information on appropriate personal protective equipment.

See Section (>>> "Section 12" Page 110) for environmental precautions.

See Section (>>> "Section 13" Page 111) for additional waste treatment information.

## Section 7 Handling and storage

### 7.1 Precautions for safe handling

|                                                |                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Safety measures:</b>                        | Put on appropriate personal protective equipment.                                                                                                                                                                                                                                                                                  |
| <b>Advice on general occupational hygiene:</b> | Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Wash thoroughly after handling. Remove contaminated clothing and protective equipment before entering eating areas.<br><br>See Section (>>> "Section 8" Page 105) for additional information on hygiene measures. |

### 7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in a dry, cool and well-ventilated area, away from incompatible materials (see Section (>>> "Section 10" Page 109)). Keep away from heat and direct sunlight. Keep containers tightly closed and sealed until usage. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Store and use only in equipment/containers designed for use with this product. Do not store in unlabeled containers.

Germany - storage class: 11

### 7.3 Specific end use(s)

Recommendations: See Section 1.2 and Exposure scenarios in annex, if applicable.

## Section 8 Limitation and monitoring of exposure / Personal protective equipment

### 8.1 Control parameters

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Occupational exposure limits:</b>      | No exposure limit value known.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Recommended monitoring procedures:</b> | If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity of using respiratory protective equipment. Reference should be made to monitoring standards, such as the following: European Standard EN 689 (Workplace atmospheres - Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy), European Standard EN 14042 (Workplace atmospheres - Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents), European Standard EN 482 (Workplace atmospheres - General requirements for the performance of procedures for the measurement of chemical agents). Reference to national guidance documents for methods for the determination of hazardous substances will also be required. |

|                                           |                           |
|-------------------------------------------|---------------------------|
| <b>Derived No Effect Level:</b>           | No DNELs/DMELs available. |
| <b>Predicted No Effect Concentration:</b> | No PNECs available.       |

■ 8.2 Limitation and monitoring of exposure

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Appropriate engineering controls:</b> | <p>Provide exhaust ventilation or other engineering controls to keep the relevant airborne concentrations below their respective occupational exposure limits.</p> <p>All activities involving chemicals should be assessed for their risks to health, to ensure exposures are adequately controlled. Personal protective equipment should only be considered after other forms of control measures (e.g. engineering controls) have been suitably evaluated. Personal protective equipment should conform to the applicable standards and be suitable for the intended purpose, be kept in good condition and be maintained in accordance with the regulations. Select personal protective equipment in compliance with the valid standards. For this purpose, please consult your supplier of personal protective equipment. Further information about standards is available from your national organization responsible.</p> <p>The final choice of protective equipment will depend on a risk assessment. It must always be ensured that all items of personal protective equipment are compatible with one another.</p> |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Personal safety precautions    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hygiene measures:</b>       | Wash hands, forearms and face thoroughly after handling chemical products and before eating, smoking or using the toilet, as well as at the end of the working day. Ensure that eye-wash stations and safety showers are close to the workstation location.                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Respiratory protection:</b> | <p>Respiratory protection is not normally required where there is adequate natural or local exhaust ventilation to control exposure.</p> <p>Where there is insufficient ventilation, wear suitable respiratory equipment.</p> <p>The correct choice of respiratory protection depends upon the chemicals being handled, the conditions of work and use, and the condition of the respiratory equipment. Safety procedures should be developed for each intended application. Respiratory protection equipment should always be chosen in consultation with the manufacturer and in accordance with the local working conditions.</p> |
| <b>Eye/face protection:</b>    | Protective goggles with side shields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Skin protection                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hand protection/General information:</b> | <p>Because specific work environments and material handling practices vary, safety procedures should be developed for each intended application. The correct choice of protective gloves depends upon the chemicals being handled, and the conditions of work and use. Most gloves provide protection for only a limited time before they must be discarded and replaced (even the best chemically resistant gloves will break down after repeated chemical exposures).</p> <p>Protective gloves should therefore be selected in consultation with the supplier/manufacturer, giving full consideration to the specific working conditions.</p> <p>Recommendation: nitrile gloves.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Breakthrough time:</b>                   | <p>Breakthrough time data are generated by glove manufacturers under laboratory test conditions and represent how long a glove can be expected to provide effective permeation resistance. It is important when following breakthrough time recommendations that actual workplace conditions are taken into account. Always consult with your glove supplier for up-to-date technical information on breakthrough times for the recommended glove type.</p> <p>Our recommendations on the selection of gloves are as follows:</p> <p>Continuous contact:</p> <p>Gloves with a minimum breakthrough time of 240 minutes, or &gt;480 minutes if suitable gloves can be obtained.</p> <p>If suitable gloves are not available to offer that level of protection, gloves with shorter breakthrough times may be acceptable as long as appropriate glove maintenance and replacement regimes are determined and adhered to.</p> <p>Short-term / splash protection:</p> <p>Recommended breakthrough times as above.</p> <p>It is recognized that for short-term, transient exposures, gloves with shorter breakthrough times may commonly be used. Therefore, appropriate maintenance and replacement regimes must be determined and rigorously followed.</p> |

| Skin protection                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Glove thickness:</b>                 | <p>For general applications, we recommend gloves with a thickness typically greater than 0.35 mm.</p> <p>It should be emphasized that glove thickness is not necessarily a good predictor of glove resistance to a specific chemical, as the permeation efficiency of the glove will be dependent on the exact composition of the glove material. Therefore, glove selection should also be based on consideration of the task requirements and knowledge of breakthrough times.</p> <p>Glove thickness may also vary depending on the glove manufacturer, the glove type and the glove model. Therefore, the manufacturer's technical data should always be taken into account to ensure selection of the most appropriate glove for the task.</p> <p>Note: Depending on the activity being conducted, gloves of varying thickness may be required for specific tasks. For example:</p> <ul style="list-style-type: none"> <li>■ Thinner gloves (down to 0.1 mm or less) may be required where a high degree of manual dexterity is needed. However, these gloves are only likely to give short duration protection and would normally be just for single use applications, then disposed of.</li> <li>■ Thicker gloves (up to 3 mm or more) may be required where there is a mechanical (as well as a chemical) risk, i.e. where there is abrasion or puncture potential.</li> </ul> |
| <b>Skin and body:</b>                   | <p>Use of protective clothing is good industrial practice.</p> <p>Personal protective equipment for the body should be selected on the basis of the task being performed and the risks involved and should be approved by a specialist before handling this product.</p> <p>Cotton or polyester/cotton overalls only provide protection against light, superficial contamination which will not soak through to the skin. Overalls should be laundered on a regular basis. When the risk of skin exposure is high (e.g. when cleaning up spillages or if there is a risk of splashing), then chemical-resistant aprons and/or impervious chemical suits and boots will be required.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Refers to the standard:</b>          | <p>Respiratory protection: EN529</p> <p>Gloves: EN420, EN374</p> <p>Eye protection: EN166</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Environmental exposure controls:</b> | <p>Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Section 9

## Physical and chemical properties

- 9.1 Information on basic physical and chemical properties

|                                                 |                                                                |
|-------------------------------------------------|----------------------------------------------------------------|
| Appearance:                                     | Grease                                                         |
| Physical state                                  |                                                                |
| Appearance:                                     | Beige [light]                                                  |
| Color                                           |                                                                |
| Odor:                                           | Not available                                                  |
| Odor threshold:                                 | Not available                                                  |
| pH value:                                       | Not available                                                  |
| Melting point/freezing point:                   | Not available                                                  |
| Boiling point and boiling range:                | Not available                                                  |
| Flash point:                                    | Closed cup: 220 °C (> 428 °F) [Estimated. Based on base oils.] |
| Evaporation rate:                               | Not available                                                  |
| Flammability (solid, gas):                      | Not available                                                  |
| Upper / lower flammability or explosion limits: | Not available                                                  |
| Vapor pressure:                                 | Not available                                                  |
| Vapor density:                                  | Not available                                                  |
| Relative density:                               | Not available                                                  |
| Density:                                        | < 1000 kg/m <sup>3</sup> (<1 g/cm <sup>3</sup> ) at 20 °C      |
| Solubility:                                     | Insoluble in water                                             |
| Partition coefficient (n-octanol/water):        | Not available                                                  |
| Auto-ignition temperature:                      | Not available                                                  |
| Decomposition temperature:                      | Not available                                                  |
| Viscosity:                                      | Not available                                                  |
| Explosive properties:                           | Not available                                                  |
| Oxidizing properties:                           | Not available                                                  |

- 9.2 Other information  
No additional information.

## Section 10

### Stability and reactivity

- 10.1 Reactivity  
No specific test data available for this product. Refer to "Conditions to be avoided" and "Incompatible materials" for additional information.
- 10.2 Chemical stability  
The product is stable.
- 10.3 Possibility of hazardous reactions  
No hazardous reactions occur under normal storage conditions and in normal use.  
No hazardous polymerization occurs under normal storage conditions and in normal use.
- 10.4 Conditions to be avoided  
No specific data.
- 10.5 Incompatible materials  
Reactive or incompatible with the following substances: oxidizing materials.

- 10.6 Hazardous decomposition products

No hazardous decomposition products should be formed under normal conditions of storage and use.

## Section 11 Toxicological information

- 11.1 Information on toxicological effects

| Acute toxicity estimates |              |
|--------------------------|--------------|
| Mode of operation:       | Oral         |
| ATE value:               | 98,666 mg/kg |

|                                               |                                                  |
|-----------------------------------------------|--------------------------------------------------|
| Information on the likely routes of exposure: | Routes of entry anticipated: dermal, inhalation. |
|-----------------------------------------------|--------------------------------------------------|

| Potential acute health effects |                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------|
| Inhalation:                    | Vapor inhalation under ambient conditions is not normally a problem due to low vapor pressure. |
| Ingestion:                     | No particular effects or risks known.                                                          |
| Skin contact:                  | Defatting to the skin. May cause skin dryness and irritation.                                  |
| Contact with eyes:             | No particular effects or risks known.                                                          |

| Symptoms related to the physical, chemical and toxicological characteristics |                                                                                  |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Inhalation:                                                                  | No specific data.                                                                |
| Ingestion:                                                                   | No specific data.                                                                |
| Skin contact:                                                                | Adverse symptoms may include the following:<br>Irritation<br>Dryness<br>Cracking |
| Contact with eyes:                                                           | No specific data.                                                                |

| Delayed and immediate effects and also chronic effects from short-term and long-term exposure |                                                                                              |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Inhalation:                                                                                   | Inhalation of oil mist or vapors at elevated temperatures may cause respiratory irritation.  |
| Ingestion:                                                                                    | Ingestion of large quantities may cause nausea and diarrhea.                                 |
| Skin contact:                                                                                 | Prolonged or repeated contact can dry out the skin and lead to irritation and/or dermatitis. |
| Contact with eyes:                                                                            | Potential risk of transient stinging or redness if accidental eye contact occurs.            |

| Potential chronic health effects |                                       |
|----------------------------------|---------------------------------------|
| General:                         | No particular effects or risks known. |
| Carcinogenicity:                 | No particular effects or risks known. |
| Mutagenicity:                    | No particular effects or risks known. |
| Developmental effects:           | No particular effects or risks known. |
| Fertility effects:               | No particular effects or risks known. |

## Section 12 Ecological information

- 12.1 Toxicity

Dangers to the environment: Not classified as dangerous.

- 12.2 Persistence and degradability  
Expected to be biodegradable.
- 12.3 Bioaccumulative potential  
Not available
- 12.4 Mobility in soil  
Soil/water partition coefficient ( $K_{OC}$ ): Not available  
Mobility: Non-volatile. Grease. Insoluble in water.
- 12.5 Results of PBT and vPvB assessment  
PBT: Not applicable  
vPvB: Not applicable
- 12.6 Other adverse effects  
No particular effects or risks known.

**Section 13****Disposal information**

- 13.1 Waste treatment methods

| Product                 |                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| Waste disposal methods: | Recycle the products if at all possible. Disposal must be carried out by an authorized waste disposal contractor. |
| Hazardous waste:        | Yes                                                                                                               |

| European Waste Catalog (EWC) |                         |
|------------------------------|-------------------------|
| Waste code:                  | 12 01 12*               |
| Waste designation:           | Used waxes and greases. |

Use of the product for purposes other than those specified and/or impurities can necessitate the use of a different waste code number for the waste producer.

| Packaging                       |                                                                                                                                                                                                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste disposal methods:         | Recycle the products if at all possible. Disposal must be carried out by an authorized waste disposal contractor.                                                                                                                                           |
| Waste code:                     | 15 01 10*                                                                                                                                                                                                                                                   |
| European Waste Catalog (EWC)    | Packaging containing the residue of hazardous materials or contaminated by hazardous materials.                                                                                                                                                             |
| Special precautionary measures: | Waste and containers must be disposed of in a safe manner. Empty containers and linings may contain product residues. Prevent released material from dispersing or flowing away and from coming into contact with soil, surface waters and drainage system. |

**Section 14****Transport information**

- 14.1 UN number

| ADR/RID       | ADN           | IMDG          | IATA          |
|---------------|---------------|---------------|---------------|
| Not regulated | Not regulated | Not regulated | Not regulated |

- 14.2 UN proper shipping name

| ADR/RID | ADN | IMDG | IATA |
|---------|-----|------|------|
| -       | -   | -    | -    |

- 14.3 Transport hazard class

| ADR/RID | ADN | IMDG | IATA |
|---------|-----|------|------|
| -       | -   | -    | -    |

■ 14.4 Packing group

| ADR/RID | ADN | IMDG | IATA |
|---------|-----|------|------|
| -       | -   | -    | -    |

■ 14.5 Environmental hazards

| ADR/RID | ADN | IMDG | IATA |
|---------|-----|------|------|
| No      | No  | No   | No   |

**Additional information:**

| ADR/RID | ADN | IMDG | IATA |
|---------|-----|------|------|
| -       | -   | -    | -    |

■ 14.6 Special precautions for user

Not available

## Section 15

### Regulatory information

- 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

|                                                                                                                                                                        |                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>EC regulation (EC) No. 1907/2006 (REACH):</b><br><br><b>Annex XIV - List of substances subject to authorization:</b><br><br><b>Substances of very high concern:</b> | None of the components are listed. |
| <b>Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles:</b>                             | Not applicable                     |

#### Other regulations

|                                                                           |                                                                                                                                                      |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>REACH Status:</b>                                                      | The company, as identified in Section (>>> "Section 1" Page 101), sells this product in the EU in compliance with the current requirements of REACH. |
| <b>US inventory (TSCA 8b):</b>                                            | All components are listed or exempted.                                                                                                               |
| <b>Australian inventory (AICS):</b>                                       | All components are listed or exempted.                                                                                                               |
| <b>Canadian inventory:</b>                                                | At least one component is not listed.                                                                                                                |
| <b>Inventory of Existing Chemical Substances in China (IECSC):</b>        | All components are listed or exempted.                                                                                                               |
| <b>Japanese inventory of Existing and New Chemical Substances (ENCS):</b> | All components are listed or exempted.                                                                                                               |
| <b>Korean Existing Chemicals Inventory (KECI):</b>                        | All components are listed or exempted.                                                                                                               |

| Other regulations                                                  |                                                                                                                                                                        |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Philippine Inventory of Chemicals and Chemical Substances (PICCS): | All components are listed or exempted.                                                                                                                                 |
| Taiwan Existing Chemical Substance Inventory (CSNN):               | Not determined.                                                                                                                                                        |
| National regulations                                               |                                                                                                                                                                        |
| Water hazard classification:                                       | 1, Annex no. 4 (classification acc. to the German Administrative Regulation on the Classification of Substances Hazardous to Waters into Water Hazard Classes (VwVwS)) |

■ 15.2 Chemical safety assessment

This product contains substances for which Chemical Safety Assessments are still required.

## Section 16

### Other information

#### Abbreviations and acronyms:

ADN: European Provisions concerning the International Carriage of Dangerous Goods by Inland Waterway

ADR: The European Agreement Concerning the International Carriage of Dangerous Goods by Road

ATE: Acute Toxicity Estimate

BCF: Bioconcentration Factor

CAS: Chemical Abstracts Service

CLP: Classification, Labelling and Packaging Regulation [Regulation (EC) no. 1272/2008]

CSA: Chemical Safety Assessment

CSR: Chemical Safety Report

DMEL: Derived Minimal Effect Level

DNEL: Derived No Effect Level

DPD: Dangerous Preparations Directive [1999/45/EC]

DSD: Dangerous Substances Directive [67/548/EEC]

EINECS: European Inventory of Existing Commercial Chemical Substances

ES: Exposure Scenario

EUH statement: CLP-specific hazard statement

EWG: European Waste Catalog

GHS: Globally Harmonized System of Classification and Labelling of Chemicals

IATA: International Air Transport Association

IBC: Intermediate Bulk Container

IMDG: International Maritime Dangerous Goods

LogPow: Decadic logarithm of the octanol/water partition coefficient

MARPOL 73/78: International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution)

OECD: Organisation for Economic Co-operation and Development

PBT: Persistent, Bioaccumulative and Toxic

PNEC: Predicted No Effect Concentration

RID: The Regulations concerning the International Carriage of Dangerous Goods by Rail

RRN: REACH Registration Number

SADT: Self-Accelerating Decomposition Temperature

SVHC: Substances of Very High Concern

STOT-RE: Specific Target Organ Toxicity - Repeated Exposure

STOT-SE: Specific Target Organ Toxicity - Single Exposure

TWA: Time-Weighted Average

UN: United Nations

UVCB: Complex hydrocarbon substance

VOC: Volatile Organic Compounds

vPvB: Very Persistent and Very Bioaccumulative

|                                                |                |
|------------------------------------------------|----------------|
| <b>Full text of abbreviated H-phrases:</b>     | Not applicable |
| <b>Full text of classifications [CLP/GHS]:</b> | Not applicable |
| <b>Full text of abbreviated R-phrases:</b>     | Not applicable |
| <b>Full text of classifications [DSD/DPD]:</b> | Not applicable |

| History                            |                     |
|------------------------------------|---------------------|
| <b>Issue date / Revision date:</b> | 19/08/2015          |
| <b>Date of previous issue:</b>     | 01/04/2015          |
| <b>Created by:</b>                 | Product Stewardship |

| Notes for the reader                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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## 11 KUKA Service

### 11.1 Requesting support

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b> | This documentation provides information on operation and operator control, and provides assistance with troubleshooting. For further assistance, please contact your local KUKA subsidiary.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Information</b>  | <p><b>The following information is required for processing a support request:</b></p> <ul style="list-style-type: none"> <li>■ Description of the problem, including information about the duration and frequency of the fault</li> <li>■ As comprehensive information as possible about the hardware and software components of the overall system</li> </ul> <p>The following list gives an indication of the information which is relevant in many cases:</p> <ul style="list-style-type: none"> <li>■ Model and serial number of the kinematic system, e.g. the manipulator</li> <li>■ Model and serial number of the controller</li> <li>■ Model and serial number of the energy supply system</li> <li>■ Designation and version of the system software</li> <li>■ Designations and versions of other software components or modifications</li> <li>■ Diagnostic package <b>KrcDiag</b>:<br/>           Additionally for KUKA Sunrise: Existing projects including applications<br/>           For versions of KUKA System Software older than V8: Archive of the software (<b>KrcDiag</b> is not yet available here.)</li> <li>■ Application used</li> <li>■ External axes used</li> </ul> |

### 11.2 KUKA Customer Support

|                     |                                                                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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